

**SOCIOECONOMIC IMPACT EVALUATION OF THE F/B PAGBABAGO
LIVELIHOOD DEVELOPMENT PROGRAM: *Evidence from a Quasi-Experimental
Study in Coastal Fishing Communities of the Philippines***

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LIST OF ABBREVIATIONS

BFAR – Bureau of Fisheries and Aquatic Resources
DOI – Durables Ownership Index
FB – Fisherfolk-Beneficiary
F/BPLDP – F/B Pagbabago Livelihood Development Program
FIMA – Fishing Income Monthly Average
DOI –Durables Ownership Index
GSIS – Government Service Insurance System
ICI – Insurance Coverage Index
IPW – Inverse Probability Weighting
LCI – Living Condition Index
MWI – Material Well-being Index
NB – Non-Beneficiary
NFE – No Formal Education
PD – Proportional Difference
PEI – Program Evaluation Index
PPS – Probability Proportional to Size
RPI – Real Property Index
SD – Standard Deviation
SLF – Sustainable Livelihoods Framework
SMD – Standardized Mean Difference
SSI – Social Services Index
SSS – Social Security System
TIMA – Fishing Income Monthly Average

EXECUTIVE SUMMARY

This study presents a comprehensive evaluation of the F/B Pagbabago Program (F/BPLDP), a large-scale asset-based livelihood intervention initiated by the Bureau of Fisheries and Aquatic Resources (BFAR) that involved the nationwide distribution of motorized fiberglass boats to small-scale fisherfolk. Using a mixed-methods design grounded in the Sustainable Livelihoods Framework (SLF), the research assessed both the material and perceived socio-economic impacts of the program on beneficiary households, comparing their current status (2024) with recalled benchmark conditions prior to receiving the boats.

Survey data from over 1,100 fisherfolk—including both beneficiaries (FBs) and non-beneficiaries (non-FBs)—provided evidence of modest but consistent improvements in key areas of household well-being. Results indicated notable gains in the ownership of consumer durables, improved access to electricity and drinking water, slight upgrading of housing structures, and increased—but still limited—access to social protection. Comparative analyses revealed that FBs had significantly better outcomes than non-FBs across several indices of socio-economic standing.

The study introduced a novel Material Well-being Index (MWI), synthesized from four sub-indices (durables, utilities, property, and insurance). Results showed a higher MWI for FBs compared to non-FBs, reinforcing the program's potential contribution to asset accumulation. A perception-based evaluation also revealed high levels of satisfaction among FBs, with the Program Evaluation Index (PEI) averaging 4.36 out of 5—interpreted as “Very Satisfactory.” Regression analysis found that shared ownership and better boat condition were strong predictors of more favorable program assessments. Notably, regional differences emerged: FBs from Visayas and Mindanao reported higher satisfaction levels and greater socioeconomic gains than those from Luzon, suggesting varying degrees of program implementation quality and local government support.

Despite these promising outcomes, persistent gaps were noted in housing tenure, long-term income sustainability, and insurance coverage. While PhilHealth remains the most accessed insurance product among FBs, coverage remains far from universal.

Additionally, the high number of boats under repair at the time of the survey signals critical issues in continuity of benefits and post-distribution support.

In response to these findings, the study advances several policy recommendations:

1. Institutionalize post-distribution maintenance support mechanisms.
2. Promote shared ownership schemes for better community cohesion.
3. Address geographic disparities with context-sensitive implementation.
4. Integrate financial literacy and insurance enrollment into future rollouts.
5. Sustain and expand monitoring and evaluation systems.
6. Support complementary interventions to enhance housing and tenure security.
7. Establish baseline data systems to enable longitudinal impact tracking.

The study concludes that the F/BPLDP is a valuable entry point into more sustainable and inclusive fisheries development. While the boats themselves offer immediate gains in productivity and dignity, the program's long-term success depends on improved governance, adaptive support systems, and holistic integration into broader rural development strategies.

1.0 INTRODUCTION

1.1 Rationale of the Project

Municipal fisheries are a critical pillar of the Philippine economy and food system, sustaining the livelihoods of millions of coastal households. However, municipal fisherfolk remain among the most economically vulnerable groups in the country. Chronic poverty, limited access to productive assets, exposure to climate and environmental risks, and inadequate institutional support continue to constrain their development. These structural challenges necessitate targeted and evidence-driven interventions to support livelihood security and improve long-term resilience.

In response, the Bureau of Fisheries and Aquatic Resources (BFAR) deployed the F/B Pagbabago Livelihood Development Program (F/BPLDP)—a national-scale initiative aimed at revitalizing small-scale fisheries through the distribution of Fiberglass Reinforced Plastic (FRP) boats to qualified municipal fisherfolk. Said program also provided technical support and training to ensure safe and productive boat usage. By reducing entry barriers to fishing and improving mobility and asset ownership, the F/BPLDP aspires to enhance income generation and promote inclusive development in vulnerable coastal areas.

While anecdotal reports suggest that the program has improved fishing productivity and community morale, empirical valuation of its socioeconomic outcomes remain limited. This project addresses this gap by conducting a systematic assessment of the F/BPLDP's socioeconomic impact. It compares fisherfolk-beneficiaries (FBs) of the program with non-beneficiaries (NBs) based on structured livelihood indicators and determine changes over

time through a benchmark-versus-current approach. The project also integrates fisherfolk perceptions to evaluate the perceived relevance, social impact, and sustainability of the F/BPLDP as an asset-based intervention mechanism.

Empirical studies from similar interventions in Southeast Asia have underscored the importance of asset-based programs in transforming small-scale fisheries. For instance, boat rehabilitation and distribution programs in Indonesia post-tsunami recovery efforts were found to significantly improve fishers' income and community cohesion, albeit only when accompanied by institutional support and technical training (Walley et al., 2019). Impact assessment of the Fisheries Development Program in Bangladesh, likewise, revealed that asset grants coupled with capacity-building led to improved productivity and household income, though long-term sustainability depended on both market access and community-based management (Haque et al., 2020). In the Philippines, earlier evaluations of community-based resource management projects (e.g., Pomeroy & Carlos, 1997; Bene et al., 2016) found that livelihood interventions are more likely to succeed when integrated with participatory governance and environmental safeguards.

Anchored in the Sustainable Livelihoods Framework (SLF), the project employed the mixed-methods design and uses weighting techniques to minimize selection bias. In doing so, it generates empirical (science-based) evidence to inform future program planning, as the same provides policy guidance for enhancing the delivery, targeting, and sustainability of livelihood assistance programs in the Philippine fisheries sector.

Building on these insights, this project positions the F/BPLDP not merely as an asset distribution exercise, but more importantly, as a potential catalyst for inclusive and resilient fisheries development—provided that the enabling systems are in place. By employing an evidence-informed evaluation, the project aims to inform BFAR and its partners on how to strategically refine the program's implementation strategies to effect broader, deeper, and more sustainable impacts.

1.2 Objectives of the Project

The general objective of the project was to assess the socioeconomic impact of the F/B Pagbabago Livelihood Development Program on fisherfolk-beneficiaries. Specifically,

The specific objectives were to:

1. Describe the fisherfolks' demographic and socioeconomic characteristics;
2. Assess the economic and material impact of the F/BPLDP on its beneficiaries using objective indicators and gain scores;
3. Evaluate the implementation and short-term outcomes of the program based on perceived relevance, social impact, and sustainability; and
4. Formulate policy recommendations to strengthen the implementation and sustainability of the program.

1.3 Theoretical Framework

This project is grounded in the Sustainable Livelihoods Framework (SLF; Figure 1.1), which conceptualizes how individuals and households navigate vulnerabilities and pursue improved well-being by leveraging various forms of capital—human, physical, financial, social, and natural. These capitals interact within a broader institutional and environmental context, shaping livelihood strategies and development trajectories.

The F/BPLDP aligns strongly with this framework, particularly by enhancing physical capital (fishing boats), human capital (training), and to a lesser extent, financial and social capital (through income gains and community cooperation). By addressing structural vulnerabilities—such as lack of assets, low income, and limited mobility—the program aims to facilitate more stable and productive livelihoods for small-scale fisherfolk.



Figure 1.1 | Theoretical foundation of the project based on the Sustainable Livelihoods Framework, illustrating the role of evaluation in informing program performance.

This project situates itself within the SLF as a critical feedback mechanism. As shown in Figure 1.1, the project assesses how program inputs and capital improvements translate into livelihood outcomes, and whether these outcomes align with sustainable development goals. By employing empirical methods to analyze objective indicators (e.g., income, asset ownership) and subjective perceptions (e.g., relevance, social impact, sustainability), this project delivers evidence-based insights that feed back into the system, thereby informing future policy, improving program design, and enhancing its overall sustainability.

1.4 Conceptual Framework

The conceptual framework (Figure 1.2) guiding this project is structured around the logic model that connects the core components of the F/BPLDP to measurable livelihood outcomes. The key inputs from the program—fiberglass boat distribution and the technical

training—are intended to address physical and human capital limitations among fisherfolk. These inputs interact with various moderating contextual variables including demographic and socioeconomic profile, livelihood experience, and specific attributes of the asset (e.g., boat design, condition, ownership model, etc.). Geographic location is likewise accounted for, acknowledging cultural and sociopolitical distinctions in implementation environments and livelihood systems.

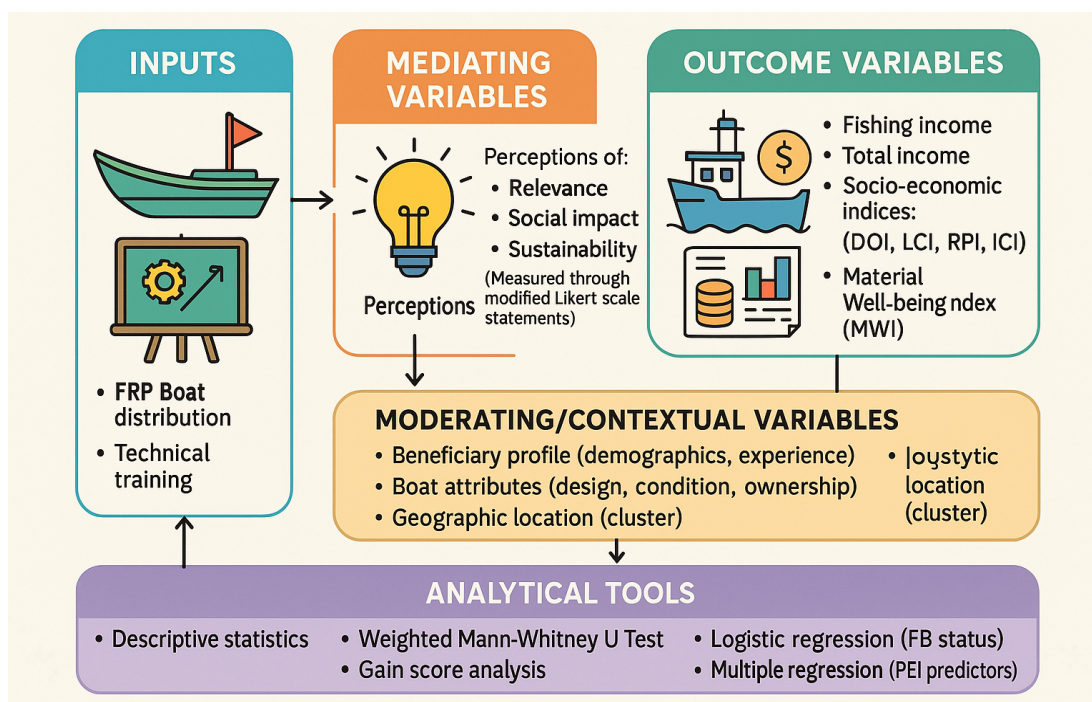


Figure 1.2 | Conceptual framework illustrating the interaction between inputs, contextual variables, perceptions, and outcomes in evaluating the F/BPLD Program.

The program’s effect (impact) on outcomes is mediated by beneficiaries’ perceptions of program quality and relevance. These perceptions were measured using an 18-item Likert instrument comprising reworded, impact-oriented statements, designed to capture three thematic dimensions: Relevance, Social Impact, and Sustainability. These attitudinal constructs offer insight into the experiential and subjective dimensions of the program’s effectiveness and overall performance.

The intended outcome variables include both economic and material dimensions of well-being. Fishing and total income reflect financial capital, whereas material well-being is measured through structured indices covering consumer durables, basic utilities, real property, and insurance coverage. These were further synthesized into a composite Material Well-being Index (MWI). Change over time was captured through gain scores comparing benchmark and current values.

To assess the program's broader performance, a Program Evaluation Index (PEI) was derived as a weighted average of the three perception-based constructs. The project used a range of statistical methods—including descriptive analysis, weighted Mann–Whitney U tests, gain score analysis, weighted logistic regression, and linear regression—to examine for differences between FBs and NBs and identify predictors of program participation and perceived effectiveness.

The framework emphasizes how livelihood inputs, contextual and perceptual factors, and outcome measures are interlinked in the evaluation of asset-based interventions. The model supports a multi-dimensional and evidence-based approach to assessing program outcomes, informing future design and implementation strategies.

2.0 METHODOLOGY

2.1 Research Design and Sample Selection

2.1.1 Research Design

This research project employed the quantitative, quasi-experimental design, utilizing a non-equivalent group comparison framework to evaluate the socioeconomic impact and perceived quality of the F/B Pagbabago Livelihood Development Program (F/BPLDP). This research focused on assessing differences between fisherfolk-beneficiaries (FBs) who are recipients of fiberglass boats through the F/BPLDP, and non-beneficiaries (NBs) as control group i.e., non-participants of the program. In the absence of random assignment, a quasi-experimental approach was employed to approximate the conditions of a counterfactual comparison. The research was cross-sectional in nature, albeit it covered benchmark and current reference periods to assess the socio-economic change among FBs.

Primary data were collected using a structured household survey, complemented by administrative records and secondary information provided by the Bureau of Fisheries and Aquatic Resources (BFAR). The research adopted the Sustainable Livelihoods Framework (SLF) as guide to structure both asset-based and perception-based analyses.

2.1.2 Sample Size Determination

The theoretical sample size (n_0) for FBs was initially calculated with Cochran's (1977) formula for proportions, with a 95% confidence level, 5% margin of error, and an assumed

population proportion of 0.5. Subsequently, a finite population correction (FPC) was used to adjust the sample size based on a known population (N) of FBs provided by BFAR.

According to BFAR records, a total of 6,452 FBs received fiberglass boats from 2016 to 2019. This constituted the statistical universe. To arrive at the accessible population, regions with fewer than 100 FBs were excluded—namely the National Capital Region (NCR, with 99 FBs) and Region 4B (25 FBs). Region 11, despite having 118 FBs on record, was not included due to logistical constraints. The exclusion of these regions reduced the eligible FB population to 5,130, which was used to adjust the minimum sample size. In this regard, applying Cochran’s formula with FPC yielded a minimum sample size (n_a) of 358, plus 10% buffer to account for potential attrition, resulting in a final target of 394 FBs.

To determine sample size allocation, sampling proportional to size (PPS) was applied at the regional level using the distribution of FBs per region. The share for each region to the 394 target was calculated and rounded to identify the minimum sample size per region. The resulting sample distribution is summarized in Table 1.

Table 1 – Sample distribution of FBs per region based on sampling proportional to size.

REGION	NO OF ELIGIBLE MUNICIPALITIES	TOTAL FBs *	% ALLOCATION	MINIMUM SAMPLE SIZE	ACTUAL SAMPLE SIZE	NO. OF COVERED MUNICIPALITIES
CAR	4	160	3	13	20	1
R1	5	163	3	13	18	1
R2	31	2,041	41	160	155	5
R3	7	242	5	19	13	1
R4A	9	312	6	25	62	4
R5	4	131	3	10	12	1
R6	13	860	17	68	172	5
R7	4	226	5	18	40	1
R10	28	579	12	46	38	3
R11	4	0	0	0	0	0
R12	11	298	6	23	32	2
TOTAL	120	5,012	100	394	562	24

* Based on the records obtained from BFAR

Municipality Selection Process. Once the minimum required sample per region was established, municipalities were selected based on the following guidelines:

1. If the regional minimum sample size was less than the average number of FBs per municipality, only one municipality was selected—the one with the highest number of FBs—to maximize response potential.
2. If the minimum sample size exceeded the average FB count per municipality, multiple municipalities were selected by choosing those with the highest FB counts. In cases of tied values for a spot, random selection was employed.
3. If the count of FBs across municipalities were relatively similar (e.g., Region 10), the required number of municipalities was randomly selected from the list.

A total of 24 municipalities across ten regions (Figure 2.1.1) were selected for field visits. This included areas from CAR, Regions 1–7 (excluding Region 4B), Region 10, and Region 12. Notably, Regions 8 and 9 did not appear in the database of BFAR and were presumed not to have been covered by the program. Chosen municipalities were scheduled for field visits. Through coordination with regional offices of BFAR and the LGUs, FBs were invited to participate in a seminar and survey. All FBs who attended on the scheduled activity were included in the study. To establish a comparison group, focal persons from the LGUs were also requested to invite an approximately number of non-beneficiary (NB) fisherfolk from the same localities. This yielded a total of 562 FBs and 760 NBs, both exceeding the minimum targets.

2.1.3 Sampling Technique

A stratified cluster sampling design with purposive elements was applied to capture geographic differences and ensure comparability between FB and NB group. The primary strata were based on geographic clusters corresponding to BFAR's operational coverage: Northern Luzon, Southern Luzon, Visayas, and Mindanao.

Within each cluster, coastal municipalities were selected purposively using sampling proportional to size (PPS), based on the regional distribution of FBs. Selection of municipalities followed structured guidelines to ensure that the required minimum sample per region were met while maximizing field efficiency and representativeness.

FBs were identified through the official BFAR lists of beneficiaries covering the years 2016 to 2019. Where possible, random selection of FBs within municipalities was applied; however, participation ultimately depended on attendance during the scheduled field visits and LGU mobilization. To establish a comparison control group, non-beneficiary fisherfolk from the same or neighboring communities, approximating the number of FBs per area to support matched group comparisons.

Although the national sample of fisherfolk-respondents was not randomly drawn, the use of stratified sampling and proportional allocation allowed for meaningful geographic representativeness. The matched comparison structure also supported the use of Inverse Probability Weighting (IPW) during analysis to correct for potential selection bias and covariate imbalance between FB and NB groups.

2.2 Survey Instrument

The primary research instrument employed in this study was a structured survey questionnaire (SQ) designed to elicit both quantitative and qualitative data from fisherfolk-respondents. The instrument was organized into five (5) major sections: (1) socio-demographic characteristics; (2) fisherfolk livelihood profile; (3) a checklist of indicators relevant to the assessment of socio-economic conditions; (4) beneficiary profiling in relation to program involvement; and (5) a set of Likert-scale items developed specifically for fisherfolk-beneficiaries (FBs) to evaluate the F/B Pagbabago Program. The instrument was carefully designed to capture both individual-level and community-level information aligned with the study's objectives.

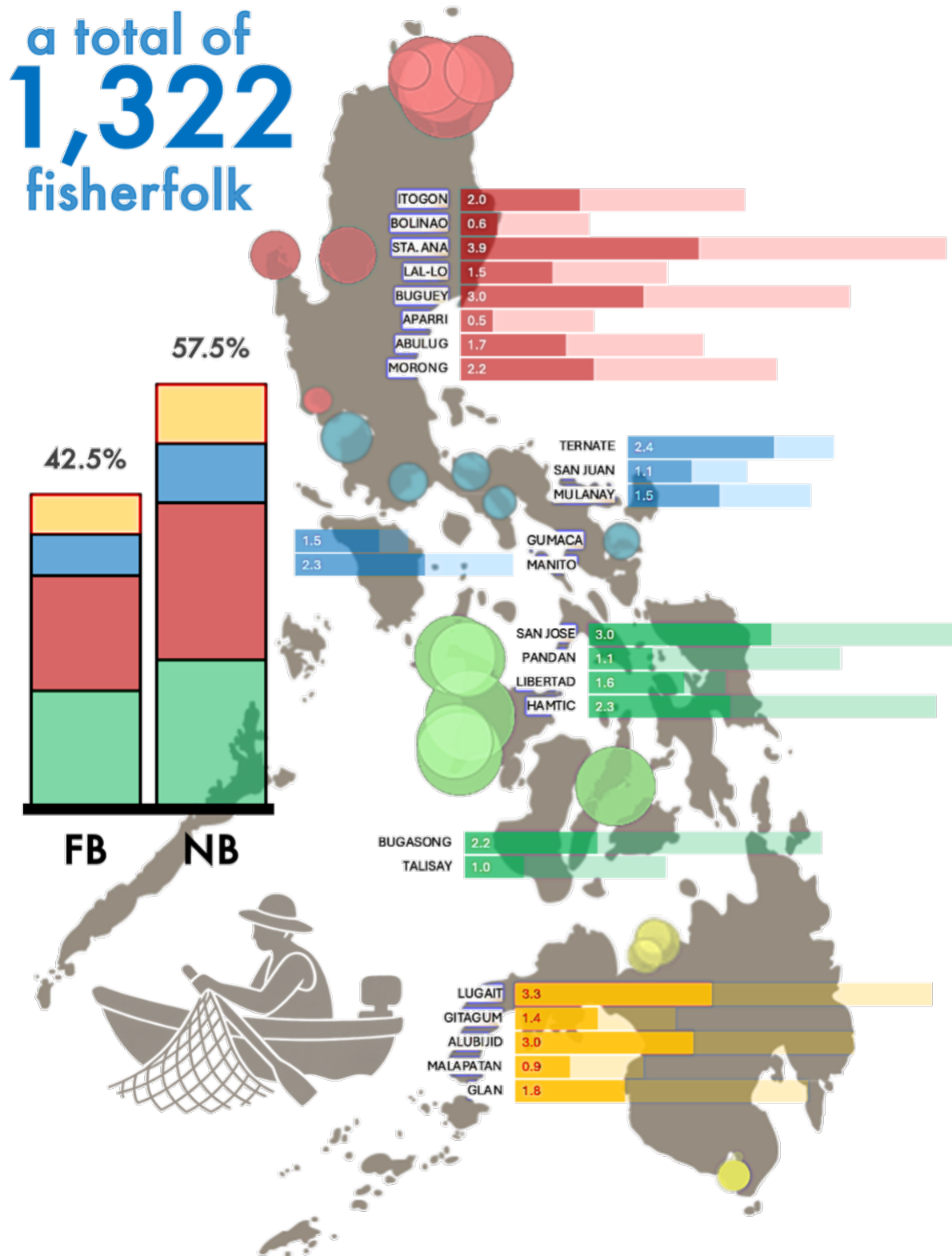


Figure 2.1.1 Distribution map of fisherfolk-respondents from 24 municipalities across nine regions spanning Northern and Southern Luzon, Visayas, and Mindanao.

To ensure clarity and accessibility, the SQ was prepared in both English and Tagalog versions. It was administered through face-to-face interviews conducted during scheduled field visits, which enabled the research team to provide necessary clarifications, facilitate accurate responses, and encourage participation even among respondents with limited literacy. A copy of the final questionnaire is provided in Appendix C.

2.2.1 Construction of Socio-economic Indices

To evaluate the socio-economic standing of fisherfolk-respondents, a set of multi-dimensional indices were derived using the data gathered through a structured checklist-type survey instrument. The indices included the Durables Ownership Index (DOI), Living Condition Index (LCI), Real Property Index (RPI), and Insurance Coverage Index (ICI). These indices served as proxy measures for different aspects of household-level capital assets and were constructed to reflect the lived realities of fisherfolk communities – consistent with the established practices in asset-based welfare assessment (Filmer & Pritchett, 2001; Vyas & Kumaranayake, 2006).

Each index was derived based on the presence or absence of relevant household assets or conditions. The DOI measured ownership of durable goods i.e., vehicles, home appliances, and digital tools. The LCI captured indicators of living condition, that is, access to basic services, such as electricity, potable water, and internet. The RPI reflected property-related attributes such as house and/or lot ownership and housing quality, while the ICI was based on enrollment in formal insurance programs related to health and retirement.

For each domain, responses were assigned weights based on the estimated average values (in Peso) or relative importance, and index scores were expressed in standardized form (0–100 scale) by min-max normalization. This method preserved relative differences in asset value while ensuring comparability across households and between indicators (Howe et al., 2012). Final scores for sub-indices reflected the socioeconomic value of household assets.

To provide a broader measure of household-level socio-economic status, the Material Well-Being Index (MWI) was computed as a weighted aggregate of the four sub-component indices. The following weights were considered based on theoretical grounding and expert consultation: DOI (30%), LCI (30%), RPI (25%), and ICI (15%). These weights underscore the relative importance of each dimension in determining the material security and asset-based capacity of fisherfolk households. The MWI, therefore, serves as a composite score that allows for the comparative assessment of material well-being between fisherfolk-beneficiaries and their non-beneficiary counterparts.

The development and utility of these indices aligns well with the SLF, particularly in operationalizing physical and financial capital dimensions. The use of the indices served as basis for statistical comparison but also provided a foundation for exploring the relationship between material well-being and program outcomes.

2.2.2 Computation of Socio-Economic Gain Scores

To assess the change in socio-economic condition of FBs following their participation in the F/BPLD Program, Gain Scores were computed based on the difference in index values between two reference points: 2016–2019 (benchmark) and 2023–2024 (current). The analysis of gain scores allowed for the quantification of asset-based improvements at the household level, while providing an empirical measure of material well-being outcome attributed to the intervention.

Gain Scores were calculated for each of the four asset-based sub-indices (DOI, LCI, RPI, and ICI). For each respondent, the benchmark score was subtracted from the corresponding current score, with the resulting difference representing the net change. Positive values indicated improvement, while negative or zero values denoted stagnation or potential decline.

In addition to individual sub-index gains, an overall MWI Gain Score was derived by aggregating the weighted sub-index changes, following the same composite structure used for the Material Well-Being Index (MWI): DOI (30%), LCI (30%), RPI (25%), and ICI (15%). This allowed for the assessment of holistic change in material well-being, with scores standardized to a 0–100 scale for comparability.

The use of Gain Scores enabled the study to evaluate the tangible socio-economic outcomes of the F/BPLD Program among its direct beneficiaries. The scores were utilized in subsequent descriptive and inferential analyses to determine the distribution and significance of observed changes across different attributes of the fisherfolk-respondents and geographic clusters.

2.2.3 Development of the Program Evaluation Index

To assess respondents' perceptions of the performance of the F/BPLDP, a structured set of statements was developed using a five-point Likert scale:

- 5 – Strongly Agree;
- 4 – Agree;
- 3 – Tend to Agree;
- 2 – Disagree;
- 1 – Strongly Disagree.

A total of 18 item-indicators were designed to evaluate perceptions across three core dimensions: Program Relevance, Social Impact, and Sustainability. These thematic areas were derived from the program's stated objectives and were contextualized to reflect the conditions of small-scale fisheries and government-led livelihood interventions. The Likert statements used to capture each construct are provided below.

Relevance (7 statements)

These statements capture awareness, appropriateness, direct value, and willingness to endorse the program:

1. I was already aware of the F/B Pagbabago Livelihood Program before becoming a beneficiary.
2. The program has brought noticeable improvements to my livelihood or personal situation.
3. My engagement in the program has been affected by recurring challenges or limitations. *(reverse-coded)*
4. The program contributes meaningfully to the promotion of sustainable livelihoods for fisherfolk.
5. Being part of the program has led to clear improvements in my income or means of living.
6. Receiving a boat from the program has not significantly enhanced my fishing productivity or financial status. *(reverse-coded)*
7. I am willing to recommend the program to other members of my fishing community.

Social Impact (6 statements)

These assess safety, social relationships, and collective benefit:

8. The fiberglass boats provided through the program have enhanced our safety and well-being at sea.
9. Since receiving the boat, my volume of fish catch has increased significantly.
10. The program has strengthened cooperation and solidarity among fisherfolk in my area.
11. Participation in the program has improved how I am regarded in my local community.
12. The program encourages the exchange of knowledge and practical skills among fisherfolk.
13. The program has contributed to overall improvements in both social and economic aspects of my life.

Sustainability (5 statements)

These statements reflect environmental impact, implementation structure, and long-term viability:

14. The boat distribution component of the program may negatively affect the marine environment or fish stocks. *(reverse-coded)*
15. The fiberglass boats are high-quality, cost-effective, and reliable for long-term fishing use.
16. The program is being implemented in an organized, consistent, and well-structured manner across levels.
17. The program is helping to build systems that support the long-term sustainability of the fishing sector.
18. The program should be discontinued due to lack of support from local government units. *(reverse-coded)*

The Likert items underwent content validation by subject matter experts in fisheries development and social science to ensure their alignment with the program's goals and the Sustainable Livelihoods Framework (SLF) underpinning this study. The instrument was pilot-tested prior to actual fieldwork to assess clarity and applicability. Reliability analysis yielded a Cronbach's alpha of 0.81, indicating high internal consistency.

For analysis, item responses were numerically coded from 1 to 5, with higher scores reflecting more favorable perceptions of the F/BPLD Program. Negatively worded items were reverse-coded to maintain uniform score directionality. Three sub-indices were computed—one for each thematic dimension. For each respondent, raw scores were derived by summing applicable item responses per domain, then standardized to a 0–100 scale using min-max normalization to account for differences in the number of items across dimensions.

To provide a unified and interpretable measure of perceived program performance, a Program Evaluation Index (PEI) was constructed. The PEI was computed as the average

of the three sub-indices (Program Relevance, Social Impact, and Sustainability), resulting in a standardized composite index ranging from 0 to 100. Higher PEI value indicated more favorable overall perceptions of the program's efficiency, benefits, and long-term viability. The PEI, along with the sub-indices, were used as key explanatory variables in subsequent inferential analyses, enabling a structured evaluation of how FBs perceived the outcomes and effectiveness of the F/BPLD Program across different dimensions and contexts.

2.3 Statistical Treatment

2.3.1 Descriptive Analyses

Descriptive statistics was used to summarize the socio-demographic characteristics of respondents and other relevant background information. For nominal variables, such as sex, marital status, and membership in fisherfolk organizations, data were summarized using frequencies and percentages. Ordinal variables, including household size, income brackets, and educational attainment, were described using frequency distributions, with median and mode reported to indicate central tendency.

For continuous variables—such as age, fishing experience, and the computed index scores (e.g., DOI, LCI, RPI, ICI, and MWI)—standard descriptive measures were applied. These included measures of central tendency (mean, median), measures of dispersion (standard deviation, variance), and range-based indicators (minimum and maximum values) to assess variability and distributional characteristics. These descriptive techniques provided an initial overview of the sample and informed subsequent statistical analyses by identifying patterns, potential outliers, and distributional assumptions.

2.3.2 Propensity Score and Inverse Probability Weighting

To address potential selection bias in comparing fisherfolk-beneficiaries (FBs) and non-beneficiaries (NBs), propensity score estimation was employed. The propensity score

reflects the predicted probability of an individual being part of the treatment group ($FB = 1$) given a set of observed covariates. Propensity scores were estimated via binary logistic regression, with group membership as the dependent variable and relevant pre-treatment characteristics (e.g., socioeconomic benchmark data) as predictors.

Once the propensity scores were obtained, Inverse Probability Weighting (IPW) was done to adjust for baseline differences between FBs and NBs. Each fisherfolk-respondent was assigned a weight inversely proportional to the probability of receiving their observed treatment status i.e., FBs received a weight of $1/PS$ and NBs a weight of $1/(1-PS)$, where PS denotes the propensity score. This approach generates a weighted pseudo-population in which the distribution of covariates is independent of treatment assignment.

To ensure that the total weighted sample size remained comparable to the original sample and to stabilize variance in downstream analyses, the IPW values were normalized. This was accomplished by dividing each individual's IPW by the mean of all IPWs within their respective treatment group. Normalization preserved the relative influence of each observation, while preventing inflation of the effective sample size.

The normalized values of IPW were applied to test for covariate balance between the FB and NB groups. Balance diagnostics were conducted using common primary metrics: Standardized Mean Difference (SMD), which assessed balance in continuous covariates, and Proportion Difference (PD), which was applied to the categorical variables. These measures allowed for the evaluation of whether the weighted groups were statistically comparable in terms of observed pre-treatment characteristics.

Cluster-level adjustments were also incorporated during weight calculation to reflect the sampling structure. The resulting normalized and cluster-adjusted IPWs were applied in downstream weighted analyses, including regression models and estimation of average treatment effects. Such method aligns with recommended practices in quasi-experimental program evaluation where randomized assignment is not feasible.

2.3.3 Inferential Statistics

Inferential statistics were applied to compare outcomes between FBs and NBs, and to test associations between program-related variables and respondent characteristics.

The weighted Mann-Whitney U Test was used to compare FB and NB groups in terms of the socioeconomic indices (DOI, LCI, RPI, ICI, and the composite MWI) and gain scores. The same inferential test was applied to compare access to social services.

Multiple linear regression was employed to determine the association between PEI and fisherfolk characteristics i.e., livelihood experiences, Fishing Income Monthly Average (FIMA), and MWI. Binary logistic regression was used to examine the relationship between group membership (FB vs. NB) and potential predictors i.e., FIMA, Total Income Monthly Average (TIMA), and the individual socio-economic sub-indices (DOI, LCI, RPI, and ICI). The MWI was excluded from this model due to multicollinearity.

2.4 Ethics Statement

This study strictly adhered to ethical standards for research involving human participants. The data collection process utilized a non-invasive survey instrument and was carried out in accordance with the *Ethical Principles and Guidelines for Filipino Scientists* published by the National Academy of Science and Technology (NAST) of the Philippines. To ensure ethical compliance, the research protocols employed in this project were independently reviewed by the *de facto* Research Ethics Committee of the College of Fisheries, Laguna State Polytechnic University.

Participation in the survey was fully voluntary. All respondents were informed of the study's purpose, their rights as participants, and the option to withdraw at any point without penalty. Written informed consent was obtained from each participant prior to the interview. All collected data were treated with strict confidentiality.

3.0 RESEARCH FINDINGS

3.1 Fisherfolk-Respondents' Baseline Profile

The respondents of this survey were fisherfolk drawn from 24 coastal municipalities, across nine regions and clustered by geographic reference into Northern Luzon, Southern Luzon, Visayas, and Mindanao. In total, 1,322 fisherfolk-respondents were included in the survey, covering 562 fisherfolk-beneficiaries (FBs) and 760 non-beneficiaries (NBs). The frequency distribution of respondents disaggregated according to their geographic cluster and region of residence is summarized below (Figure 3A)

The baseline profile of the fisherfolk depicts an overview of their socio-demographic characteristics and economic–livelihood conditions. The same descriptive statistics serve as contextual anchors, offering insight into the general attributes of the sample population and establishing a measurable basis for comparability between program beneficiaries and non-beneficiaries prior to the livelihood intervention.

3.1.1 Socio-demographic Characteristics

The fisherfolk population was predominantly composed of individuals in the middle-aged to elderly brackets. Among FBs, the majority (41.6%) were aged 53–69, followed by 39.6% in the 36–52 bracket. In contrast, NB group was more distributed, but most of them were also concentrated in the 36–52 group (25.34%). NBs had a higher representation in the 18–35 bracket (12.56%) compared to FBs (13.17%) as shown in Figure 3B.

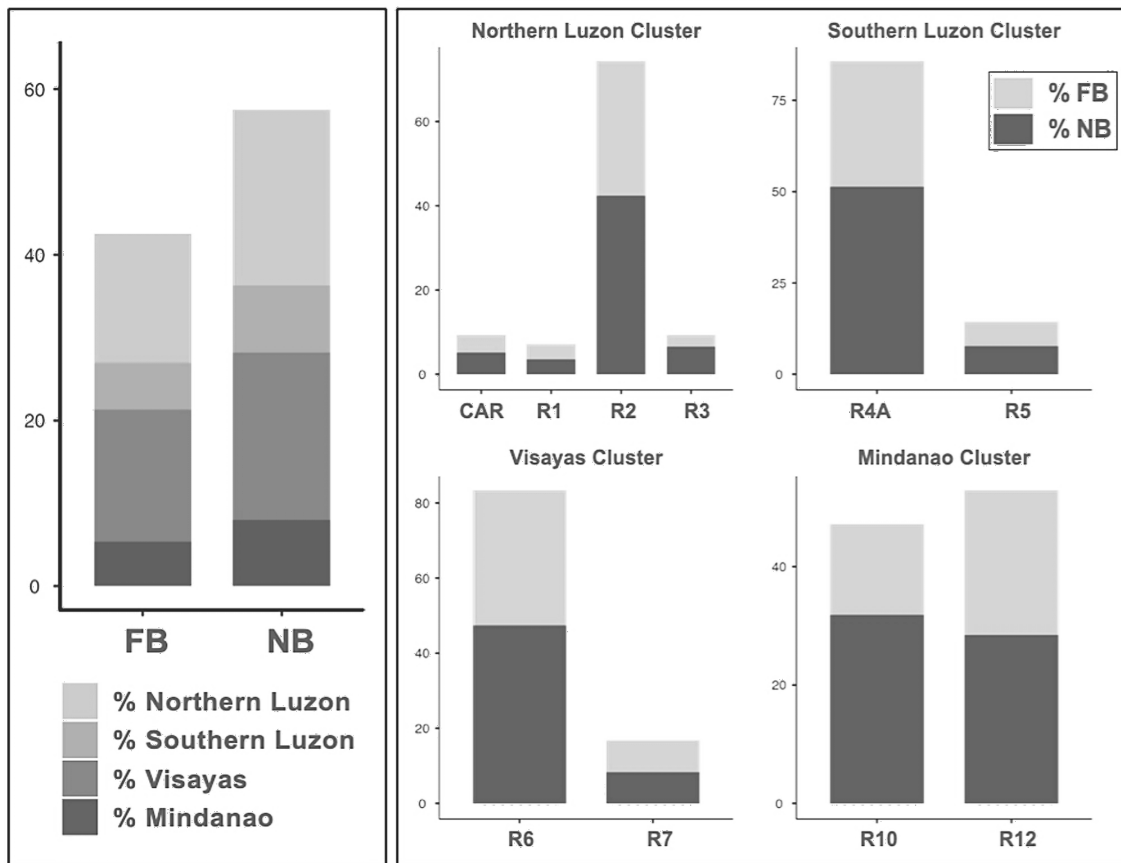


Figure 3.1.1a – Percent distribution of FB and NB disaggregated according to geographic cluster (% of the total) and region (% within the cluster).

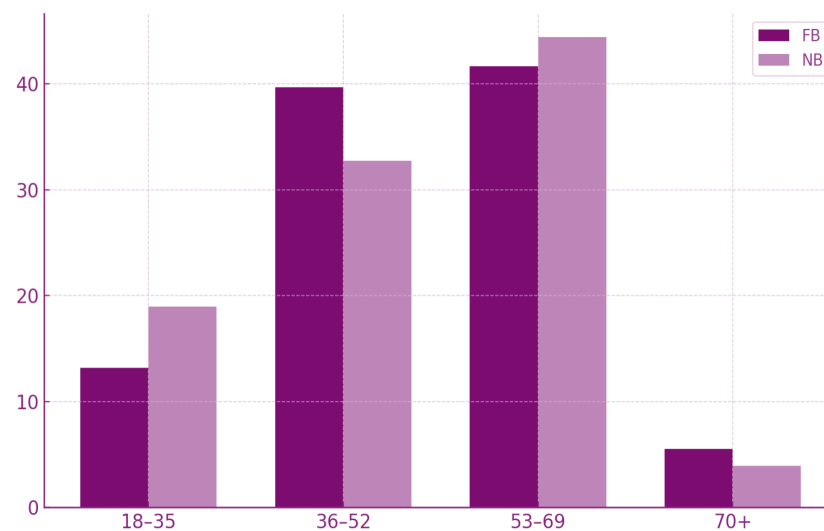


Figure 3.1.1b – Percentage distribution of FB and NB respondents by age bracket.

Beyond categorical groupings, measures of central tendency further highlight the demographic difference between the two groups. FB respondents had a mean age of 50.87 years, while NBs had a mean of 46.61 years. This apparent difference in the average age supports the observation that the FB segment of the fisherfolk population skews older. As visualized in Figure 3C, the age distribution curves of both groups reveal overlapping yet distinct patterns, with FBs tending toward older age intervals and NBs exhibiting a slightly broader spread across younger and middle-aged ranges. The advanced age profile among FBs appears to reflect program eligibility dynamics and a signal of limited generational succession within the sector.



Figure 3.1.1c – Density histogram comparing FB and NB group in terms of actual age.

The sex distribution of respondents reveals that both fisherfolk-beneficiaries (FBs) and non-beneficiaries (NBs) were predominantly male. As shown in the side-by-side pie charts (Figure 3D), 94.1% of FBs identified as male, while only 5.9% were female. Similarly, among NBs, 85.8% were male and 14.2% were female. This reflects the continuing male-dominated nature of small-scale fisheries in the Philippines, where men typically assume

primary fishing roles, while women often participate in ancillary activities such as fish processing, vending, or household support. The slight increase in female representation among NBs may also reflect broader livelihood diversification strategies in non-beneficiary households, where women are more engaged in alternative income-generating roles.

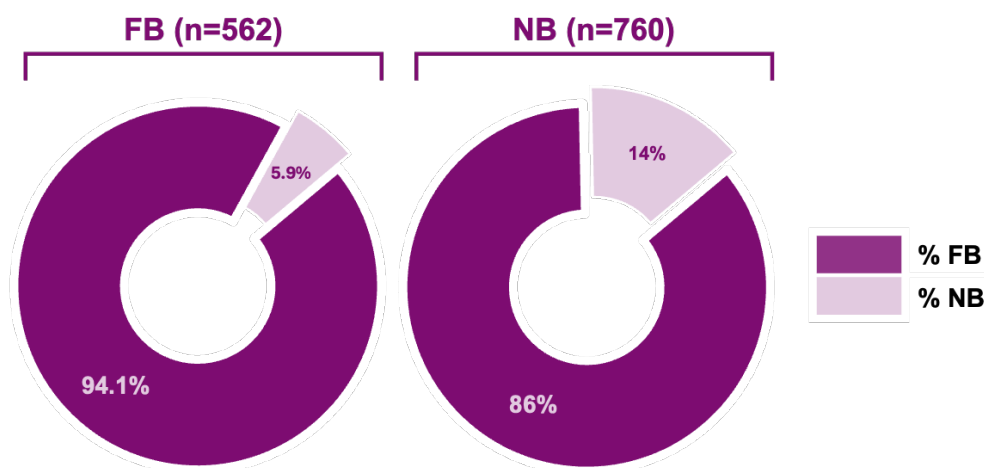


Figure 3.1.1d – Percentage distribution of FB and NB disaggregated by sex.

The marital status profile of fisherfolk respondents reveals that the vast majority were in a partnered relationship—either formally married or living with a partner. As shown in Figure 3E, 82.6% of FBs and 81.1% of NBs reported being either married or cohabiting, underscoring the family-oriented nature of small-scale fishing communities. A small proportion of respondents were categorized as single, comprising 13.0% of FBs and 15.1% of NBs. Meanwhile, only a minor share of both groups were widowed or separated—together accounting for 4.4% of FBs and 3.8% of NBs. These figures point to a general pattern of relational stability across both groups, with FBs exhibiting a slightly higher proportion of partnered respondents. This may be linked to the selection dynamics of the program, which likely favored household heads or individuals perceived as primary breadwinners. The data also suggest that livelihood interventions such as the F/B Pagbabago Program are being implemented within the broader context of family structures, with potential implications for intra-household decision-making, labor allocation, and the distribution of benefits.

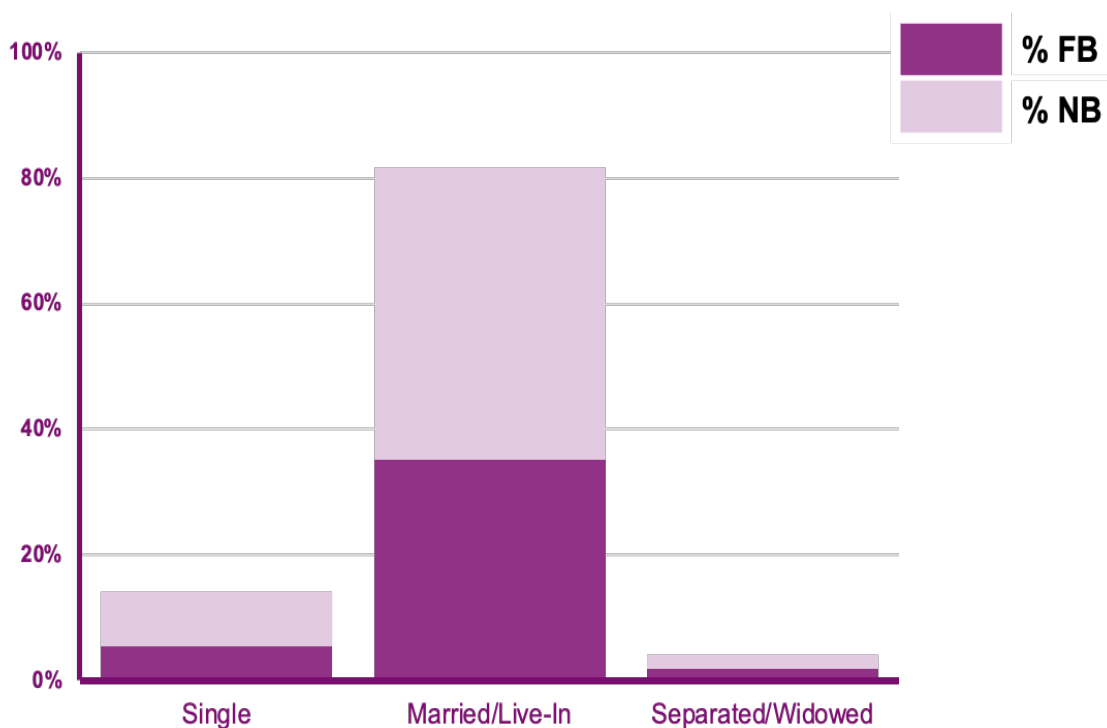


Figure 3.1.1e – Percentage distribution of FB and NB respondents categorized according to marital status.

Household sizes among respondents further support the characterization of fishing households as family-oriented but moderately sized. A significant portion of fisherfolk households contained 4 to 6 members—52.8% among FBs and 56.1% among NBs. Only a small number of respondents reported living in households with 10 or more members (4.1% FBs; 2.4% NBs), indicating that extended family living arrangements are less common in this context.

The educational profile of the respondents highlights notable similarities and slight differences between fisherfolk-beneficiaries (FBs) and non-beneficiaries (NBs). As illustrated in Figure 3F, both groups predominantly attained elementary and high school education levels, underscoring the widespread prevalence of basic education among coastal communities. Specifically, a substantial proportion of both FBs and NBs completed elementary education, followed closely by high school graduates.

However, a closer comparison reveals that NBs had a slightly higher proportion of respondents with at least a high school diploma, while FBs showed a relatively even distribution between elementary and high school levels. The proportion of college-educated individuals was modest across both groups, although slightly more prevalent among NBs. Post-graduate attainment remained low overall, reflecting limited access to higher education opportunities in coastal barangays. Notably, both groups recorded a small number of respondents with no formal education (NFE), indicating persistent educational vulnerability among a minority of fisherfolk households.

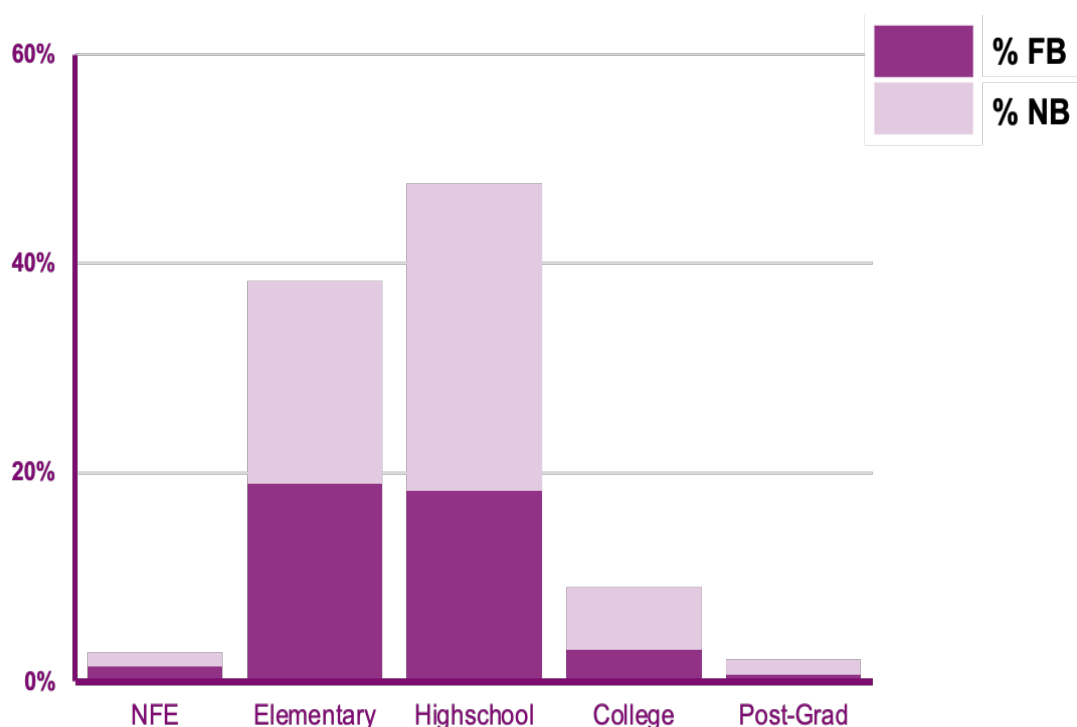


Figure 3.1.1f – Percentage distribution of FB and NB respondents by educational attainment.

These results emphasize the importance of strengthening educational access and lifelong learning programs in fisherfolk communities. Despite the overall attainment of basic education, limited representation in higher education may constrain access to diversified livelihood options and reduce resilience to socio-economic shocks.

3.1.2 Economic-Livelihood Indicators

Income-based variables painted a sobering picture of the financial precarity experienced by many fisherfolk. Among FBs, nearly one-third (69.4%) earned below PHP 10,000 per month from fishing alone, while the figure was even higher for NBs at 87.1%. Only a marginal fraction of respondents from either group reported average earning over PHP 30,000 monthly, which underscores the limited profitability of capture fisheries as a primary livelihood.

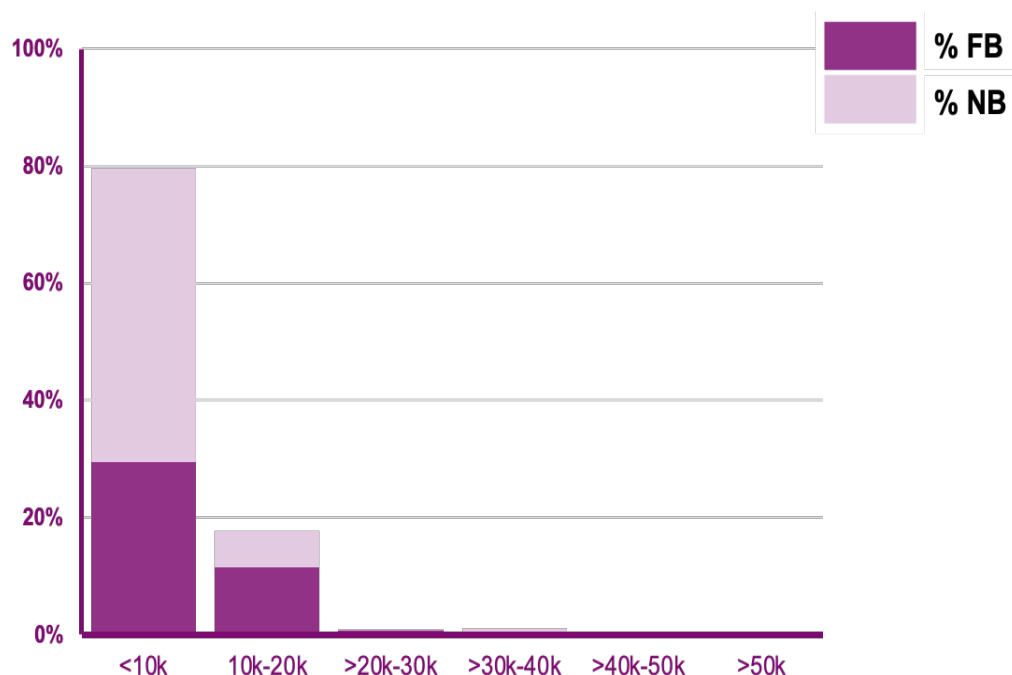


Figure 3.1.2a – Percentage distribution of FB and NB respondents categorized according to fishing income monthly average.

When other sources of income were considered, the total household income (Figure 3.8) improved somewhat. Still, the largest income bracket remained PHP 10,001 to PHP 20,000, which covered 62.6% of FBs and 79.5% of NBs. Slightly more FBs than NBs reported incomes in the PHP 20,001 to PHP 30,000 range (27.6% vs. 12.6%), hinting at modest income diversification or possible program spillovers in FB households.

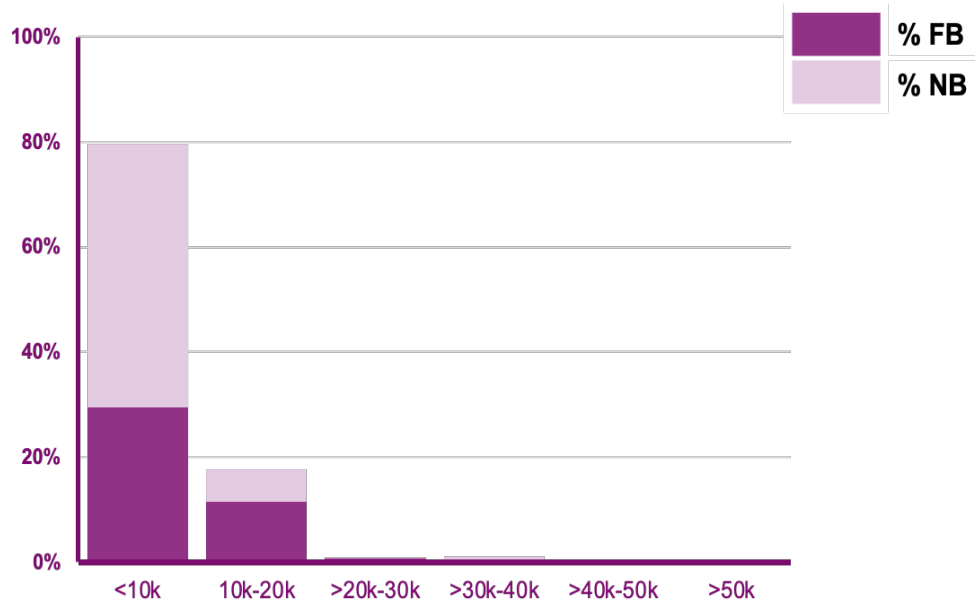


Figure 3.1.2b – Percentage distribution of FB and NB respondents categorized according to total income monthly average.

Years in fishing correlated strongly with these income patterns. Over 31% of FBs and 40% of NBs had been fishing for more than a decade, demonstrating long-term occupational commitment. Interestingly, a small but present proportion of respondents in both groups had been fishing for less than five years, suggesting either new entrants into the sector or returnees displaced from other livelihoods.

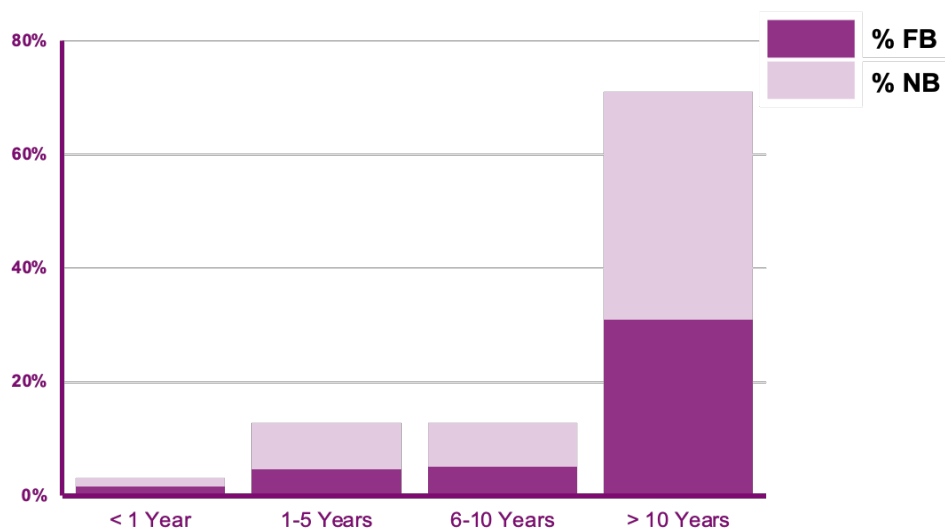


Figure 3.1.2c – Percentage distribution of FB and NB respondents classified based in the number of years engaged in fishing livelihood.

Organizational affiliation emerged as a potential differentiator in access to support services. About 25.8% of FBs and 34.5% of NBs reported being members of a fisherfolk organization, while the remaining respondents were unaffiliated. This gap in membership could affect awareness of programs and eligibility for certain interventions. Membership in formal fisherfolk groups often serves as a conduit for accessing training, credit, and livelihood support, as government and NGO-led initiatives are frequently channeled through organized associations. The lower organizational participation among FBs may indicate missed opportunities for collective action, resource pooling, and sustained engagement with institutional support systems.

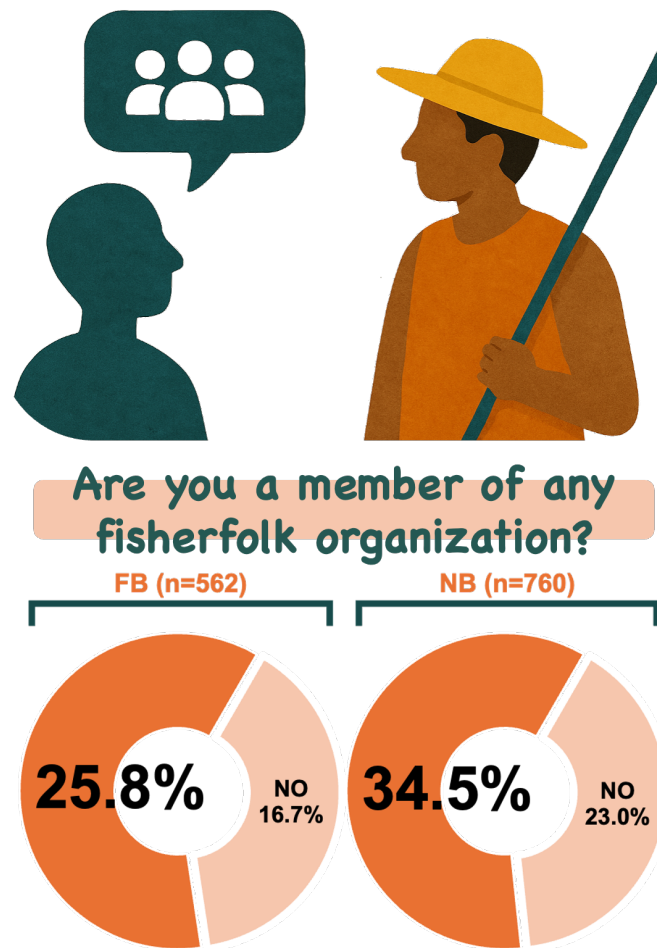


Figure 3.1.2d – Percentage of “Yes” and “No” response among FBs and NBs with regard to their membership in any fisherfolk organization.

The distribution of fishing operations underscores the predominance of small-scale, artisanal fishing across both groups. As shown in Figure 3.1.2e, a vast majority of both fisherfolk-beneficiaries (FBs) and non-beneficiaries (NBs) reported engaging primarily in artisanal fishing — comprising 33.3% of FBs and 51.1% of NBs. This reinforces the deeply entrenched role of municipal fisheries as the backbone of coastal livelihoods. Meanwhile, only a small proportion of FBs (7.1%) and NBs (6.7%) were involved in medium-scale fishing operations. Engagement in commercial-scale fishing was markedly rare among respondents, with less than 1% of both groups reporting participation.

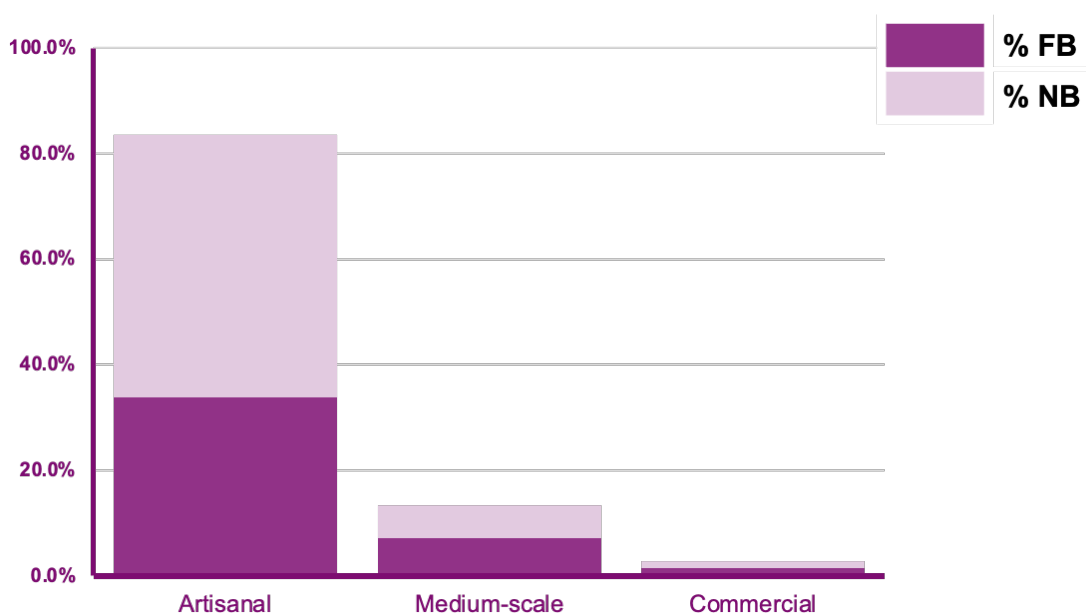


Figure 3.1.2e – Percentage distribution of respondents by type of fishing operations.

The minimal existence of both medium- and commercial-scale operations seemingly suggests certain forms of regulatory constraints and/or geographic inaccessibility of deeper fishing grounds. The type of fishing vessel used by respondents closely reflects this trend. Small-motorized boats were widespread, utilized by 25.1% of FBs and 27.0% of NBs. This aligns with the earlier findings that artisanal fishing was the dominant mode of operation among both groups. Interestingly, a higher proportion of FBs (11.5%) than NBs (10.1%) reported using medium-motorized boats. Large-motorized boats or trawlers

were rarely used, cited by only 0.2% and 0.6% among the FBs and NBs, respectively. This suggests limited engagement in capital-intensive, commercial-scale fishing.

A notable difference emerged in the use of non-motorized boats (*bangka*): 2.9% of FBs reported using such vessels, compared to a higher 8.4% among NBs, which may reflect differences in access to motorized fishing assets. Additionally, 2.9% of FBs and a considerably higher 11.3% of NBs reported not using any boat, indicating that a segment of respondents, particularly NBs, may engage in shore-based activities such as gleaning or operate as fish-workers rather than primary fishers.

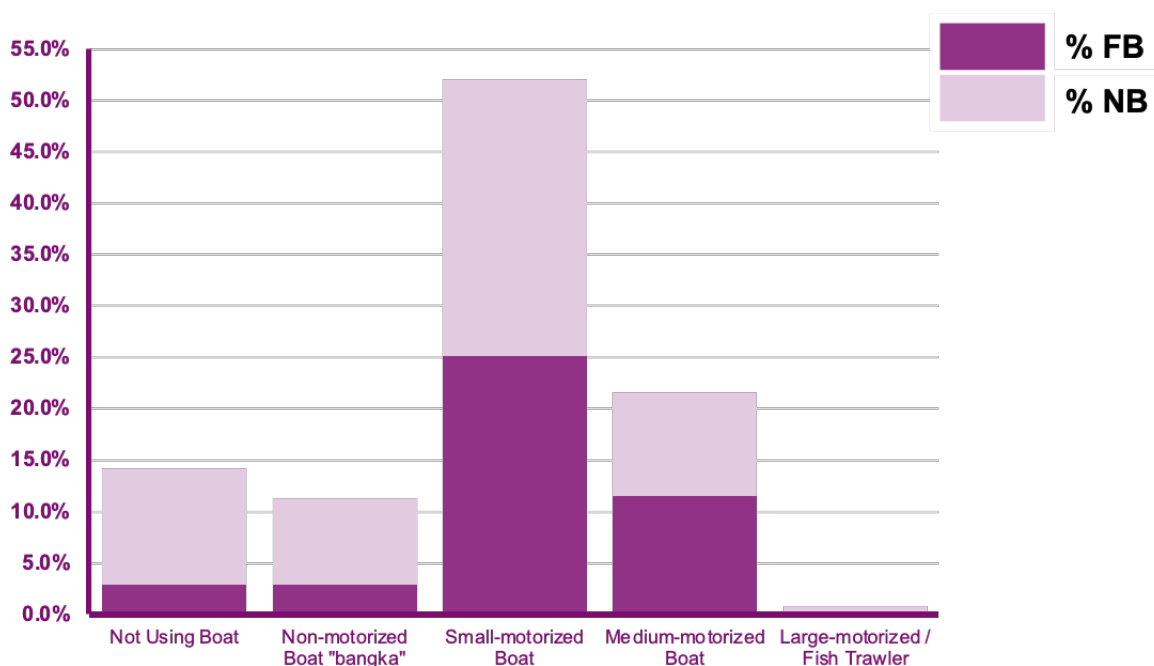


Figure 3.1.2f | Percentage distribution of respondents by type of sea vessel used for fishing.

The observed patterns on in terms of operations and vessel used illustrate the highly localized and low-capital nature of the fishing livelihoods in the surveyed areas. However, these insights also reinforce the appropriateness of BFAR's boat distribution approach in targeting artisanal fishers, whose mobility and productivity are directly associated with the level of access they have to functional, small-scale fishing vessels.

Fishing gear utilization demonstrated consistency in practice, with gill nets and hook-and-line emerging as the most commonly used technique. Among FBs, 22.5% used gill nets and 25.6% used hook-and-line, while among NBs, the figures were 30.4% and 36.2%, respectively. Fish traps and push-pull nets were utilized by both groups, yet less frequently. These findings may reflect localized gear preferences likely influenced by target species, access to capital, and traditional knowledge.

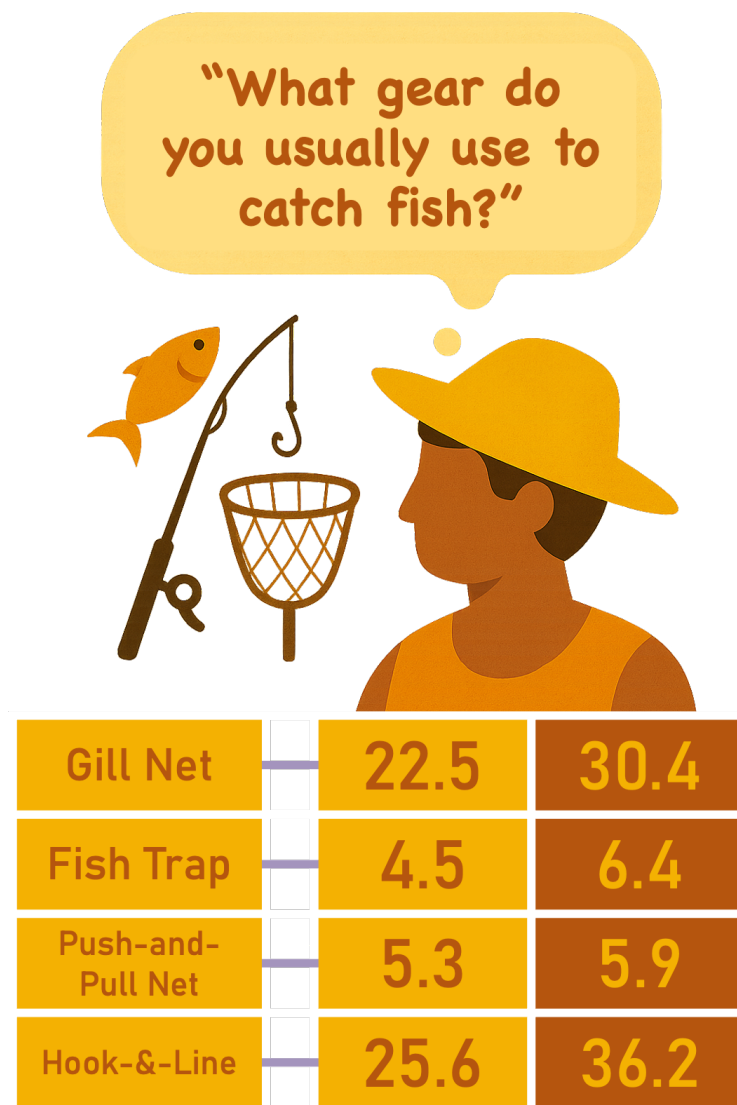


Figure 3.1.2g – Percentage of “Yes” and “No” response among FBs and NBs with regard to the main type of gear used in fishing activity.

On safety equipment use, flashlights were the most widely adopted safety tool, reported by 35.02% of FBs and 49.02% of NBs. Life jackets and buoys were less commonly used, suggesting that basic safety preparedness may still be lacking among fisherfolk. Notably, fewer FBs than NBs reported using cellphones as part of their safety or communication practices (12.78% vs. 21.18%), a trend that may reflect differences in access to technology or mobile signal coverage in areas where FBs are concentrated.

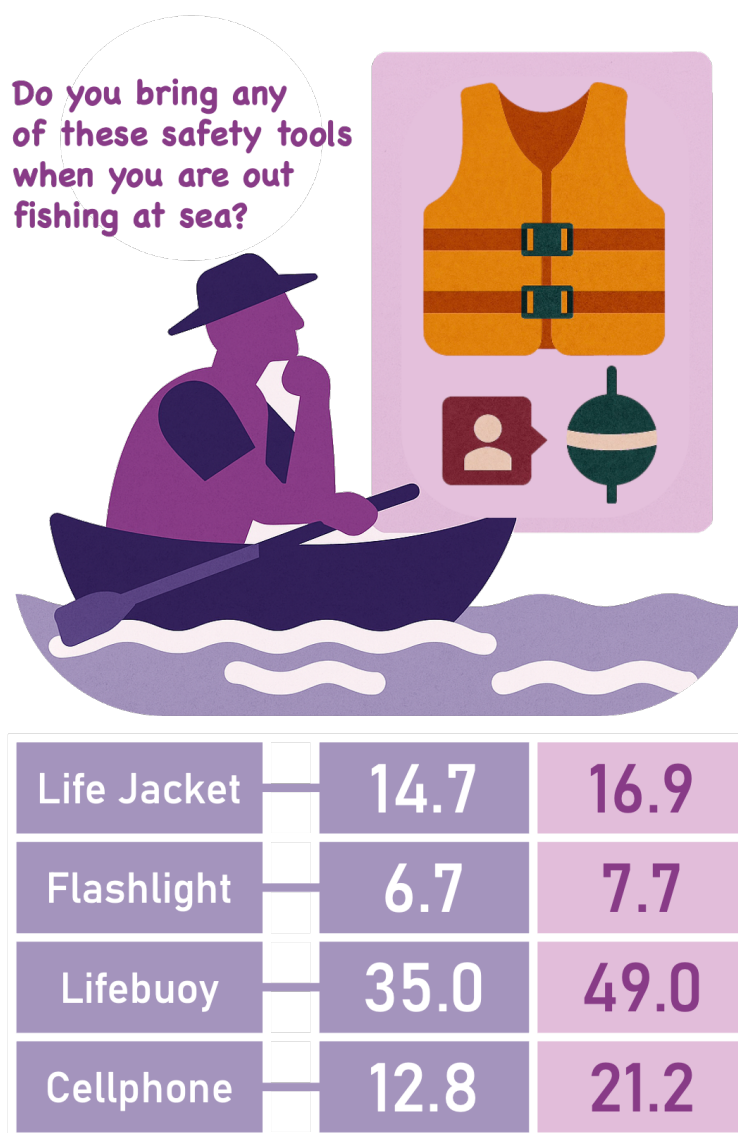


Figure 3.1.2h – Percentage of “Yes” and “No” response among FBs and NBs with regard to the usage of safety equipment when fishing.

Fisherfolk respondents across both groups reported a range of persistent operational challenges. Bad weather was the most frequently cited issue—36.38% of FBs and 53.25% of NBs reported it as a major concern, highlighting the vulnerability of small-scale fishers to climate and environmental risks. Faulty boat engines were another widespread problem (24.96% FBs; 31.39% NBs), which may directly affect fishing efficiency and safety.

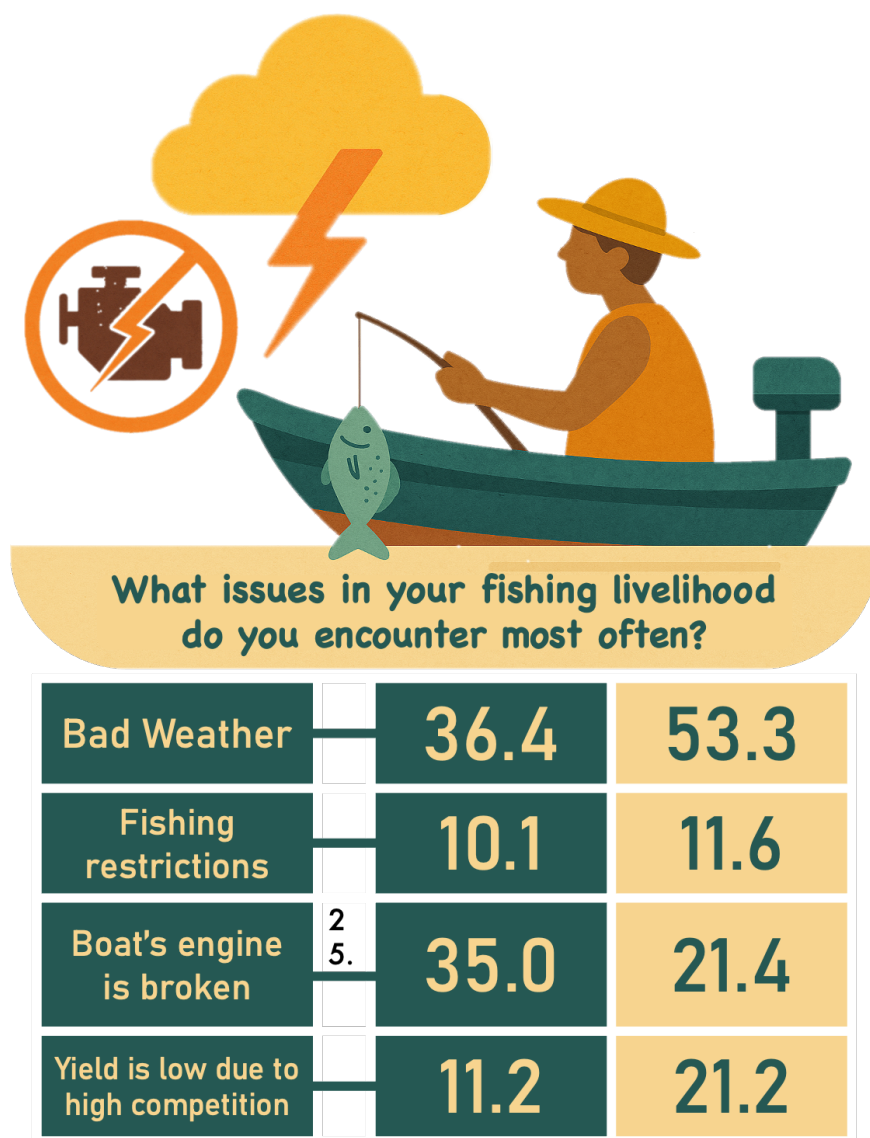


Figure 3.1.2i – Percentage of “Yes” and “No” response among FBs and NBs with regard to the common issues they may have encountered as fisherfolk.

Other frequently reported issues included fishing restrictions and high competition. Interestingly, 11.2% of FBs reported facing high competition in fishing grounds, compared to 21.2% of NBs, suggesting that beneficiaries of the F/BPLD Program may have benefited from enhanced mobility or access to alternative areas. These findings point to systemic constraints, not only impacting productivity but also influence how fisherfolk perceive the viability of their livelihood over time.

3.1.3 Access to Social Services

Access to social services was evaluated through a checklist of eight items reflecting key government programs on social welfare related to education, health, and financial assistance. Fisherfolk-respondents were asked to indicate whether they or any household member had accessed each service as enumerated in Figure 3.1.2a, showing the relative proportions of FB and NB who answered “Yes” in specific items.

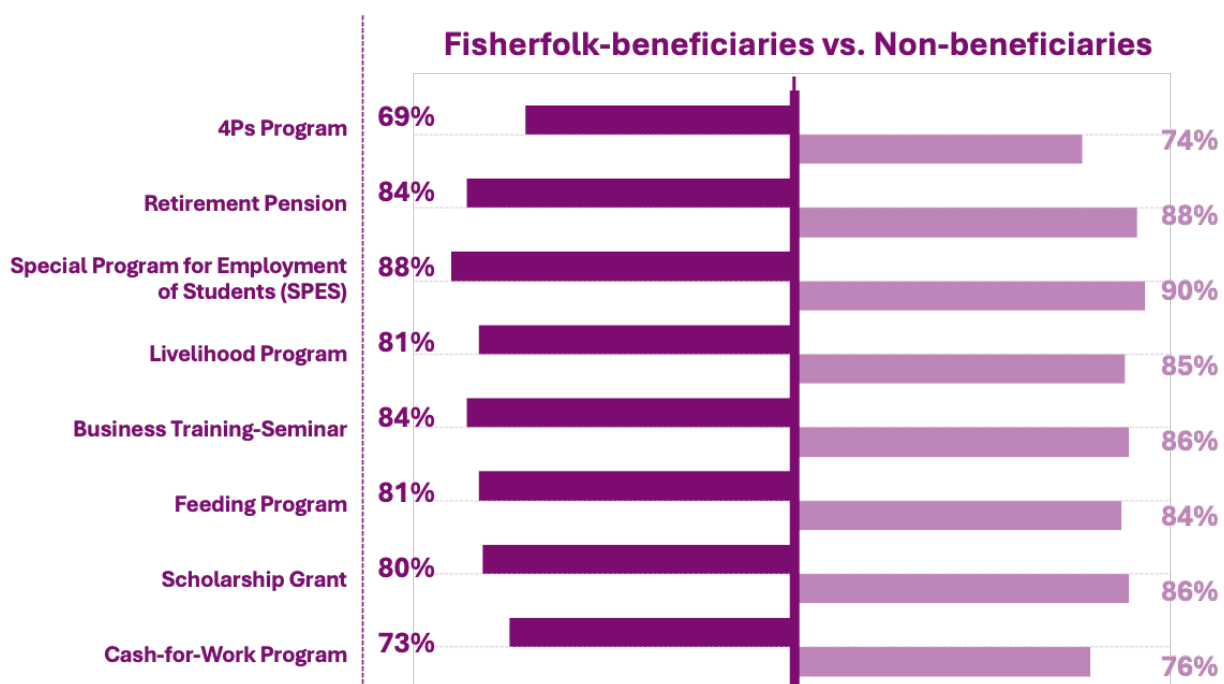


Figure 3.1.3a – Proportions of FB and NB who signified their access to certain social services.

Results show that both FBs and NBs reported high frequency of uptake across most services. “Yes” responses predominated in both groups, particularly for public health and education programs, indicating widespread awareness and utilization. “Maybe” and “No” responses were far less common, and typically concentrated around less familiar or irregularly implemented services, such as emergency financial assistance or LGU-specific interventions. The results were used to construct a composite Social Services Index (SSI), scaled from 0 to 100, representing the overall level of service uptake.

At the summative level, the SSI of the fisherfolk-respondents in general was 82.8 out of 100 (weighted mean score,), indicating that the available social welfare programs were highly accessible among the sampled population. Analysis of disaggregated data by group revealed that FBs and NBs posted nearly equivalent weighted mean scores—82.0 for FBs and 83.5 for NBs—suggesting that both groups enjoy relatively high and comparable rates of access to available social support mechanisms. This equivalency is visually apparent in the overlaid histograms (Figure 3.#), characterized by the overlapping distribution curves. The same was confirmed through the application of a weighted Mann–Whitney Test, which revealed no significant difference between the two groups ($t = -1.06$, $p = 0.289$).

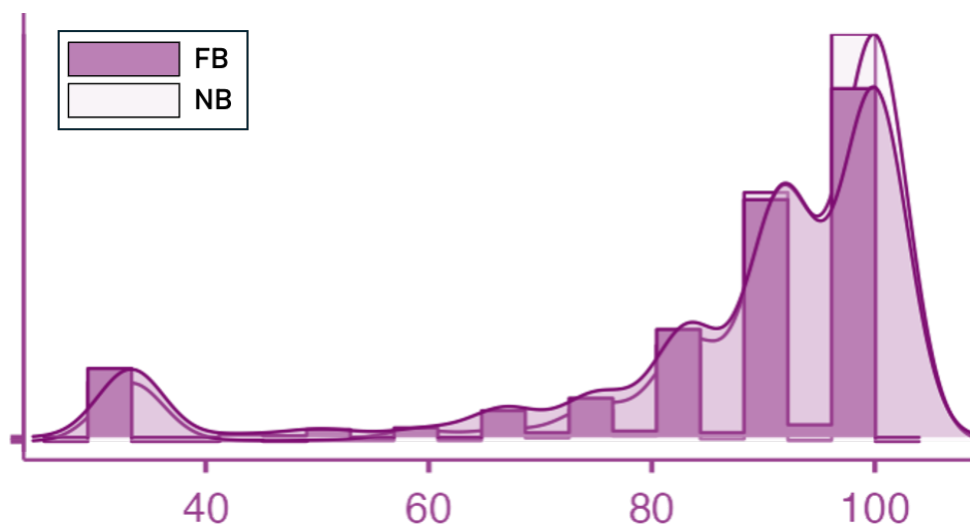


Figure 3.1.3b – Overlaid SSI density histograms of FBs and NBs characterized by the overlapping distribution curves. Values along the x-axis represent SSI scores.

However, regional breakdowns revealed important nuances. In Northern Luzon and Southern Luzon, NBs exhibited significantly higher SSI scores than FBs ($p = 0.003$ and $p = 0.016$, respectively), possibly reflecting stronger institutional support systems or more consistent local program delivery. In contrast, differences in Visayas and Mindanao were statistically insignificant ($p = 0.075$ and $p = 0.333$, respectively), suggesting equitable service delivery in these areas.

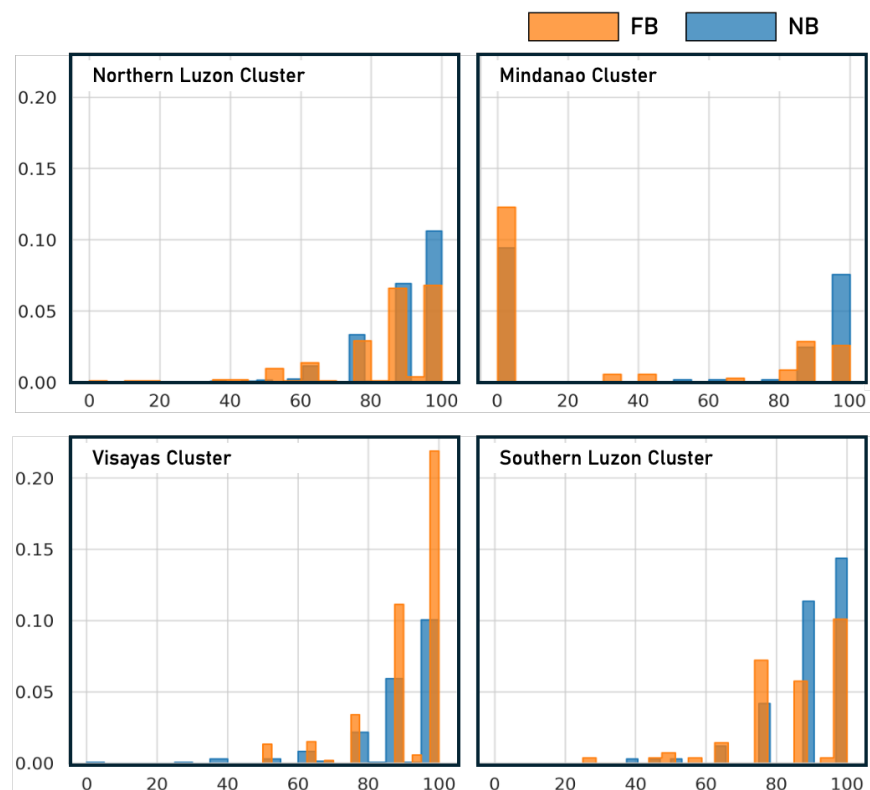


Figure 3.1.3c – Overlaid SSI density histograms of FBs and NBs disaggregated according to geographic location.

Taken together, the findings reflect a broader pattern of relatively equitable delivery of basic public services in coastal areas, independent of program participation. These services—while vital to household well-being—appear to be distributed through mechanisms external to the F/B Pagbabago Program. As such, the program’s added value likely lies not in altering access to these support systems, but in its direct livelihood-enhancing interventions such as the provision of fiberglass boats and fishing inputs.

3.2 Living Standards of FB Households

Presented in the following section is a descriptive account of the living standards of FBs based on the four dimensions comprising the Material Well-Being Index (MWI): (1) ownership of consumer durables, (2) housing and living conditions, (3) real property assets, and (4) insurance coverage. These indicators, anchored in the Philippine context, provide tangible proxies for assessing improvements in material well-being.

3.2.1 Consumer Durables

The Consumer Durables Ownership was quantified based on the self-reported data on asset ownership at two points in time—prior to receiving the F/B boat (benchmark) and at the time of the survey (current)—to study the trajectory in household material progress. Prior to receiving the motorized fishing boats through the BFAR distribution program, the majority of FB households had limited access to transport and appliance assets, reflecting their constrained purchasing power and modest economic status. Most notably, less than 11% reported owning a motorcycle (143 out of total FBs), whereas the ownership of other vehicles such as bicycles (7.03%), tricycles (1.89%), or private cars (0.23%) was even less common. The situation was similar for motorized utility vehicles like jeeps or trucks, which were nearly absent from FB households' pre-program inventories.

In terms of appliances, ownership patterns were slightly more encouraging, albeit still limited. Just under 22% had a television, and about 9.6% had access to a washing machine—both considered semi-essential for information access and domestic convenience. Even more modest were the ownership rates of air conditioners (2.27%), refrigerators (8.32%), or electric stoves (9.53%). Only 4.31% of FB households had basic furniture like a sofa or dining set, suggesting that many lived with minimal comfort in their domestic spaces. The digital divide was also evident: just over a quarter (25.19%) owned a cellphone, but access to computers (1.06%) and landlines (0.30%) was very rare.

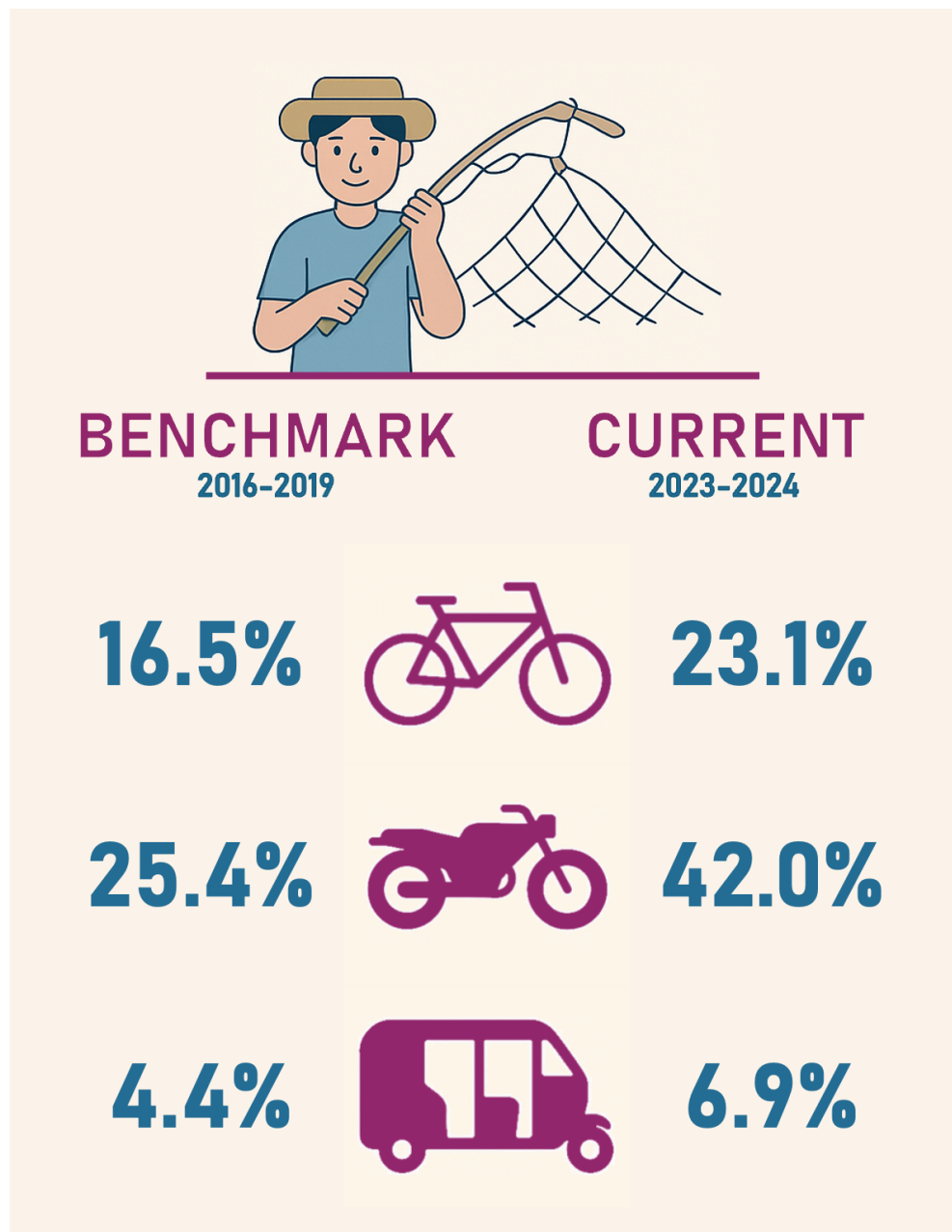


Figure 3.2.1a –

However, the inventory for the current period reveals a noticeable and encouraging uptick in the material living standards of FB households. Ownership of motorcycles rose sharply to 17.85%, making them the most common motor vehicle among beneficiaries—likely reflecting the importance of mobility for both livelihood and access to services.

Bicycle ownership also improved from 7.03% to 9.83%, while tricycle ownership more than doubled from 1.89% to 2.95%, albeit still low in absolute terms. Car, jeep, and truck ownership remained virtually unchanged—expected, given their high cost and maintenance burdens.

The most dramatic improvements were recorded in household appliances. The proportion of FBs with televisions grew to 28.74%, up from 21.86%, and washing machine ownership jumped significantly from 9.61% to 16.26%, indicating a shift toward better time and labor-saving devices in the home. Refrigerator ownership followed a similar upward trend, climbing to 12.71%, as did electric fans (now at 35.17%)—the most widely owned appliance among FBs. Stove ownership also rose slightly to 12.78%, a modest gain that may reflect better access to electricity or improved kitchen facilities. Furniture ownership increased to 6.20%, suggesting some effort toward domestic comfort and dignity.

On the communication front, ownership of mobile phone had notable gains, reaching 33.66% of FB households—an increase of more than 8 percentage points. On the other hand, computer ownership remained low at 1.97%, and landline connections, which have been largely phased out in favor of mobile services, remained rare at 0.76%.

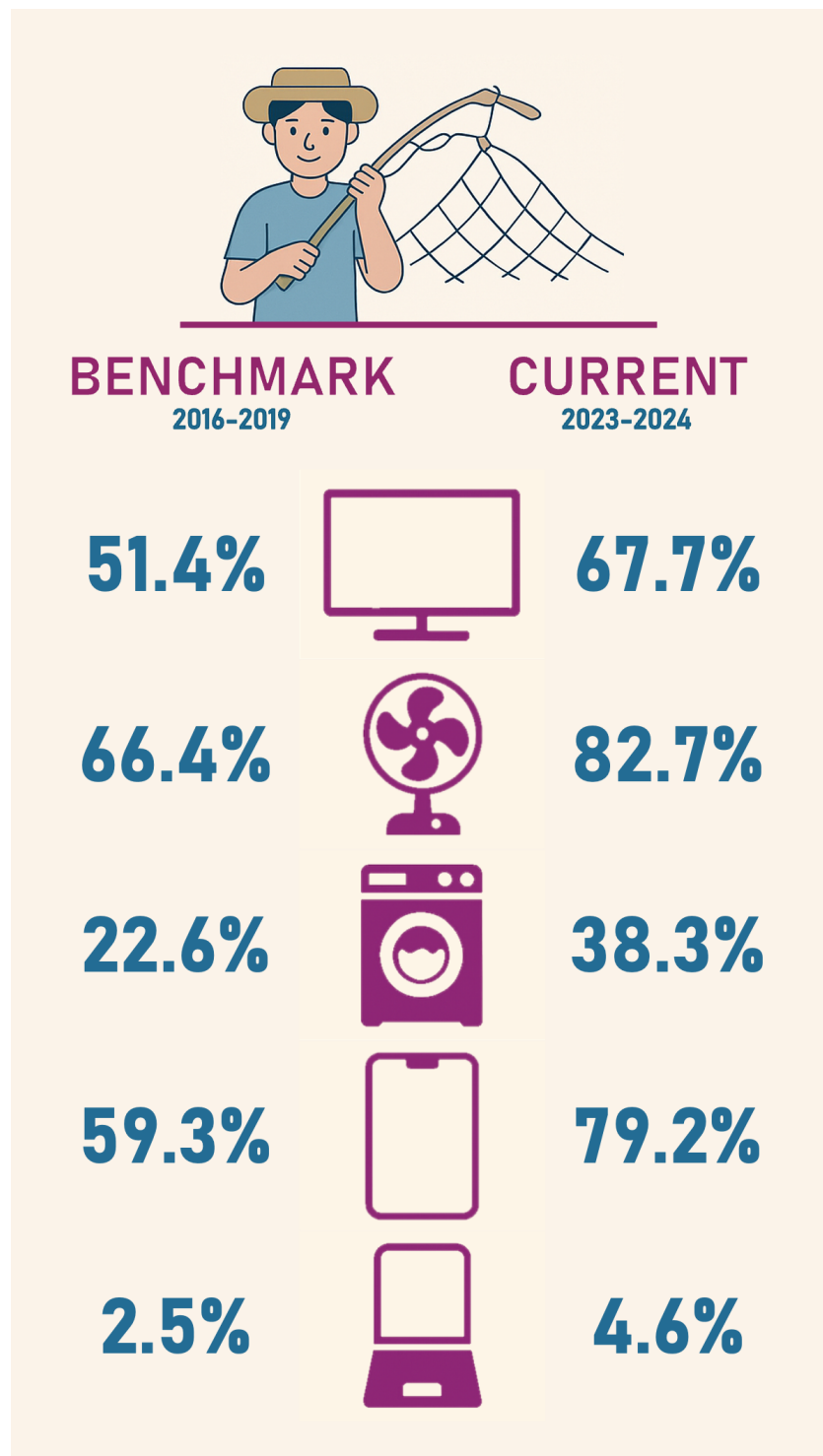


Figure 3.2.1b –

3.2.2 Basic Utilities

Over the years, fisherfolk-beneficiaries (FBs) have reported modest yet significant improvements in their access to essential utilities, reflecting a gradual shift in the quality of their living conditions. While structural limitations persist in coastal settlements, the data suggest a noticeable transition toward more reliable and formal utility sources during the current period compared to their benchmark situation prior to receiving the boats.

In terms of drinking water, the proportion of FB households relying on mineral or bottled water increased from 25.3% at benchmark to 35.6% at present. Likewise, those with direct access to faucet water rose from 32.0% to 38.1%, while reliance on deepwell sources dropped significantly from 24.0% to 11.0%. These shifts suggest growing access to safer, regulated drinking water, although nearly one in six households still rely on “other” sources—including rainwater collection and mobile vendors—which declined only slightly from 18.7% to 15.3%.

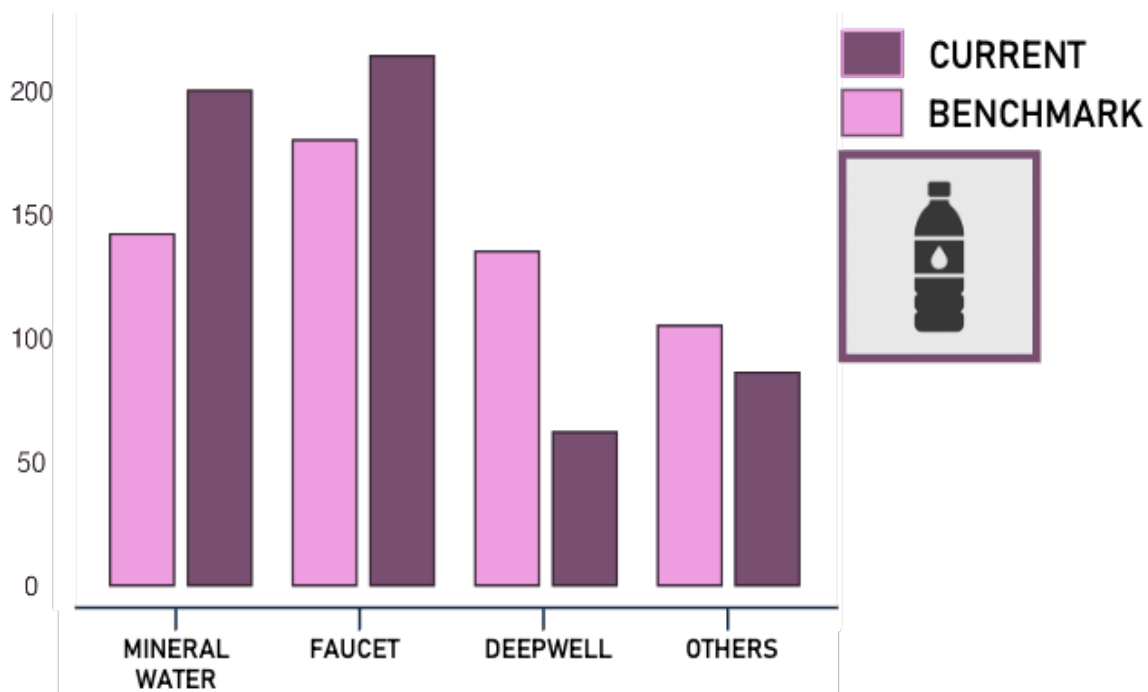


Figure 3.2.2a – Frequency Distribution of Drinking Water Source.

For domestic water use, trends remained fairly stable but positive. The share of FB households using commercially supplied water increased from 31.9% to 38.8%, while those relying on deepwell sources decreased from 36.5% to 31.0%. Use of surface water sources (e.g., lakes and rivers) also declined marginally from 3.7% to 3.0%, and the proportion depending on other sources—such as shared facilities or hand-carried supply—remained nearly unchanged at 27.9% and 27.2%, respectively. These figures reflect a cautious shift away from informal and labor-intensive sources toward more reliable, paid supply.

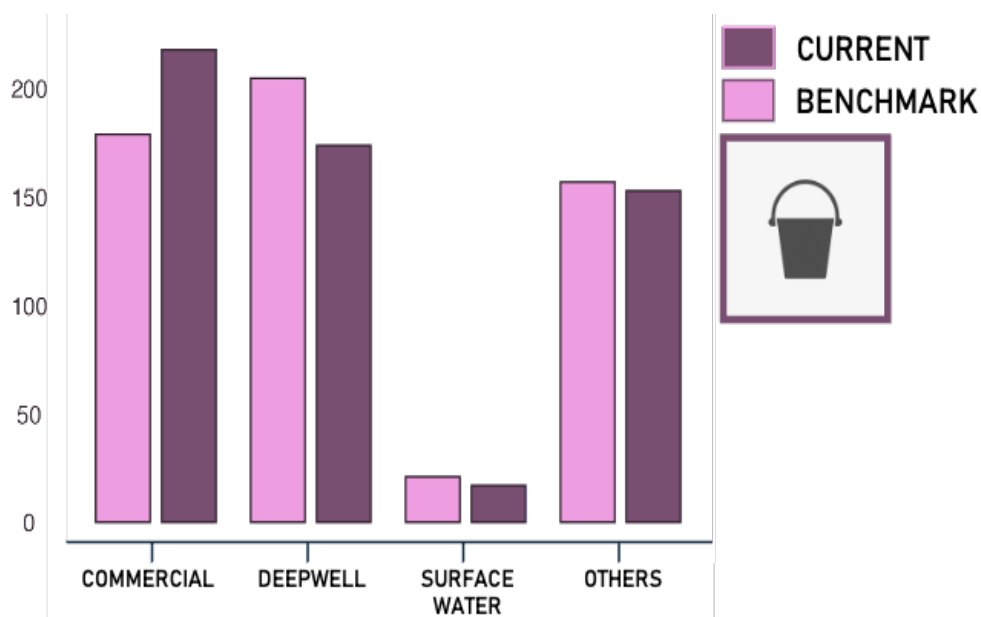


Figure 3.2.2b – Frequency Distribution of Water for Domestic Use.

Access to electric power improved sharply. The share of FB households connected to commercial electricity providers jumped from 68.7% at benchmark to 86.2% at present. Conversely, reliance on solar panels dropped from 14.8% to 6.6%, likely due to increased grid penetration. The share of FBs with no access to electricity declined from 12.5% to 5.5%, while those using other power sources (e.g., batteries or generators) decreased from 4.1% to 1.8%. These trends point to more secure and stable energy access across surveyed coastal households.

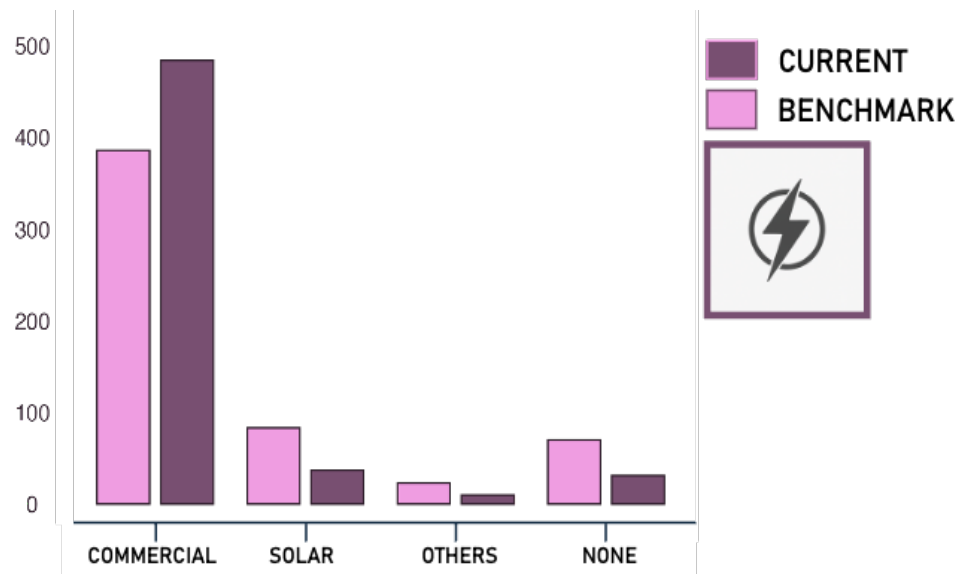


Figure 3.2.2c – Frequency Distribution of Power Supply.

In terms of cooking fuel, there was a clear but modest shift toward cleaner energy. The use of LPG or electric cooking appliances rose from 18.7% to 24.9%, while dependence on firewood—the dominant source—decreased from 66.2% to 53.7%. Use of coal remained almost stable (from 12.5% to 13.9%), and those using other fuels (such as kerosene or mixed alternatives) increased slightly from 2.7% to 7.5%. While biomass continues to dominate, the trend suggests a gradual transition toward safer and more efficient cooking methods.

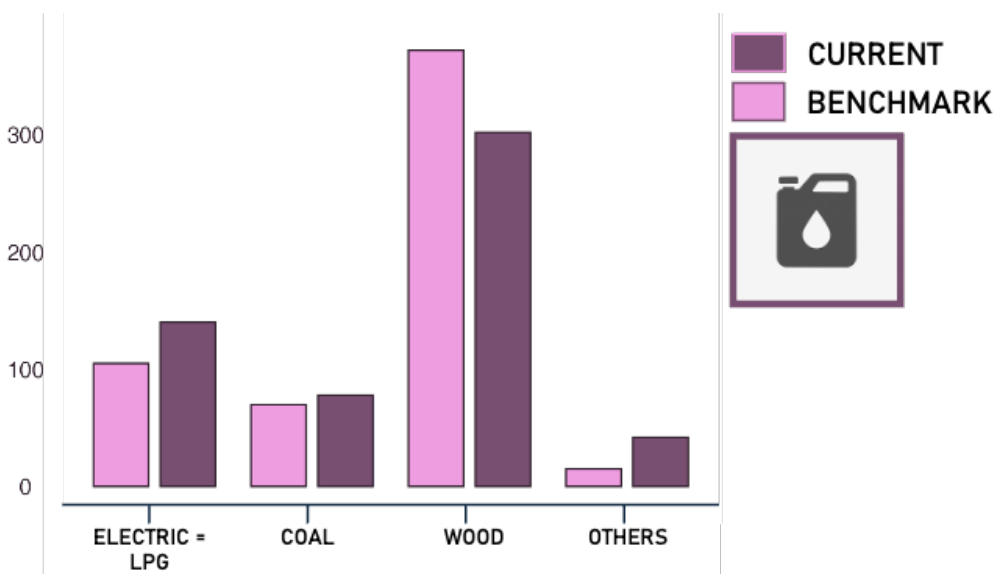


Figure 3.2.2d – Frequency Distribution of Fuel for Cooking.

Finally, though still very limited, internet access among FB households improved slightly. The proportion reporting an active subscription rose from 6.9% to 9.6%, suggesting early stages of digital inclusion. This may be driven by increased mobile connectivity, though the majority (over 90%) still lack fixed or formal access at home.

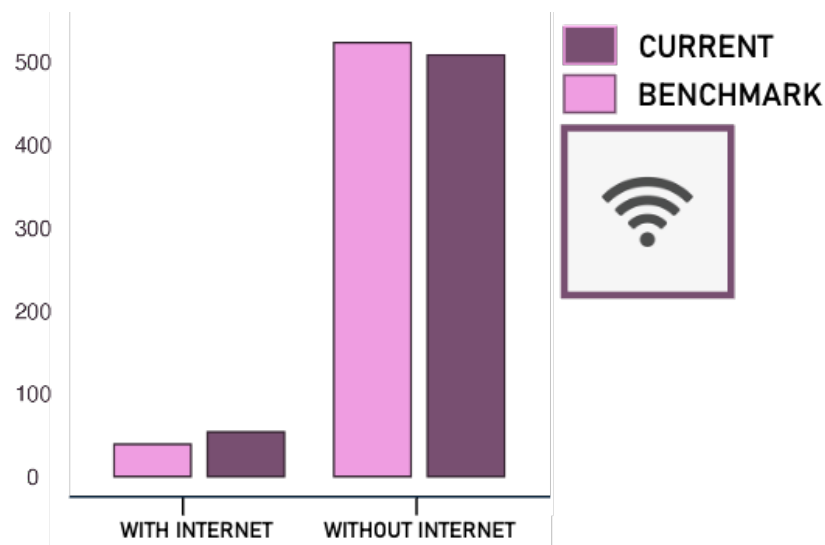


Figure 3.2.2e – Frequency Distribution of Internet Subscription.

Altogether, these updated indicators show clear movement toward improved living conditions, especially in areas of electricity and water access. However, gaps remain in digital connectivity and in the continued reliance on traditional fuels. The gains observed may be attributed in part to the broader economic and infrastructural benefits that followed the boat distribution program.

3.2.3 Real Properties

The real property profile of fisherfolk-beneficiaries (FBs) paints a picture of relative stability, marked by incremental gains in homeownership and structural quality of dwellings, but continued limitations in formal tenure and property acquisition. While some positive movement was observed, the broader pattern suggests that access to secure, formalized housing remains a structural constraint for many coastal households.

In terms of house tenure, the proportion of FBs who reported owning their house rose slightly from 71.7% at benchmark to 73.1% at present. This modest increase may indicate some consolidation in permanent residence and housing security. The proportion of households renting their homes remained almost unchanged, dipping slightly from 8.7% to 8.5%, while those who reported “other” terms of tenure—likely encompassing informal occupancy, familial transfer, or caretaking—declined from 19.6% to 18.3%.

With respect to the mode of acquisition, more than half of the FBs acquired their homes through full payment or inheritance, which rose modestly from 56.2% to 56.8%. The share of households that gained access through partial payment increased slightly from 11.9% to 12.5%, suggesting minor growth in installment-based housing ownership. Meanwhile, those who obtained their houses via informal arrangements declined from 20.1% to 19.8%, while those who reported not owning their current home dropped from 11.7% to 11.0%. These figures signal a gradual, if uneven, movement toward more secure and possibly formalized property acquisition.

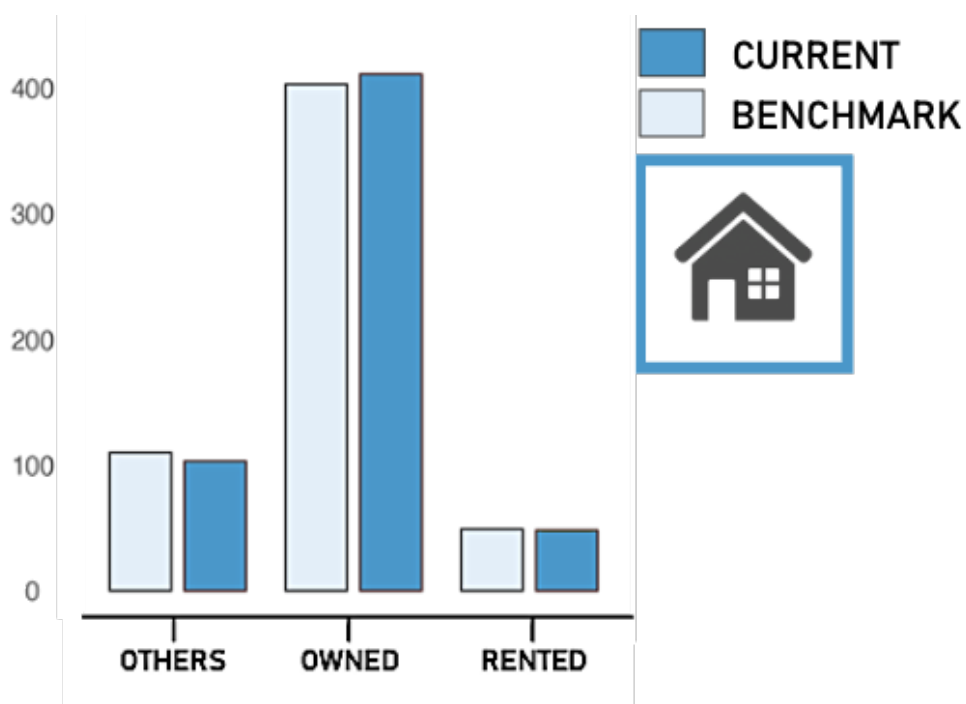


Figure 3.2.3a – Frequency Distribution of House Tenure Arrangement.

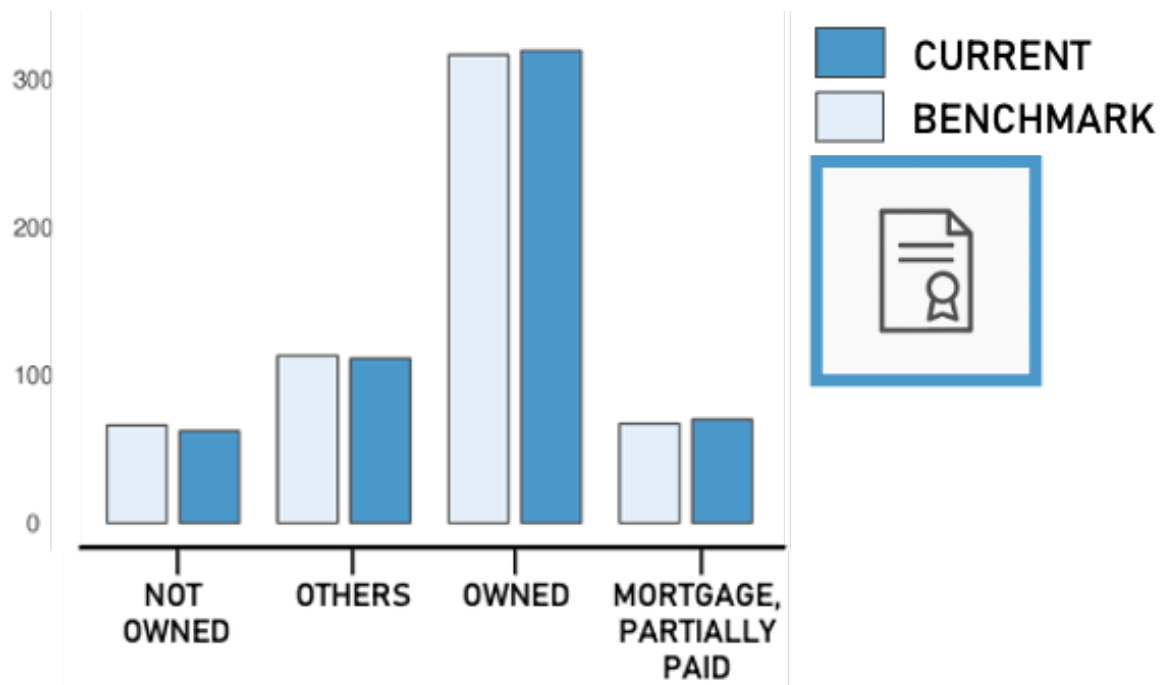


Figure 3.2.3b – Frequency Distribution of Mode of Property Acquisition.

In terms of house structure, small but meaningful improvements were observed. The share of households residing in simple concrete houses increased from 50.9% to 52.1%, while those in fancier concrete dwellings rose from 5.7% to 6.0%. Meanwhile, the proportion living in non-concrete structures—including makeshift or temporary homes—decreased from 43.4% to 41.8%. These shifts point to gradual upgrading in housing materials and construction quality, likely resulting from incremental home improvements over time rather than large-scale housing interventions.

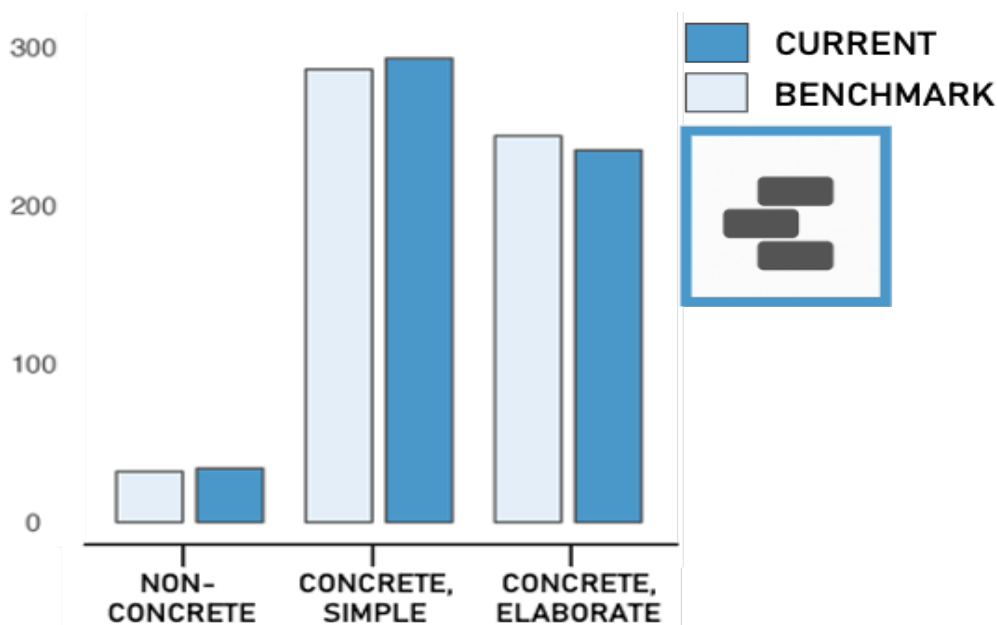


Figure 3.2.3c – Frequency Distribution of House Structure.

When it comes to ownership of additional real properties, a relatively stable pattern was noted. The proportion of FBs who reported owning a house only (without land) remained constant at 33.1%, while ownership of a house and lot decreased slightly from 33.3% to 31.1%. Meanwhile, ownership of other forms of real property—such as undeveloped residential lots or land in another area—increased from 16.4% to 18.9%, and those reporting no real property holdings declined from 17.3% to 16.9%. These data suggest a subtle diversification in household assets, with some FBs acquiring secondary or off-site properties.

The findings above underscore the persistent challenges that FB households face in attaining formal, secure housing and property rights. While positive developments in home structure and ownership were observed, many respondents still rely on informal or hybrid arrangements. This highlights the importance of long-term strategies focused on tenure regularization, targeted housing assistance, and asset-building support tailored to the realities of coastal and fishing-dependent communities.

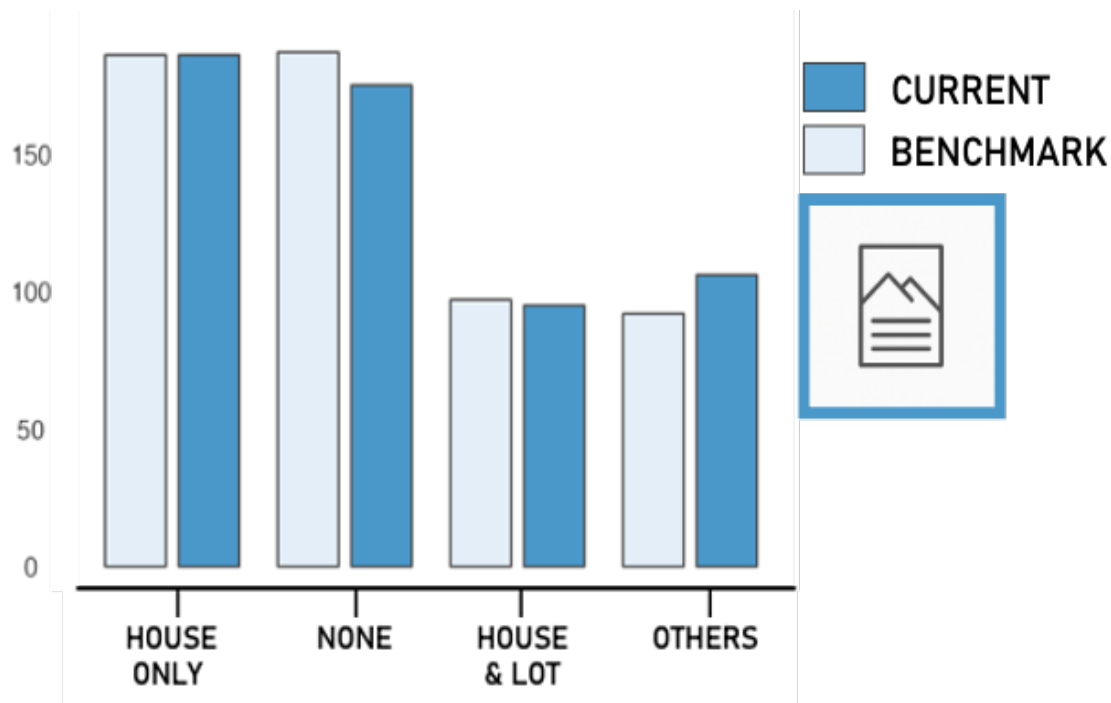


Figure 3.2.3d – Frequency Distribution of Other Real Property Ownership.

3.2.4 Insurance Coverage

Insurance coverage among fisherfolk-beneficiaries (FBs) reflects a complex but clarifying picture of modest gains, persistent gaps, and evolving access to social protection. The data show that while some risk protection mechanisms have expanded, others—particularly those historically associated with informal workers—remain inconsistently accessed or misunderstood.

As combined measure, enrollment in retirement plans—defined here as participation in either the Social Security System (SSS) or the Government Service Insurance System (GSIS)—grew from 22.9% (129 FBs) during the benchmark period to 49.6% (279 FBs) at present. This remarkable increase suggests that FB households have become increasingly connected to formal retirement mechanisms, likely through employed household members, voluntary enrollment, or spillovers from public sector-linked eligibility. Disaggregated data show that SSS enrollment declined slightly, from 19.9%

(112 FBs) to 18.1% (102 FBs), pointing to challenges in maintaining consistent participation among informal earners. Meanwhile, GSIS membership rose from 4.4% (25 FBs) to 5.2% (29 FBs)—still modest in absolute terms but indicative of growing links to state employment or eligibility via family members.

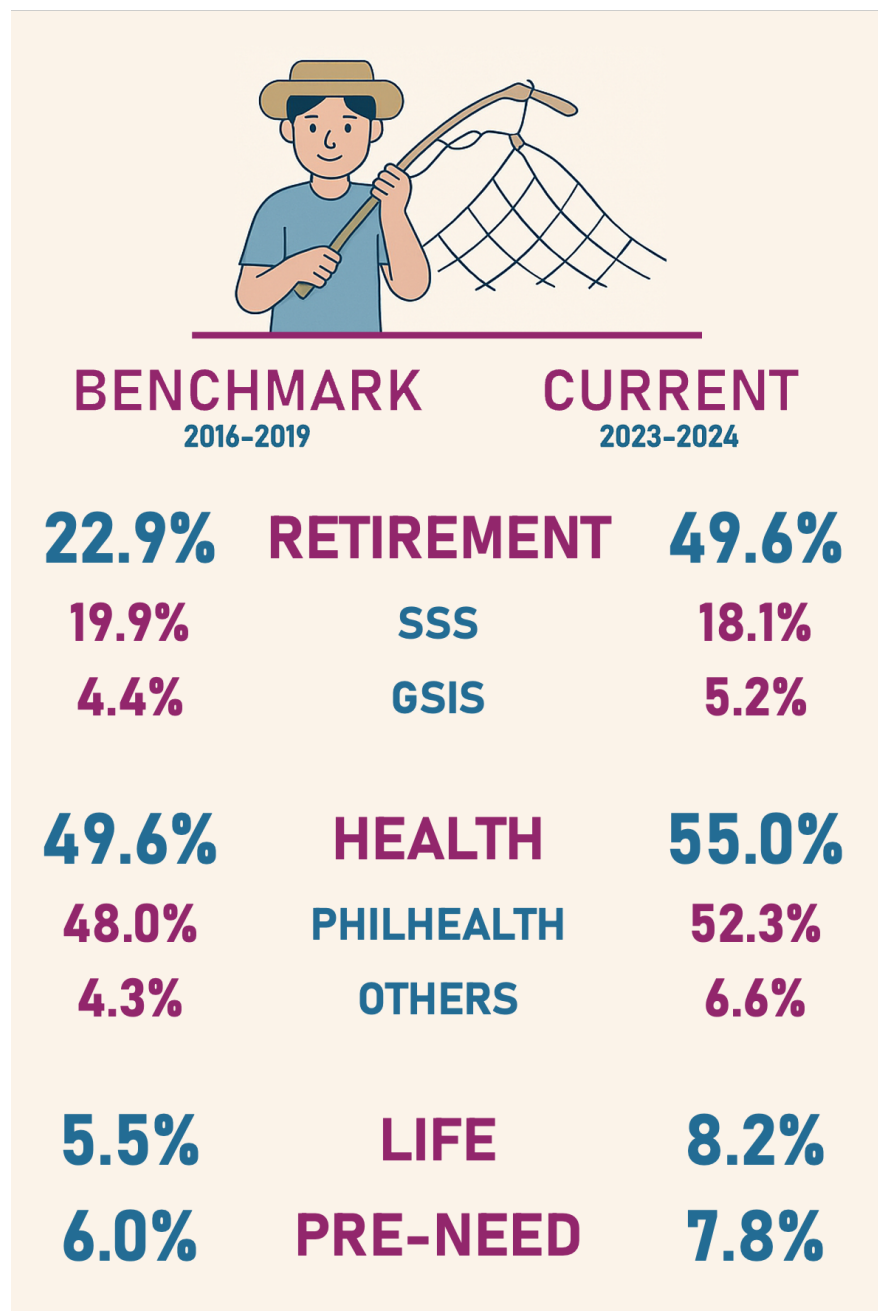


Figure 3.2.4a – Benchmark and Current Proportion of FB Households with Access to Selected Insurance Products.

Health insurance coverage also improved significantly. Combined access to PhilHealth and other health insurance products increased from 49.6% (279 FBs) at benchmark to 55.0% (309 FBs) at present. The vast majority of this coverage came from PhilHealth, which covered 48.0% (270 FBs) during the benchmark period and 52.3% (294 FBs) currently. This upward trend aligns with the goals of the Universal Health Care (UHC) Act and suggests expanded public insurance access, though gaps still persist. In contrast, other health insurance—comprising cooperative-based, employer-provided, or microinsurance—showed a small increase from 4.3% (24 FBs) to 6.6% (37 FBs). While encouraging, this remains a marginal portion of the total, suggesting limited reach of private or alternative schemes in coastal areas.

Other types of insurance continue to be lightly adopted. Life insurance coverage rose from 5.5% (31 FBs) to 8.2% (46 FBs), while pre-need insurance—typically used for memorial or educational plans—increased from 6.0% (34 FBs) to 7.8% (44 FBs). These gains, while positive, highlight that a majority of fisherfolk households remain outside formal risk mitigation products, likely due to affordability, financial literacy, or competing household needs. Altogether, these findings reflect a cautiously optimistic trajectory in the insurance profile of FBs. The consistent expansion in PhilHealth and the substantial uptake in retirement plans underscore the role of institutional efforts in extending social protection to coastal populations. However, the low rates of life, pre-need, and private health insurance coverage point to the continued vulnerability of these households in the face of health, livelihood, or mortality risks.

As FB households remain exposed to a range of socioeconomic uncertainties, strengthening the reach, continuity, and literacy around insurance mechanisms remains a critical policy imperative. Interventions should especially aim to bridge informal workers into voluntary insurance schemes, institutionalize community-level enrollment drives, and clarify perceived barriers around documentation, premiums, and coverage portability.

3.2.5 Benchmark vs. Current: A Synthesis

The preceding subsections present an integrated account of the material living standards of fisherfolk-beneficiaries (FBs), viewed through four key dimensions: consumer durables, basic utilities, real properties, and insurance coverage. This synthesis consolidates the evidence to surface overarching patterns of change between the benchmark period—referring to respondents’ recalled conditions prior to receiving the BFAR-distributed motorized boats—and their current situation in 2024.

Across all four domains, the results suggest incremental but consistent gains in the material well-being of FB households. Although the pace and scale of improvement vary by indicator, the combined trajectory signals a pattern of cautious socioeconomic mobility, likely influenced in part by expanded livelihood capacities brought about by the boat distribution program.

In the area of consumer durables, households showed marked improvements in the ownership of essential appliances and small vehicles. Ownership of mobile phones increased from 25.2% to 33.7%, washing machines from 9.6% to 16.3%, and motorcycles from 10.8% to 17.9%. These gains reflect improvements not only in domestic comfort and labor-saving capacities but also in personal mobility and communication—critical enablers of both economic and social participation.

Progress in basic utilities further affirms this upward shift. Household access to commercial electricity rose from 68.7% to 86.2%, while reliance on solar panels and off-grid alternatives decreased. There was also a notable increase in access to safe drinking water, with use of bottled or mineral water rising from 25.3% to 35.6% and tap water from 32.0% to 38.1%. Although internet access remains limited, the increase from 6.9% to 9.6% hints at early movement toward digital connectivity. These changes suggest that more FB households now benefit from stable, regulated, and modern utility services.

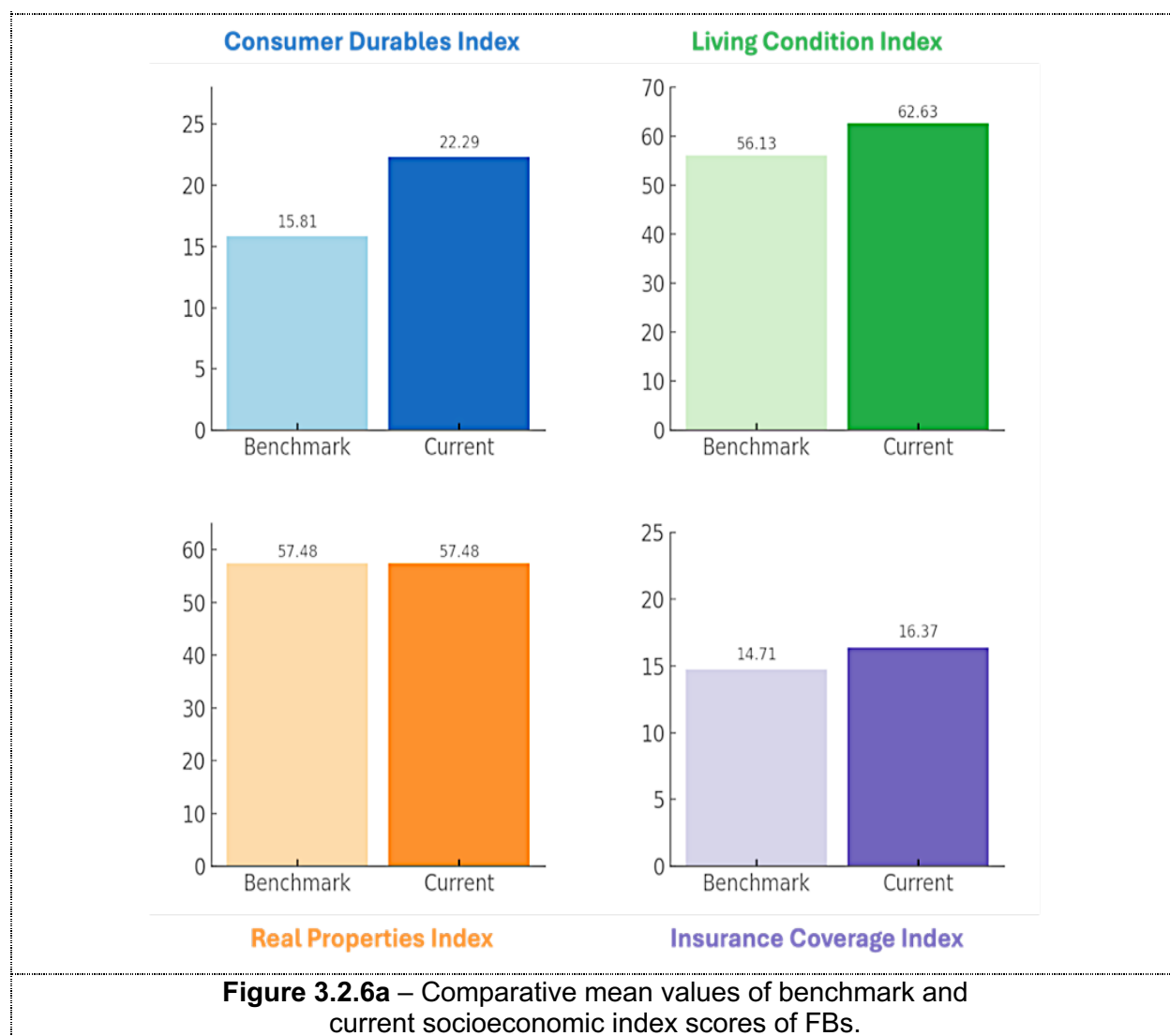
The picture is more mixed when it comes to real property. Homeownership rose modestly from 71.7% to 73.1%, and more households now live in concrete structures—up from 56.6% (simple + fancy) to 58.1%. Acquisition through full payment or inheritance remains the dominant mode, while informal and partial payment arrangements persist. Meanwhile, ownership of other types of property (e.g., residential lots or land elsewhere) rose slightly, signaling asset diversification. These incremental improvements point to long-term investment in security and permanence, albeit constrained by the structural challenges typical of coastal informal settlements.

In terms of insurance coverage, the results present a more complex story. There was a substantial increase in enrollment in retirement plans (SSS + GSIS), from 22.9% to 49.6%, suggesting expanded access to formal social protection. Health insurance coverage also improved, rising from 49.6% to 55.0%, with PhilHealth covering over half of the surveyed households. However, life insurance (from 5.5% to 8.2%) and pre-need plans (from 6.0% to 7.8%) remain underutilized. These mixed results reflect both the progress and persistent gaps in integrating FBs into risk protection systems, especially those operating outside the public sector.

Taken together, the data reveal a general pattern of upward movement—one defined less by dramatic transformation and more by steady progress in essential material assets, access to services, and basic financial protections. While the BFAR boat distribution program was not designed to be a comprehensive poverty alleviation intervention, the improvements seen across all four dimensions suggest that the boats may have served as a catalytic asset, reinforcing economic agency and enabling FB households to build on their livelihoods incrementally. These findings underscore the importance of anchoring asset-based interventions within a broader ecosystem of public support, especially in structurally vulnerable communities.

3.2.6 Material Well-being

Summarized herein is the fisherfolk-beneficiaries' (FBs) living standards based on component subindices—Durables Ownership, Living Conditions, Real Property, and Insurance Coverage—and culminates in the composite Material Well-being Index (MWI). Each index captures a distinct dimension of socio-economic quality of life and was assessed at two points in time: the Benchmark period (prior to receiving the BFAR-distributed boat) and the Current period (at the time of the 2024 survey).



The DOI reflects FB households' access to vehicles, appliances, and digital devices. The index improved significantly from a benchmark mean of 15.81 to a current mean of 22.29, as confirmed by the Wilcoxon signed-rank test ($W = 33,961$, $p < .001$). This increase points to modest but meaningful gains in household asset ownership, most notably in items that enhance mobility (e.g., motorcycles) and domestic convenience (e.g., washing machines and televisions).

The LCI measures access to basic utilities such as water, electricity, cooking fuel, and internet connectivity. Between the benchmark and current periods, the average LCI score rose from 56.13 to 62.63, a statistically significant difference ($W = 7,559$, $p < .001$). These results suggest modest infrastructure improvements, especially in access to tap water, commercial electricity, and clean cooking fuels, although digital connect remains limited.

Unlike other domains, the RPI remained unchanged at 57.48 across both time periods, with no statistically significant difference detected ($p = 0.791$). This stability indicates that improvements in property ownership or structural housing quality were either absent or insufficiently widespread to influence index values. Many FB households continued to occupy homes of modest build and tenure, with only incremental gains in formal ownership.

The ICI captures enrollment in public and private insurance programs, including retirement, life, and health plans. While coverage remained low overall, the index increased from 14.71 to 16.37, a statistically significant gain ($W = 2,019$, $p < .001$). Improvements were observed in PhilHealth, pre-need plans, and other health-related products, although large-scale underinsurance persists, reflecting systemic barriers to financial risk protection.

Integrating all four component indices, the MWI serves as the study's composite measure of FB households' material progress. The mean MWI score rose from 38.16 at benchmark to 42.30 at present ($W = 25,534$, $p < .001$), indicating statistically significant overall improvement in socio-economic standing. Notably, while most households remained in the “moderately low” well-being category, the share of those reaching “moderately high” status grew from 14.8% to 22.2%, and those categorized as “low” fell by half—from 12.8% to 6.0%.

In summary, FB households experienced steady, empirically validated gains in their material conditions, especially in the domains of asset ownership, basic services, and limited financial inclusion. However, challenges remain in expanding formal housing access and securing more robust insurance coverage. The data reflect an encouraging, if gradual, trajectory toward socio-economic uplift, underscoring the complex but potentially transformative role of the BFAR boat distribution program in enhancing livelihood resilience and household welfare.

The descriptive findings in Section 3.2 present a coherent picture of gradual but meaningful improvement in the material well-being of fisherfolk-beneficiaries (FBs) over a five- to eight-year period. These results align with broader development literature affirming that targeted livelihood support can stimulate upward mobility among marginalized rural populations, particularly when such programs reduce capital constraints and stimulate household-level investments (Barrett & Carter, 2010; Béné et al., 2016).

The documented increases in ownership of small appliances, personal transport (e.g., motorcycles), and mobile communication devices echo findings from rural livelihood studies which suggest that incremental asset accumulation often signals increased resilience and productive capacity (De Janvry & Sadoulet, 2001; Filipinski & Taylor, 2012). In fishing communities, improved mobility has been linked to greater market access, reduced transaction costs, and enhanced participation in off-boat economic activities

(Pomeroy et al., 2007; Brummett et al., 2011). Likewise, improved access to energy, water, and basic services reflects trends observed in infrastructure-led poverty reduction, where even modest improvements in utilities can yield disproportionately positive impacts on household health, time use, and schooling outcomes (Banerjee et al., 2012; UNDP, 2016).

The stability observed in real property ownership echoes earlier studies suggesting that tenure and housing quality are often the slowest to change among capital assets, particularly in coastal or informal settlements where legal ownership remains contested or weakly enforced (Kiddle, 2011; Ballesteros & Egana, 2016). Nonetheless, even marginal structural improvements may signal incremental formalization or internal resource mobilization for home upgrading (Rakodi, 2002).

In the area of financial protection, the persistent underinsurance gap mirrors findings across the Global South, where informal workers—especially in fisheries and agriculture—remain excluded from contributory social protection systems (ILO, 2017; Sumner & Mallet, 2013). As observed in this study, PhilHealth remains the most accessible insurance product for low-income Filipinos, but participation in retirement or private insurance schemes remains limited, constrained by affordability, low financial literacy, and limited trust in institutions (Habito & Briones, 2005; GIZ, 2015).

The steady increase in the composite Material Well-being Index (MWI), though modest in magnitude, aligns with studies emphasizing the role of livelihood asset-building in enhancing household resilience to shocks and future uncertainty (Ellis, 2000; Scoones, 1998). The Sustainable Livelihoods Framework (SLF), which underpins this evaluation, posits that improvements in one or more forms of capital—physical, financial, social—can have multiplier effect on long-term well-being (Chambers & Conway, 1992; DFID, 1999).

While this study stops short of attributing causality, the alignment between the timing of BFAR’s boat distribution and subsequent material improvements—particularly in assets

directly related to productivity, mobility, and energy—supports the notion that productive asset transfers can stimulate positive change when embedded in the local socioeconomic context (Sabates-Wheeler & Devereux, 2011; Davis et al., 2016). The findings also resonate with Philippine-based research that underscores the importance of asset ownership and productive resources in enhancing income diversification, poverty reduction, and community cohesion in coastal communities (Israel & Llorin, 2003; Cruz-Trinidad, 2003).

In synthesis, the data reflect a pattern of cautious but encouraging uplift in FB households' material living standards. While more work is needed to close gaps in property security and social protection, the results affirm the relevance of asset-focused interventions—like the boat distribution program—as part of a broader strategy for building resilient, inclusive, and dignified rural livelihoods.

3.3 Material-based Socioeconomic Impact

3.3.1 Comparative Analysis of Socioeconomic Domains

To assess the socio-economic impact of the F/BP program, the study compared program beneficiaries (FBs) and non-beneficiaries (NBs) using a combination of inferential and descriptive analyses. The evaluation considered both current household status and change over time (gain scores) across five key dimensions: consumer durables, basic utilities, real property, insurance coverage, and fishing income. The findings offer a multi-layered view of program relevance and effectiveness in improving material well-being and livelihood stability.

A comparison of socio-economic indicators between fisherfolk beneficiaries (FBs) and non-beneficiaries (NBs) revealed compelling evidence of the F/BPLD Program's contribution to the livelihood outcomes of recipient households. Across all domains, FBs consistently reported higher weighted mean scores, with several outcome variables showing statistically significant advantages—particularly for income and resilience.

As summarized in Table X, FBs exhibited significantly higher values in both fishing income ($W = 215,175.0$, $p < 0.001$) and total income ($W = 221,273.5$, $p < 0.001$), confirming the program's direct impact on improving the financial capital of beneficiary households. A statistically significant difference was also observed in the insurance coverage index ($W = 259,388.5$, $p = 0.0172$), suggesting improved access to formal risk-reduction mechanisms among FBs. These findings held true across both weighted and unweighted statistics, lending robustness to the observed patterns.

Visual comparison of outcome distributions further substantiates these findings. As shown in Figure #, boxplots for each index reveal that FBs tend to outperform their non-beneficiary counterparts across all dimensions of material well-being. Frequency

distributions for income brackets also illustrate a clear shift toward higher income categories among FBs (Figures # and #), affirming the positive trajectory of income-based outcomes.

Table 2. Weighted Mann–Whitney U Test Results (FB vs. NB)

Variable	U Statistic	p-value	Interpretation
Durables Ownership Index	219351.50	0.398	Not Significant
Living Condition Index	217518.00	0.560	Not Significant
Real Properties Index	216066.50	0.713	Not Significant
Insurance Coverage Index	229263.00	0.014	Significant
Material Well-being Index	220974.50	0.280	Not Significant
Fishing Income Monthly Average	250866.50	0.000	Significant
Total Income Monthly Average	247267.00	0.000	Significant

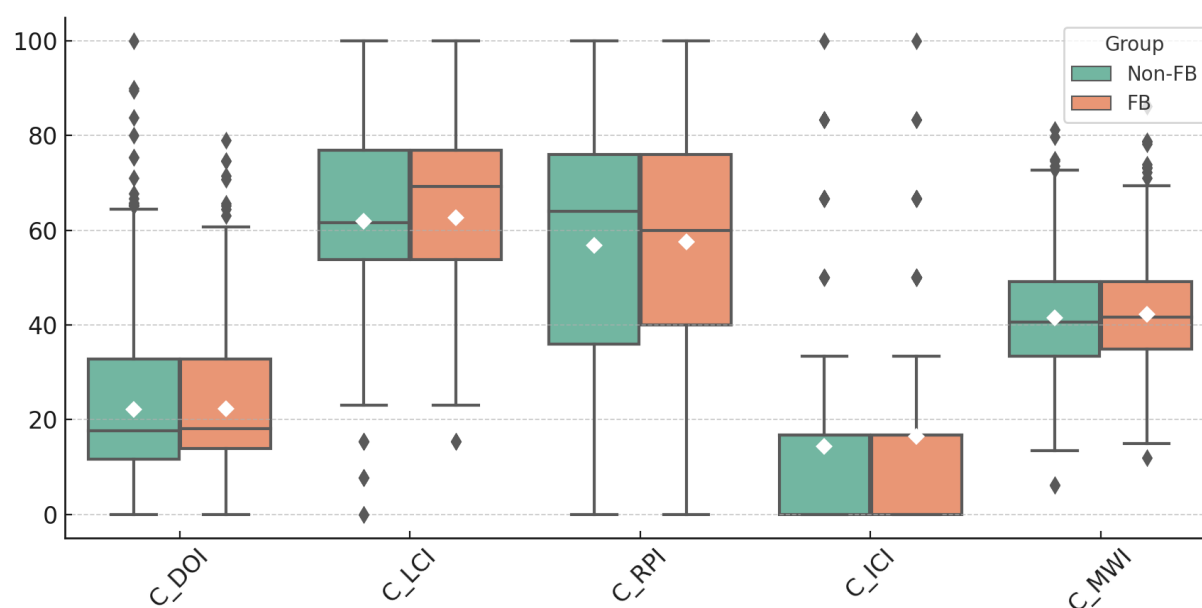


Figure 3.3.1a – Comparative central tendencies of the socio-economic indices.

Disaggregated results by geographic cluster reinforce these overall conclusions. In Northern Luzon, FBs posted significantly higher fishing income ($W = 30,638.0$, $p < 0.001$)

and total income ($W = 31,215.0$, $p = 0.0005$). While other indicators were not statistically significant, FBs maintained higher mean scores across all indices, indicating directional gains. Cluster-specific visual summaries (boxplots for indices and frequency distributions for FIMA and TIMA) are appended.

In Southern Luzon, FBs again showed significantly higher fishing income ($W = 17,730.0$, $p = 0.0034$) and total income ($W = 17,918.5$, $p = 0.0009$), along with the only significant improvement in the real property index ($W = 21,712.5$, $p = 0.0498$), suggesting emerging gains in physical assets among recipients. In Mindanao, the pattern was similarly consistent: fishing income ($W = 36,465.5$, $p = 0.0018$) and total income ($W = 36,746.5$, $p = 0.0002$) were both significantly higher among FBs, accompanied by directionally positive—but non-significant—differences in all other indices.

In contrast, the **Visayas** region did not register statistically significant differences in any variable. However, FBs still consistently posted higher mean values in both income and non-income indicators, indicating a positive trend that may become more pronounced with a longer implementation timeline or larger sample.

Despite the absence of statistical significance in the indices of durables ($W = 265,917.0$, $p = 0.2867$), living conditions ($W = 281,952.0$, $p = 0.2517$), real property ($W = 270,013.0$, $p = 0.6815$), and the composite material well-being index ($W = 268,490.5$, $p = 0.4676$), FBs consistently outperformed NBs in terms of average values. These findings suggest that while income improvements are immediate and measurable, gains in structural and resilience-related indicators may follow a slower trajectory—a trend commonly observed in post-intervention livelihood studies (Béné et al., 2016; Porio, 2019; World Bank, 2018).

These results align with the Sustainable Livelihoods Framework (SLF), which positions financial and physical capital as foundational components of sustainable household well-being. By directly enhancing income and enabling modest but steady

improvements in asset accumulation and resilience, the F/B Pagbabago Program demonstrates both short-term impact and long-term potential.

3.3.2 Descriptive Analysis of Gain Scores

To complement the comparative evaluation, the study examined gain scores, defined as the difference between current and benchmark values for each index. This approach captures the relative improvement experienced by households over time. FBs consistently demonstrated higher mean gain scores than NBs across all indices, with particularly large differentials observed in the DOI (6.48 vs. 1.34), LCI (6.50 vs. 3.36), and MWI (4.14 vs. 1.22). These findings further validate the program's role in facilitating upward socio-economic mobility.

In contrast, the mean gain in the RPI was negligible for both groups (0.00 for FBs; –0.64 for NBs), indicating limited change in real property assets during the observation period. Gains in ICI were modest (1.66 for FBs; –0.22 for NBs), reflecting incremental improvements in formal risk protection mechanisms such as insurance coverage.

Further disaggregation by geographic cluster revealed location-specific patterns. For instance, FBs in Mindanao posted the highest gains in durables (14.01), while those in Southern Luzon and the Visayas excelled in LCI (9.98 and 9.07, respectively). MWI gains were most pronounced in Southern Luzon (5.18) and the Visayas (4.61). In contrast, NBs in Mindanao showed negative or minimal gain scores in several domains. Comparative cluster-based Gain Scores for the various socio-economic indices is appended.

These regional patterns underscore the importance of contextualizing livelihood outcomes, as variations may stem from localized implementation dynamics, socio-economic baselines, or market access conditions.

3.3.3 Logit Analysis: Are FBs Better Off Socioeconomically?

To determine which socio-economic changes most strongly predict program participation, a weighted binary logistic regression was conducted with FB status as the dependent variable. The model included five independent variables: gain scores in durables (GS_DOI), living condition (GS_LCI), real property (GS_RPI), insurance coverage (GS_ICI), and fishing income monthly average (FIMA). These indicators represent primary, non-redundant dimensions of socio-economic well-being.

The model results indicate that gains in durables ownership (OR = 1.04, $p < 0.001$), living condition (OR = 1.02, $p < 0.001$), and fishing income (OR = 2.27, $p < 0.001$) were significant predictors of FB status. These findings highlight the salience of tangible asset accumulation and income improvement in differentiating program beneficiaries from non-beneficiaries. While positive, gain scores for real property and insurance coverage did not reach statistical significance, suggesting that these aspects may require a longer horizon or additional intervention layers to produce substantial effects.

Table 3 – Binary logistic regression coefficients and result of significance testing,

Predictor	B (Logit)	Odds Ratio	p-value	95% CI Lower (OR)	95% CI Upper (OR)
const	-1.502	0.223	0.000	0.1606	0.3086
Gains in Durables	0.035	1.036	0.000	1.0245	1.0468
Gains in Basic Utilities	0.018	1.018	0.000	1.0084	1.0281
Gains in Real Properties	0.014	1.014	0.152	0.9948	1.0341
Gains in Insurance	0.009	1.009	0.185	0.9959	1.0212
Fishing Income Average	0.797	2.219	0.000	1.7394	2.8311

The overall model achieved a McFadden R^2 of 0.0702, indicating modest but meaningful explanatory power. The results reinforce that the F/BP program was associated with targeted improvements in key socio-economic domains directly aligned with its core objectives.

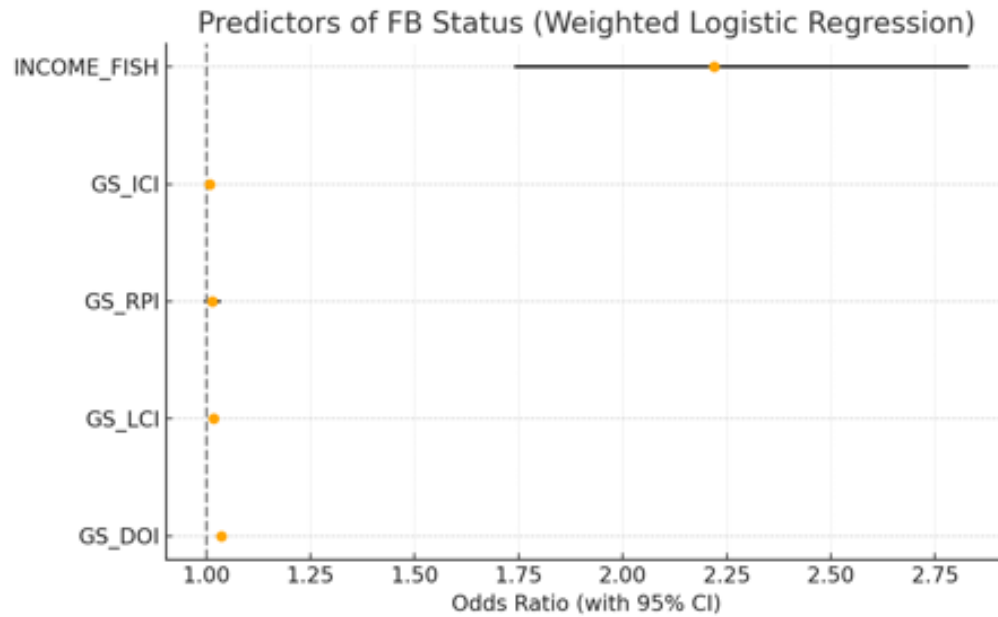


Figure 3.3.3a – Visual depiction of odds ratios for the test of association between FB status and predictor variables (FIMA and socioeconomic Gain Scores).

3.4 Perception-based Socioeconomic Impact

3.4.1 Fisherfolk-Beneficiaries' Experiences

The program experience of fisherfolk-beneficiaries (FBs) sheds light on the implementation reach, operational challenges, and overall delivery quality of the boat distribution initiative. The data highlights both the program's regional scope and the technical realities experienced by FBs on the ground.

Geographic Coverage. The FB program had a broad geographic footprint, with most boats distributed in the Visayas (37.72%) and Northern Luzon (36.65%). Smaller shares were recorded in Southern Luzon (13.17%) and Mindanao (12.46%). The stronger presence in the Visayas may reflect the region's heavy reliance on coastal livelihoods, though the comparable percentage in Northern Luzon suggests a strategic effort to expand support in traditionally underserved coastal provinces.

Year of Receipt. Program rollout occurred in four waves between 2016 and 2019, with a striking concentration in 2017 (48.40%) and 2018 (36.83%). This means that more than 85% of FBs had received their boats within a two-year window—suggesting a rapid scaling of distribution during this period. However, only 4.63% received boats in 2019, indicating a steep decline, which may correspond to shifts in funding cycles or implementation priorities.

Ownership Scheme. The program predominantly promoted a shared ownership model. A significant 74.56% of FBs received boats that were jointly used, while only 25.44% received boats for sole use. This collective approach may have been intended to promote cooperation and cost-sharing. However, it may also introduce complexities in management, scheduling, and maintenance responsibilities—issues that are not uncommon in resource-sharing arrangements in small-scale fisheries.

Boat Design. Most boats distributed were motorized (93.59%), with non-motorized units accounting for a mere 6.41%. This high ratio of motorization represents a substantial upgrade for many traditional fishers, likely enhancing their mobility and reach. In terms of design, the majority of recipients (84.34%) received the standard FPR Fishing Boat model, followed by the FPR Pakura (13.35%) and a handful of multi-purpose or speedboat units. Only 0.53% fell into the “other” design category. These figures show that the program was highly standardized in technical design, which may have simplified procurement and training, but also risks overlooking local boat preferences or sea conditions.

Current Condition. Just over half of the distributed boats were reported to be in optimal condition (54.63%), while the remaining 45.37% were under maintenance. This is a substantial portion—nearly 1 in 2 boats—facing operability issues. It raises important questions about durability, quality control, or post-distribution support mechanisms.

Maintenance Issues. Among the boats under maintenance, the most frequently cited cause was a broken engine (15.84%), followed by typhoon damage (13.88%) and general deterioration (10.32%). A small portion (5.34%) cited other reasons. Interestingly, 54.63% of respondents marked “N/A” in this category, consistent with those whose boats remain in good condition. The prevalence of engine failure and weather-related damage suggests a need for more robust pre-distribution orientation and possibly stronger after-sales servicing or insurance coverage mechanisms.

3.4.2 Perceived Quality of F/BPLD Program

3.4.2.1 Perceptions on Program Relevance

Relevance. This construct captures the extent to which the F/B Pagbabago Program is perceived by FBs as appropriate, responsive, and beneficial to their actual livelihood needs. It also reflects beneficiaries’ awareness, engagement, and willingness to promote

the program within their community. Seven impact-oriented statements were used to assess this construct, and responses were captured using a five-point Likert scale. The distribution of responses is shown in Figure XX.

Respondents expressed generally favorable views on the relevance of the program. A majority (63.70%) strongly agreed and 19.04% agreed with the statement, **“I was already aware of the F/B Pagbabago Livelihood Program before becoming a beneficiary.”** This suggests that the program had a strong presence in the community prior to actual engagement. Perceived outcomes were similarly positive. 64.06% of FBs strongly agreed and 24.91% agreed that **“The program has brought noticeable improvements to my livelihood or personal situation,”** indicating widespread recognition of its immediate value. Regarding implementation challenges, most FBs disagreed with the statement, **“My engagement in the program has been affected by recurring challenges or limitations.”** Specifically, 43.06% strongly disagreed and 24.91% disagreed, suggesting a relatively smooth program experience for most FBs.

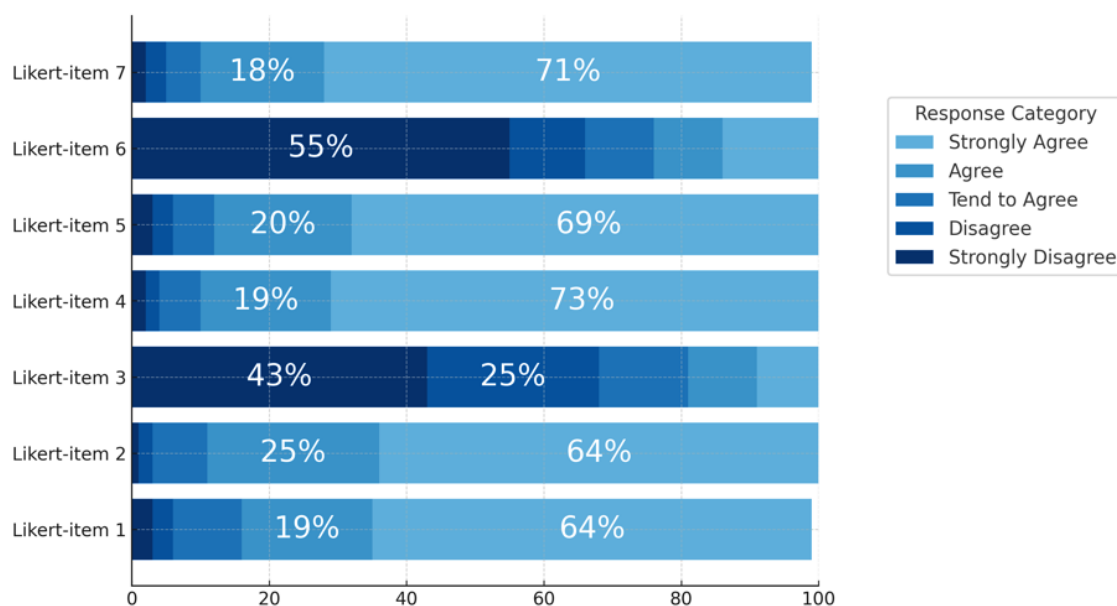


Figure 3.4.2a Distribution of fisherfolk-beneficiaries' responses to Relevance Likert statements.

The program's contribution to long-term livelihood was also affirmed. 72.60% strongly agreed and 18.86% agreed that **“The program contributes meaningfully to the promotion of sustainable livelihoods for fisherfolk.”** In a related item, **“Being part of the program has led to clear improvements in my income or means of living,”** was strongly agreed to by 68.68% of respondents, with an additional 19.57% agreeing. When asked whether **“Receiving a boat from the program has not significantly enhanced my fishing productivity or financial status,”** 54.98% strongly disagreed and 10.85% disagreed, suggesting that most respondents rejected the idea of program ineffectiveness. Lastly, program endorsement was high, with 71.35% strongly agreeing and 18.33% agreeing that **“I am willing to recommend the program to other members of my fishing community.”**

3.4.2.2 Perceptions on Social Impact

Social Impact. This construct reflects the program's contribution to the broader social well-being of fisherfolk, including safety at sea, improvements in livelihood collaboration, social status, and knowledge sharing within the fishing community. It highlights how program participation influences beneficiaries' roles and relationships within their social environment. Figure XX presents the distribution of responses.

The FBs viewed the program's social impact positively. Most of them (63.88%) strongly agreed and 26.87% agreed that “The fiberglass boats provided through the program have enhanced our safety and well-being at sea.” With respect to productivity, 58.36% strongly agreed and 25.09% agreed that “Since receiving the boat, my volume of fish catch has increased significantly.” The program also promoted social capital. 51.07% strongly agreed and 27.59% agreed with the statement, “The program has strengthened cooperation and solidarity among fisherfolk in my area.”

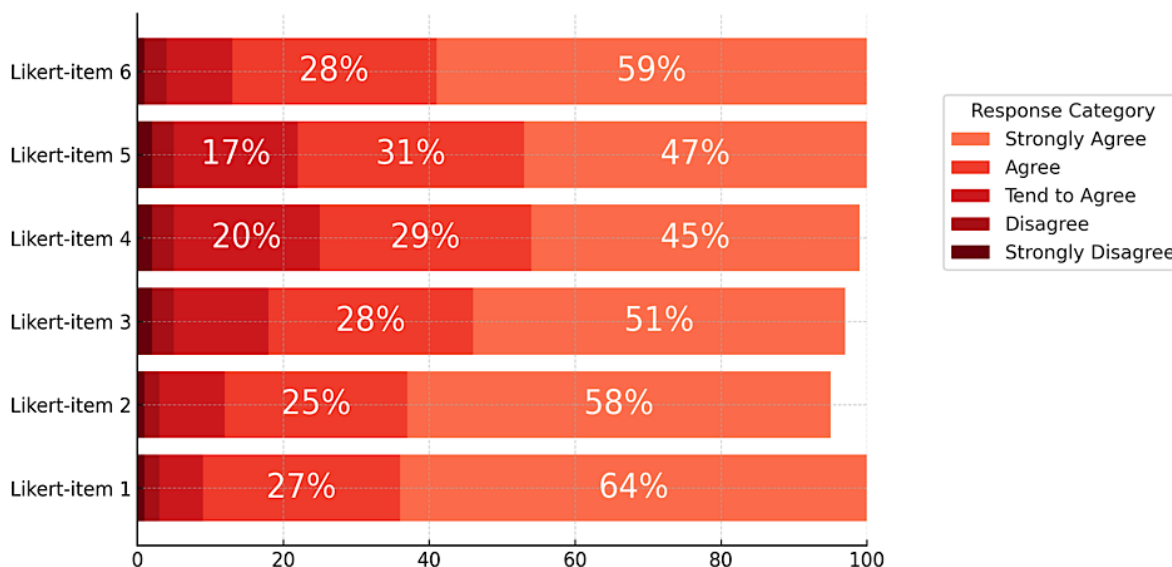


Figure 3.4.2b – Distribution of fisherfolk-beneficiaries’ responses to Social Impact Likert statements.

Social recognition was also noted. 44.84% strongly agreed and 29.36% agreed that “Participation in the program has improved how I am regarded in my local community.” When asked whether “The program encourages the exchange of knowledge and practical skills among fisherfolk,” 45.19% strongly agreed and 30.96% agreed. Finally, a combined 86.14% of respondents affirmed the broader benefit of the program, agreeing with the statement, “The program has contributed to overall improvements in both social and economic aspects of my life.”

3.4.2.3 Perceptions on Program Sustainability

This construct captures fisherfolk-beneficiaries’ perceptions of the long-term viability of the program, including its environmental impact, quality and durability of its assets, institutional implementation, and systemic support for sustainability. It also gauges their views on whether the program should be continued. The overall response pattern is illustrated in Figure XX.

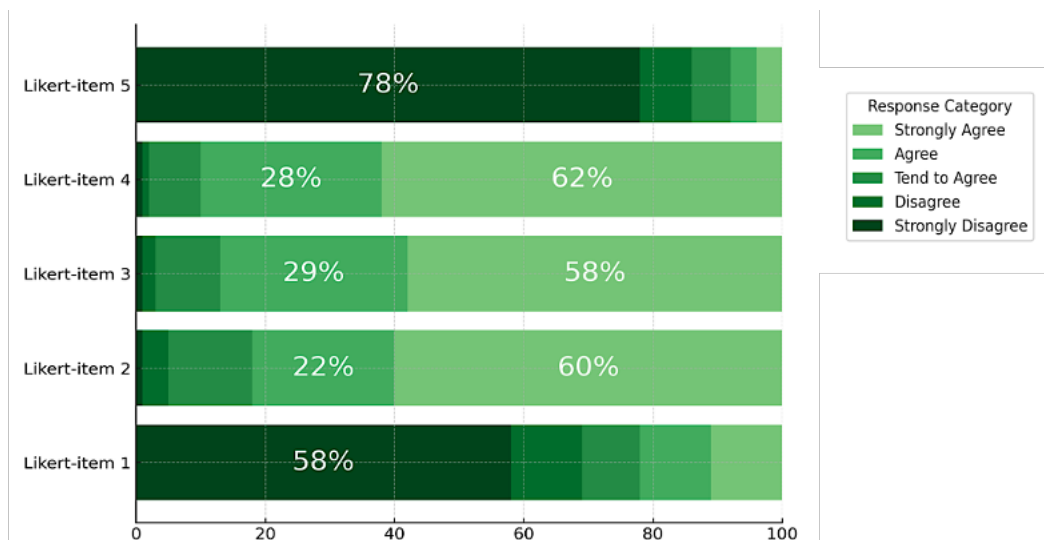


Figure 3.4.2c – Distribution of fisherfolk-beneficiaries’ responses to Sustainability Likert statements.

While overall perceptions of sustainability were favorable, responses were more varied than in earlier dimensions. Regarding the statement, “**The boat distribution component of the program may negatively affect the marine environment or fish stocks,**” 28.29% strongly disagreed and 18.15% disagreed, while 30.78% expressed agreement — denoting environmental concerns among some respondents. The quality of the boats was highly rated: 52.67% strongly agreed and 29.72% agreed that “**The fiberglass boats are high-quality, cost-effective, and reliable for long-term fishing use.**” On the implementation structure, 49.82% strongly agreed and 28.65% agreed that “**The program is being implemented in an organized, consistent, and well-structured manner across levels.**”

Most respondents also believed in the program’s long-term contribution, with 55.87% strongly agreeing and 29.36% agreeing that “**The program is helping to build systems that support the long-term sustainability of the fishing sector.**” Finally, when asked whether “**The program should be discontinued due to lack of support from local government units,**” 67.62% strongly disagreed and 14.59% disagreed—demonstrating overwhelming support for program continuation.

3.4.2.4 Composite Perceptions Scores Analysis

The boxplots of the composite scores distribution on the relevance, social impact, and sustainability of the F/BP Program based on perception is illustrated in Figure ##. The composite score for Relevance ranged from 1.57 to 5.00, with a mean of 4.30 (SD = 0.59). This indicates that the program is Highly Relevant (according to the empirically-derived rating scale), characterized by the tight concentration of the composite mean values around the upper end of the scale (see boxplot).

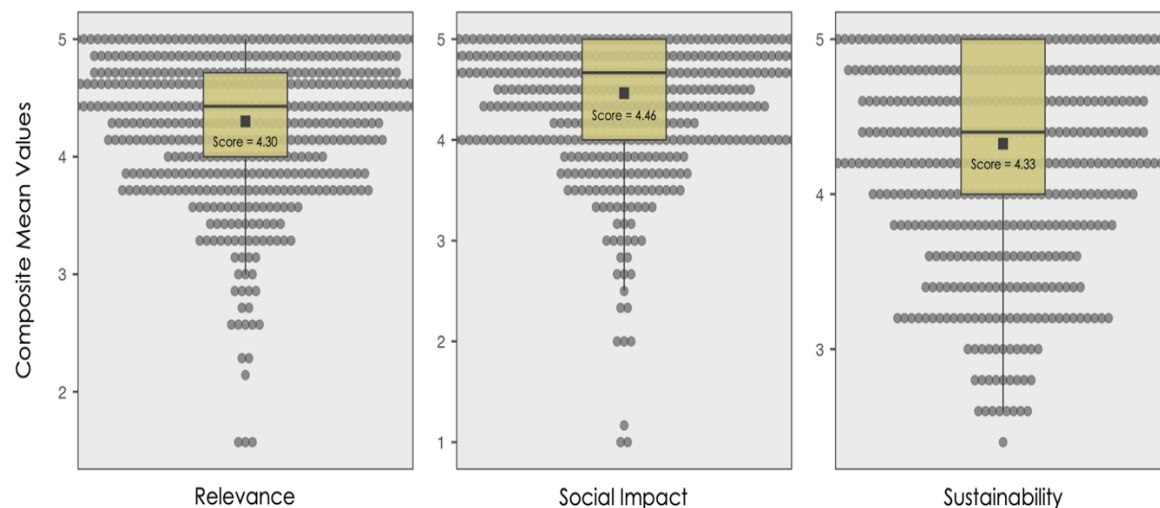


Figure 3.4.2d – Distribution of composite mean values on program relevance, social impact, and sustainability as perceived by fisherfolk-beneficiaries.

The composite score for Social Impact of the program was 4.46 (SD = 0.64), interpreted as Very Strong Social Impact based on the empirical scale. The boxplot depicts a left-skewed distribution, suggesting that most FBs rated the social outcomes of the program near the upper bound. The composite score for Sustainability was 4.33 (SD = 0.66), which translates to the adjectival rating of Sustainable. The boxplot demonstrates a positive distribution of the composite mean values, with some dispersion around the upper-middle quartile

The present findings suggest that FBs hold a generally positive perception of the boat distribution program administered by BFAR in terms of its relevance, social impact, and sustainability. Respondents strongly agreed that the program contributed to livelihood improvement, safety at sea, and community collaboration—key components of a well-functioning small-scale fisheries support system.

These perceptions are consistent with previous evaluations of boat distribution and livelihood programs in Southeast Asia, where material support combined with institutional engagement significantly improved fisherfolk satisfaction and community stability (Pomeroy et al., 2010). Similarly, a study by Walley et al. (2019) on post-disaster boat rehabilitation programs in Indonesia found that when beneficiaries were involved in planning and received operational support, their perceived relevance and satisfaction improved markedly. Similarly, the perceived impact of F/BPP on social well-being aligns with evidence from Armitage et al. (2009), emphasizing that the social capital generated through community-based resource initiatives strengthens resilience and promotes collective agency. It appeared that the FBs appreciate not only the material assets provided but also the enhanced coordination and interaction it fostered within their local groups.

However, some skepticism was reflected in responses to sustainability-focused Likert items, particularly regarding ecological concerns and institutional support. While the overall perception of program sustainability was positive, a notable share of respondents—approximately one-third—expressed agreement with concerns about the potential ecological impact of increased fishing effort. It is plausible that the large-scale distribution of fiberglass boats contributed to this perceived increase in effort, as the intervention enhanced fisherfolk's access to fishing grounds and potentially encouraged more frequent or extended trips. This aligns with previous findings in the literature, where researchers have cautioned that poorly regulated asset distribution programs can inadvertently contribute to resource overexploitation or ecosystem stress (Béné et al.,

2016). This matter should be considered in future evaluations of program sustainability. Similarly, Gutiérrez et al. (2011) stressed the need for co-management and ecological safeguards in fisheries development initiative to ensure sustainable outcomes.

This mixed but largely positive perception underscores the importance of continuous capacity-building, institutional support, and environmental safeguards alongside asset-based interventions (Ratner et al., 2014). The program's favorable ratings in relevance and social impact provide a strong foundation for long-term support, but sustainability-related concerns must be addressed through monitoring and adaptive governance.

3.4.3 Program Evaluation Index

The Program Evaluation Index (PEI) serves as an integrative measure capturing fisherfolk-beneficiaries' (FBs) overall assessment of the program. It consolidates responses across three major performance dimensions—Relevance (30%), Social Impact (30%), and Sustainability (40%)—thereby providing a multidimensional snapshot of perceived program value and effectiveness.

As shown in Figure 3.4.3a, the computed PEI values ranged from 1.72 to 5.00, with a mean of 4.36 (± 0.54). This average falls within the upper bounds of the rating scale and corresponds to a “Very Satisfactory” interpretation using the established adjectival rubric. The boxplot further illustrates that the distribution is positively skewed, with the majority of observations concentrated near the upper range and only a few isolated low outliers. The central tendency is tightly clustered, and the median closely aligns with the mean, suggesting consistency in favorable program assessments across respondents.

This interpretation is fortified via frequency histogram of PEI scores. The distribution displays a distinct rightward skew, with the tallest bars positioned between the 4.5 to 5.0 range. The curve overlay indicates that while the distribution is not strictly normal, it is unimodal and leans heavily toward the upper end of the scale. This visual pattern suggests

that the overwhelming majority of FBs rated the program positively, and only a small subset expressed moderate or low satisfaction.

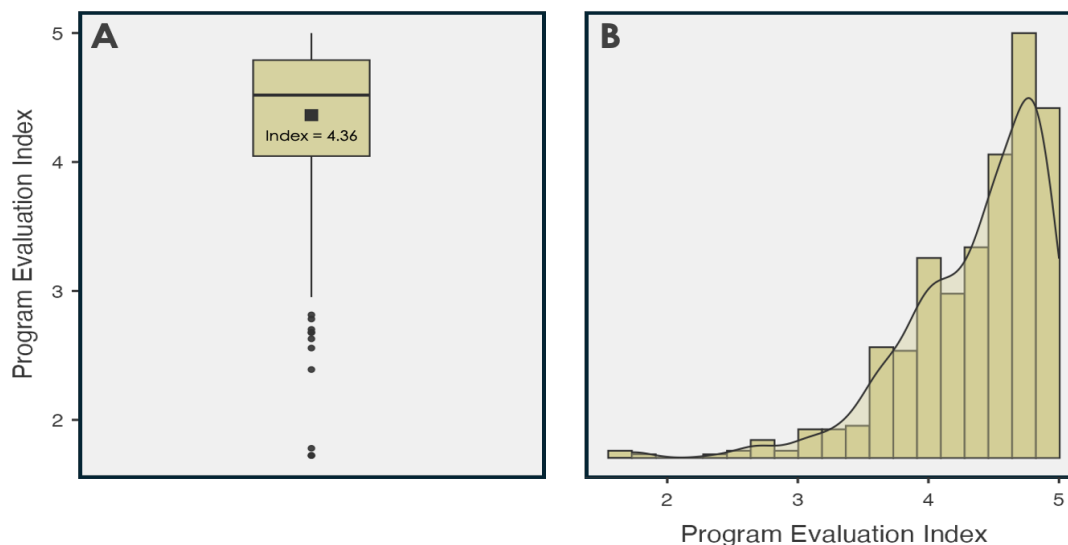


Figure 3.4.2e – Visual summary of the Performance Evaluation Index (PEI) derived based on perceived relevance, social impact, and sustainability of the F/B Pagbabago Program.

These findings underscore the high level of perceived relevance, tangible social contributions, and perceived sustainability of the program from the viewpoint of FBs. The index captures more than general satisfaction—it encapsulates how the program resonated with the needs of target beneficiaries, its effectiveness in producing lasting impacts, and the extent to which it is seen as viable in the long term. The skew toward the higher end also implies a degree of positive consensus among respondents, reinforcing the credibility of the program’s perceived performance.

Taken together, the results of the PEI analysis provide strong empirical support for the program’s value as perceived by its direct beneficiaries. While the presence of a few lower-end scores warrants attention to possible pockets of dissatisfaction or implementation gaps, the overall index affirms that, for most FBs, the program met or exceeded expectations in terms of design, outcomes, and prospects for continuity.

The Program Evaluation Index (PEI) provides a unified measure of program performance from the perspective of fisherfolk-beneficiaries. It was computed as a weighted index of the three thematic dimensions (Relevance = 30%, Social Impact = 30%, Sustainability = 40%). The PEI ranged from 1.72 to 5.00, with a mean of 4.36 ± 0.54 (Figure ##A). With reference to the adjectival rating, this index translates to a Very Satisfactory perception of the program. This is also visually evident with the histogram in Figure 3.4.2e, which shows a right-skewed distribution of individual indices (with few lower outliers) derived from each respondent.

3.4.3 Predictors of Perceived Program Quality

To further refine the understanding of factors that influence fisherfolk-beneficiaries' perception of the F/B Pagbabago Program, a multiple linear regression model was constructed using a set of potential predictors. The model included categorical indicators including cluster, fisherfolk category, boat ownership, boat condition, boat design, fishing income bracket, and a continuous variable for material well-being index. The results of the regression analysis is summarized in Table 4.

Table 4 – Multiple regression coefficients and result of significance testing,

Predictor Variable	Reference	Comparison	Estimate	β Coefficient	P-value
X = CLUSTER	Mindanao	Northern Luzon	- 0.1954	- 0.3626	0.0110
	Mindanao	Southern Luzon	- 0.1315	- 0.2440	0.1511
	Mindanao	Visayas	- 0.0292	- 0.0542	0.6990
X = FF CATEGORY	Atisanal	Non-artisanal	0.1007	0.1868	0.0729
X = BOAT OWNERSHIP	Shared	Sole owner	- 0.1264	- 0.2346	0.0198
X = BOAT CONDITION	Under maintenance	Optimal Condition	0.1841	0.3415	< 0.0001
X = BOAT DESIGN	Other boat designs	FPR fishing boat	0.1039	0.1928	0.1002
X = FISHING INCOME	Below ₱10,000	₱10,000–₱20,000	0.0746	0.1384	0.1568
		Above ₱20,000	0.0261	0.0484	0.8348
X = MW INDEX	Covariate	–	- 0.0008	- 0.0175	0.6817

The model was statistically significant, $F(10, 551) = 4.33$, $p < .001$, with an R^2 of 0.0728, indicating that approximately 7.3% of the variation in PEI scores could be attributed to the predictors included. Although the overall explained variance remained modest, several variables showed noteworthy effects.

Cluster (i.e., geographic area) was a significant predictor of PEI ($p = 0.0107$). Pairwise contrast revealed that BFs from Northern Luzon reported significantly lower PEI than those in Mindanao ($p = 0.0110$), and in the Visayas ($p = 0.0032$). No other cluster comparisons were significant. Boat Ownership was statistically significant ($\beta = -0.1264$, $p = 0.0198$), with respondents under a shared ownership scheme reporting significantly higher PEI scores than those classified as sole owners. Boat Condition remained the strongest predictor in the model ($\beta = 0.1841$, $p < 0.001$), with those reporting their boats in optimal condition rating the program higher than those with boats classified as “under maintenance.”

Other predictors—namely boat design, fishing income, material well-being index, and fisherfolk classification—were not found to be statistically significant at 0.05 confidence level. Although the ₱10,000–₱20,000 bracket under fishing income showed a positive trend ($\beta = 0.0746$) relative to the reference category (below ₱10,000), it did not reach significant level ($p = 0.1568$). Similarly, MWI was not significant ($p = 0.6817$).

These findings support earlier insights that functional utility (boat condition), as well as certain organizational and/or contextual factors (location and ownership scheme) may play a more prominent influence in shaping program evaluation, compared to broader socio-economic classifications from the perspective of the fisherfolk-beneficiaries.

The multiple linear regression results revealed that the condition of the fishing boat, the nature of boat ownership, and regional location were the most significant predictors of how the FBs rated the program. Specifically, FBs with boats in good working condition

reported significantly higher PEI, as did those in shared ownership arrangements. Additionally, FBs in Northern Luzon reported significantly lower program evaluation scores as compared to their counterparts in Mindanao and the Visayas.

The present findings affirm the central role of asset functionality in shaping program perception. According to Allison and Ellis (2001), the physical quality and usability of the livelihood assets directly influence beneficiaries' assessment of program effectiveness. A boat in poor condition limits income potential and reinforces dissatisfaction, whereas a fully operational asset reinforces feelings of empowerment and progress.

The significance of shared ownership as a positive predictor may appear counterintuitive, but similar outcomes have been observed in participatory models. As Pomeroy and Andrew (2011) note, shared arrangements can foster a greater sense of accountability and group solidarity, especially when beneficiaries are organized through fisherfolk associations. This social cohesion may buffer resource-related stress and elevate program satisfaction.

Meanwhile, the area-based disparity in PEI reflects broader spatial inequalities in fisheries development. It is likely that FBs from Mindanao and the Visayas benefited from better program implementation or more responsive local governance—both of which have been identified as critical enablers of success in decentralized livelihood programs (Bueno & Basurto, 2009). On the other hand, Northern Luzon's lower ratings may indicate logistical or institutional gaps that undermined program outcomes in that region.

While fishing income and material well-being were expected to influence PEI, their lack of significance in the regression model suggests that subjective program evaluation may rather hinge more on immediate asset usability and local context, in contrast to generalized socio-economic status. The present finding complements those of Chambers and Conway (1992), who argued that the success of livelihoods is often perceived through practical, short-term lenses rather than long-term capital accumulation.

3.5 General Discussion

The results of the socio-economic impact evaluation of the F/B Pagbabago Program (F/BPLDP) offer robust empirical support for the value of asset-based livelihood interventions in improving the well-being of small-scale fisherfolk. In line with the Sustainable Livelihoods Framework (SLF), the study demonstrates that access to productive assets (i.e., boats), when combined with appropriate institutional mechanisms, can significantly enhance the financial, physical, and social capital of marginalized communities (Chambers & Conway, 1992; DFID, 1999).

The observed increase in fishing and household income among FBs affirms previous findings that capital inputs contribute directly to livelihood productivity in resource-dependent sectors (Allison & Ellis, 2001; Béné et al., 2003). Comparable results were reported in evaluations of post-disaster boat replacement programs in Sri Lanka and Indonesia, where access to fishing assets accelerated economic recovery (Walley et al., 2019; Subasinghe & Sumanadasa, 2016). Regional disparities in income and material outcomes also support the observations of Bueno and Basurto (2009), who emphasized the role of governance quality and institutional engagement in shaping regional variation in fisheries development outcomes.

Improvements in material well-being, as captured through the Durables Ownership Index (DOI), Living Condition Index (LCI), and Insurance Coverage Index (ICI), were statistically significant and indicate positive shifts in household resilience. The ownership of essential consumer durables such as mobile phones, washing machines, and motorcycles increased notably between the benchmark and current periods, reflecting the enabling effects of asset-based programs on household mobility, productivity, and domestic welfare. These gains resonate with empirical studies linking asset accumulation to poverty reduction and resilience (Filmer & Pritchett, 2001; Ellis, 2000).

The improvement in living conditions was likewise evident, particularly in the domains of electricity access and potable water supply. As supported by Banerjee et al. (2012) and UNDP (2016), small-scale infrastructure enhancements—such as better access to commercial electricity, regulated water, and cleaner fuels—can disproportionately benefit rural poor households by reducing health risks and time burdens. However, while marginal gains in internet access were documented, digital connectivity remains low overall, underscoring a persistent infrastructure gap.

By contrast, the Real Property Index (RPI) did not improve remarkably, reinforcing the structural limitations faced by FBs in securing formal tenure and upgrading housing in informal or hazard-prone coastal settlements. This stagnation aligns with broader evidence on the slow evolution of property ownership among low-income households in peri-urban and coastal zones (Ballesteros & Egana, 2016; Kiddle, 2011).

The analysis of insurance coverage revealed modest but notable gains in enrollment, especially in publicly funded schemes such as GSIS and PhilHealth. Enrollment in retirement plans more than doubled between benchmark and current periods, while health insurance coverage also improved. Nevertheless, participation in private insurance options—including life and pre-need plans—remains low. This pattern is consistent with regional studies highlighting affordability and trust barriers in voluntary insurance uptake among informal sector workers (Habito & Briones, 2005; ILO, 2017; GIZ, 2015). While PhilHealth has often served as the most accessible entry point to social protection, uptake remains to be fragmented, echoing the concerns raised by Sumner and Mallet (2013).

That beneficiaries rated the program highly in terms of relevance, social impact, and sustainability further reinforces the empirical outcomes. The Program Evaluation Index (PEI), with a mean score of 4.36 (interpreted as “Very Satisfactory”), reflects a high level of perceived program value. The significance of boat condition and ownership arrangement in predicting PEI underscores that perceived utility and operational fairness

are critical to user satisfaction (Pomeroy & Andrew, 2011; Ratner et al., 2014). This finding is also congruent with co-management literature, which emphasizes collective agency and shared responsibility as foundations for socially embedded resource governance (Armitage et al., 2009; Gutiérrez et al., 2011).

Perception-based ratings also varied across geographic area. Visayas and Mindanao recorded higher satisfaction levels, which may be attributable to more responsive LGU facilitation or stronger community engagement during implementation. These results align with the findings from India and Southeast Asia, where decentralized interventions yielded better outcomes in areas with stronger local governance (Cruz-Trinidad, 2003; Davis et al., 2016). Conversely, lower ratings in Northern Luzon reflect logistical and administrative bottlenecks, as documented in fragmented archipelagic contexts (FAO, 2022).

By incorporating benchmark vs. current comparisons, the study operationalizes SLF's emphasis on dynamic livelihood trajectories (Scoones, 1998; Carney, 2003). Rather than relying solely on static indicators—such as current income or asset count—the study adopts a dynamic impact evaluation approach by computing gain scores across time. This allows for a more nuanced understanding of resilience and program responsiveness, in line with Sabates-Wheeler and Devereux's (2011) call for adaptive governance framework. Metrics can obscure trends and volatility; in contrast, gain scores illuminate progress, stagnation, or regression within specific livelihood capitals over time.

This research contributes to the global literature on inclusive rural transformation by emphasizing that asset-based interventions, though impactful, must be embedded within broader institutional ecosystems that address maintenance, continuity, and risk reduction (Barrett et al., 2021). The lack of baseline data collection posed a methodological constraint, prompting the use of recall-based benchmarks—a pragmatic but imperfect substitute. Future iterations of the F/BPLDP would benefit from routine tracking of targeted fisherfolk, baseline documentations, and longitudinal follow-up, in accordance with global best practices in impact evaluation (Gertler et al., 2011; World Bank, 2020).

4.0 SYNTHESIS and IMPLICATIONS

This section presents a synthesis of key findings drawn from the various components of the evaluation, highlighting their broader implications for policy, program design, and community-level development. By integrating empirical results with theoretical insights, it offers a critical interpretation of the F/B Pagbabago Program's performance and outlines recommendations for enhancing the sustainability and inclusivity of similar livelihood interventions.

4.1 Summary of Findings

- 4.11. A total of 1,322 fisherfolk respondents participated in the survey, comprising 562 FBs and 760 NBs. FBs were older on average (mean age = 50.87) and the majority across both groups were male. Typical household size was 4–6 members. Fishing income was low for both FBs and NBs, with nearly 30% of FBs and over 50% of NBs earning below ₱10,000/month. Slightly higher total income was noted due to supplemental sources. Most respondents were long-time fishers using small, motorized boats. Challenges included bad weather, engine issues, and fishing bans. Fewer FBs reported high competition, hinting at improved reach post-intervention;
- 4.12. Access to social services was relatively similar for both FBs and NBs, as indicated by comparable Social Services Index scores (FB = 82.0; NB = 83.5, $p = 0.289$). Both groups reported strong access to education, health, and financial programs. Cluster-level analysis revealed higher access scores among NBs in Northern and Southern Luzon, with no significant difference in the Visayas and Mindanao—pointing to localized implementation variation rather than a programmatic gap;
- 4.13. Nearly half of FBs received boats in 2017 (48.40%) and another 36.83% in 2018, showing rapid program roll-out. Most units were distributed in the Visayas (37.72%) and Northern Luzon (36.65%), and 74.56% of the boats were

under shared ownership. The boats were mostly motorized (93.59%) and built using the standard FPR design (84.34%). At the time of the survey, 54.63% of boats were in good condition, while 45.37% required maintenance, primarily due to mechanical damage or weathering;

- 4.14. FB households reported significant improvements in durable asset ownership. Motorcycle ownership increased from 10.82% to 17.85%, and cellphone ownership from 25.19% to 33.66%. Other gains were seen in appliances such as washing machines (from 9.61% to 16.26%). These trends indicate enhanced domestic comfort and improved mobility, especially in daily household and livelihood functions;
- 4.15. FBs showed marked improvement in access to basic utilities. Access to mineral or faucet water increased, while dependence on deepwells and surface water sources declined. Electricity access improved substantially, with grid connection rising from 68.7% to 86.2%. While firewood remained the dominant cooking fuel, the share using LPG or electric stoves grew. Internet access remained limited but showed a small increase from 6.9% to 9.6%;
- 4.16. Gains in real property ownership were marginal. Homeownership rose from 71.7% to 73.1%, and ownership via full payment or inheritance slightly increased. Informal acquisition declined. The share of FBs living in concrete houses improved modestly, while those in non-concrete dwellings dropped slightly. Ownership of house-and-lot combinations declined, while ownership of undeveloped lots increased, suggesting a shift in asset diversification;
- 4.17. Insurance enrollment remained low but improved slightly. Combined enrollment in retirement plans (SSS + GSIS) increased from 23.0% to 49.6%, driven mainly by GSIS expansion. Health insurance coverage (PhilHealth + others) grew from 49.6% to 55.0%. Life insurance (2.8% to 8.2%) and pre-need

plans (3.0% to 7.8%) also saw modest gains. However, most FBs remained underinsured or lacked sustained access to financial protection;

- 4.18. The Material Well-being Index (MWI) showed a significant increase from 38.16 to 42.30 ($p < 0.001$). DOI rose from 15.81 to 22.29, LCI from 56.13 to 62.63, and ICI from 14.71 to 16.37—all statistically significant. The RPI remained static at 57.48. The proportion of FBs with 'moderately high' well-being rose from 14.8% to 22.2%, while those in the 'low' category decreased by half, signaling meaningful progress in material conditions;
- 4.19. FBs outperformed NBs in all socio-economic indices. Significant differences were confirmed in fishing income ($p < 0.001$), total income ($p < 0.001$), and ICI ($p = 0.0172$). Although not statistically significant, FBs had higher mean scores in DOI, LCI, RPI, and MWI. These trends underscore the program's positive impact on income and asset accumulation;
- 4.110. Cluster-level analysis revealed significant differences favoring FBs in fishing and total income in Northern Luzon, Southern Luzon, and Mindanao. Southern Luzon also posted significantly higher real property scores for FBs. No statistically significant differences were observed in the Visayas, though FBs still led in average index scores;
- 4.111. Visual distribution plots illustrated that FBs shifted into higher income and well-being categories. These figures showed a higher frequency of FBs in moderate-to-high income brackets and better MWI categories. This affirms that tangible improvements in FB conditions are taking place in line with the Sustainable Livelihoods Framework (SLF), especially in financial and physical capital domains;

- 4.112. FBs reported higher mean gain scores across all material well-being indices. DOI (6.48 vs. 1.34), LCI (6.50 vs. 3.36), and MWI (4.14 vs. 1.22) posted the highest differentials compared to NBs. Gains in ICI were moderate, while RPI changes were negligible. Notable regional differences were observed, with Mindanao excelling in DOI and Southern Luzon and Visayas showing strong LCI and MWI performance;
- 4.113. Binary logistic regression identified gain in DOI (OR = 1.04, $p < 0.001$), gain in LCI (OR = 1.02, $p < 0.001$), and FIMA (OR = 2.27, $p < 0.001$) as significant predictors of FB status. Gains in RPI and ICI were not significant. The model's McFadden $R^2 = 0.0702$ suggests moderate predictive strength, confirming that physical and financial asset gains define program participation;
- 4.114. The Program Evaluation Index (PEI), computed as a weighted score of Relevance (30%), Social Impact (30%), and Sustainability (40%), had a mean of 4.36 ± 0.54 . This corresponds to a 'Very Satisfactory' evaluation. The distribution was right-skewed, with most respondents assigning high satisfaction ratings and only a few expressing dissatisfaction; and
- 4.115. Multiple linear regression results showed that boat condition ($\beta = 0.1841$, $p < 0.001$), ownership scheme ($\beta = -0.1264$, $p = 0.0198$), and geographic cluster ($p = 0.0107$) were significant predictors of PEI. Respondents with functional boats and those under shared ownership reported higher satisfaction. Regional differences also influenced evaluation, with lower scores in Northern Luzon.

4.2 Concluding Remarks

The study evaluated the socio-economic impact of the F/B Pagbabago Livelihood Development Program on fisherfolk-beneficiaries (FBs) through an integrated framework combining objective indicators and perception-based assessments. Utilizing comparative statistics, gain score analysis, and regression modeling, the results confirm that the program contributed positively to the material well-being, livelihood resilience, and overall satisfaction of FB households. These effects were observed across multiple domains, suggesting the program's capacity to generate both tangible and perceived benefits.

At the structural level, fisherfolk-beneficiaries (FBs) demonstrated significant improvements in income and multiple dimensions of material well-being. FBs consistently outperformed non-beneficiaries in fishing income, total household income, and insurance coverage, with statistically significant differences in select indices such as ICI and TIMA. Gain score analysis confirmed that FBs posted higher gains across nearly all indices, especially in consumer durables and basic utilities—critical components of household quality of life. Although changes in real property ownership were more limited, and insurance coverage remained generally low, the overall Material Well-being Index (MWI) rose significantly from the benchmark to the current period. These trends confirm incremental socio-economic gains consistent with the program's intended goals.

Regression analyses further reinforced the connection between program participation and socio-economic uplift. Gains in Consumer Durables Ownership Index (DOI) and Living Condition Index (LCI), along with higher fishing income, emerged as statistically significant predictors of FB status, while changes in real property and insurance coverage did not show significant influence. These results emphasize the critical role of tangible and functional livelihood assets in supporting household recovery and resilience. Consistent with the Sustainable Livelihoods Framework (SLF), the findings

highlight that access to financial and physical capital remains foundational to sustaining rural livelihoods and promoting inclusive development outcomes.

Complementing the objective data, fisherfolk-beneficiaries gave the program a “very satisfactory” overall rating based on their assessment of relevance, social impact, and sustainability. The Program Evaluation Index (PEI) averaged 4.36 and exhibited a right-skewed distribution, indicating that the majority of respondents reported high levels of satisfaction. Satisfaction was significantly influenced by the condition of the boat and the ownership scheme, with higher PEI scores among those with functional boats and those under shared arrangements. Regional variation in PEI further reflected the role of local governance quality and field-level facilitation—factors that appeared to enhance community trust and the perceived success of the program.

The research findings of this project affirm that the F/B Pagbabago Program provided a meaningful entry point for material recovery and livelihood improvement among small-scale fisherfolk. While limitations remain—particularly in housing security, long-term asset ownership, and sustained insurance coverage—the results validate that even modest, well-targeted capital infusions can produce significant gains across key domains of economic well-being. Moving forward, the program’s quality may be further enhanced by tailoring its implementation to the unique needs of each geographic cluster, ensuring post-distribution technical support, and fostering sustained collaboration among LGUs, related agencies and community stakeholders beyond the initial point of asset transfer.

4.3 Recommendations

Based on the key findings of this study, the following recommendations are advanced to improve the implementation, sustainability, and equity of the F/B Pagbabago Program and similar asset-based fisheries interventions:

1. Institutionalize Post-Distribution Maintenance and Support Mechanisms

The study revealed that many distributed fiberglass boats required repair or maintenance at the time of the survey, raising concerns about asset durability and operational sustainability. It is therefore recommended that BFAR develop localized maintenance support systems, in partnership with LGUs or organized fisherfolk groups, to ensure access to timely repairs, spare parts, and technical services. Strengthening aftercare will help extend the lifespan of the boats and minimize economic disruptions among beneficiaries.

2. Promote and Strengthen Shared Ownership Models

Beneficiaries under shared boat ownership arrangements consistently reported higher satisfaction levels than those with sole ownership. Shared models were found to foster greater cooperation, reduce individual burdens, and promote collective accountability. Future program rollouts should actively encourage organized group ownership schemes, with clear guidelines and capacity-building mechanisms to support equitable access and co-management.

3. Address Regional Disparities Through Context-Sensitive Implementation

The results highlighted geographic variation in program outcomes, with some regions—particularly Northern Luzon—experiencing lower levels of satisfaction and fewer livelihood gains compared to others. These differences may reflect varying capacities in local governance, infrastructure, or community engagement.

To address this, BFAR is encouraged to adopt more flexible, context-sensitive implementation strategies tailored to local realities, and to institutionalize feedback mechanisms that support adaptive program management.

4. Integrate Financial Risk Protection in Program Design

Despite improvements in income and material assets, most fisherfolk remained underinsured or entirely excluded from formal protection mechanisms. While some gains were observed in state-based retirement plans, access to health and life insurance remained extremely limited. Future implementations should integrate financial literacy training and proactive assistance in registration, enabling beneficiaries to access inclusive and sustainable risk protection instruments.

5. Sustain and Expand Monitoring and Evaluation Systems

The study affirmed the value of robust, multi-dimensional impact evaluation in guiding policy decisions. It is recommended that BFAR institutionalize regular monitoring and evaluation (M&E) processes that include both objective indicators and beneficiary perceptions. M&E frameworks should be participatory, context-aware, and designed to generate timely insights that support effective program targeting, delivery, and refinement.

6. Support Complementary Programs to Enhance Housing Tenure Security

While improvements were observed in income and material possessions, indicators related to property ownership and house structure showed limited progress. Many fisherfolk households continue to face tenure insecurity and lack the resources for housing upgrades. It is, therefore, recommended that BFAR explore partnerships with national housing and land tenure agencies to deliver complementary support that promotes long-term household stability.

7. Establish Baseline Data Systems to Enable Longitudinal Impact Tracking

One of the key challenges encountered in this study was the absence of pre-intervention baseline data, which necessitated reliance on recall-based evaluation. To enhance the validity and comparability of future impact assessments, BFAR is strongly advised to institutionalize the collection of baseline information at the point of program enrollment. These data can be integrated into digital registries of the beneficiaries and linked with existing platforms like the FishR to support continuous impact monitoring over time.

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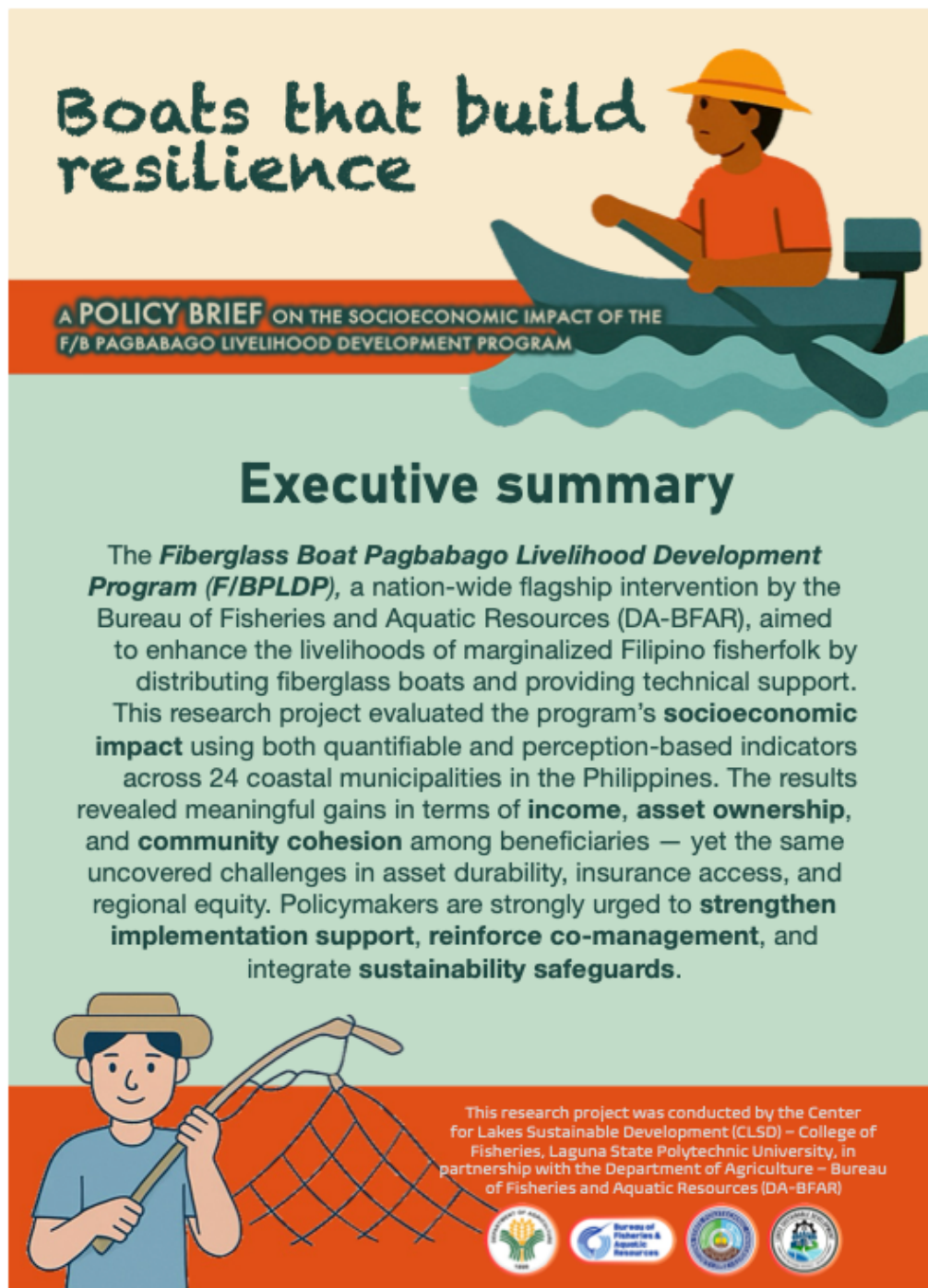
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APPENDICES

APPENDIX A. POLICY BRIEF (DRAFT)



Context & Purpose

Municipal **fisherfolk** remain one of the most **socio-economically** vulnerable groups in the country. The **F/BPLDP** was envisioned as an **asset-based** solution to address these long-standing **gaps**. However, qualifying its true **effectiveness** and **sustainability** has been proven difficult in the absence of a **systematic, empirical** evaluation.

This research fills that gap by assessing both **tangible improvements** in **material well-being** and the **perceived quality** of program delivery from the **fisherfolk's perspective**.



About the study

COVERAGE



1,322 fisherfolk from 24 coastal municipalities across 10 regions

COMPARISON



562 fisherfolk-beneficiaries vs. 760 non-beneficiaries

APPROACH



Quasi-experimental, with inverse probability weighting

INDICATORS



Income, material well-being indices, access to social services, perception-based ratings

SCIENTIFIC BASIS



Anchored in the Sustainable Livelihoods Framework

TIMELINE



2016–2019 (benchmark period), survey conducted in 2024





Key findings

Boosted Income, Measurable Gains



FBs reported significantly higher fishing and total household income (monthly average) compared to Non-FBs. Gains were most substantial in Mindanao and Southern Luzon.



Modest but Worthwhile Asset Accumulation

Significant improvements were observed in ownership of household appliances and mobility assets. However, gains in real property ownership were minimal.

Underinsurance Remains a Barrier



While PhilHealth coverage slightly improved, overall access to formal insurance remained low. The majority of FBs are still financially vulnerable.



Perceptions Are Strongly Positive

FBs rated the program as Highly Relevant, having Very Strong Social Impact, and broadly Sustainable. Boat condition and shared ownership emerged as key predictors of satisfaction.

Implementation Matters



Regional disparities were evident. FBs from Visayas and Mindanao gave higher ratings compared to those from Northern Luzon. Asset condition strongly influenced perception of program effectiveness.

Recommendations

Institutionalize Maintenance Support Mechanisms. High rates of boat breakdowns (over 45%) call for a formal aftercare system – possibly through LGUs or fisherfolk cooperatives.

1

2 **Expand and Support Shared Ownership Models.** Shared arrangements yielded higher satisfaction and social capital. Encourage cooperatives or small fisherfolk associations to promote collective stewardship.

Strengthen Local Implementation Capacities. Address disparities in regional outcomes by localizing program delivery and ensuring better alignment with LGU support systems.

3

4 **Link Fisherfolk to Inclusive Financial Protection.** Facilitate access to PhilHealth, SSS, and microinsurance schemes by embedding registration support and education in future program cycles.



APPENDIX B. SAMPLE SURVEY QUESTIONNAIRE



Republic of the Philippines
LAGUNA STATE POLYTECHNIC UNIVERSITY
COLLEGE OF FISHERIES
Center for Lakes Sustainable Development



SURVEY QUESTIONNAIRE (BENEFICIARY)

PART A. PAHINTULOT at KUMPIDENSYALIDAD NA KLAUS

Pagbati!

Ako po si [____], isang mananaliksik mula sa College of Fisheries, Laguna State Polytechnic University (COF-LSPU). Sa kasalukuyan, kami ay nagsasagawa ng isang survey para sa pag-aaral na pinamagatang "Socioeconomic Impact Evaluation of the F/B Pagbabago Livelihood Development Program of the Department of Agriculture - Bureau of Fisheries and Aquatic Resources," sa ilalim ng implementasyon ng Center for Lakes Sustainable Development.

Nais po naming humingi ng ilang impormasyon. Maaari niyo po bang kumpirmahin kung si [____] ay isang benepisaryo ng Programang F/B Pagbabago? Kung gayon, maaari ko po bang hingin ang inyong oras para sa isang panayam?

Ang lahat ng impormasyong matatanggap mula sa panayam na ito ay mahigpit na kumpidensyal at gagamitin lamang para sa layunin ng pag-aaral na ito. Walang impormasyon na maaaring magpahayag ng inyong pagkakakilanlan ang ilalathala sa anumang anyo. Hindi rin namin ibabahagi ang inyong personal na impormasyon sa anumang ikatlong partido. Bukod dito, ang inyong pakikilahok sa survey na ito ay hindi magiging basehan kung kayo ay makatatanggap o hindi ng mga benepisyo o serbisyo. Ang inyong pakikilahok sa survey na ito ay lubos na boluntaryo. May karapatan kayong tumutol sa pagproseso ng inyong personal na datos, karapatan sa pag-access sa inyong personal na datos, at karapatan na ipaayos ang inyong personal na datos.

SUMASANG-AYON BA KAYO NA MAKAPANAYAM?

OO

HINDI

PART B. DEMOGRAPHIC PROFILE OF THE RESPONDENT

B01. BUONG PANGALAN

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B02. KASALUKUYANG
TIRAHAN

APRILYED		PANGALAN		GITNANG PANGALAN (bung mayam)	
HOUSE NO.	STREET	PURCH	BIGY.	MUNICIPALITY/CITY	PROVINCE

B03. EDAD

B04. KAARAWAN

PARA SA MGA SUMUSUNOD NA ITEM, I-TSEK (✓) ANG KAHON NA TUGMA SA INYONG SAGOT

B05. KASARIAN 1 ☐ Lalaki 2 ☐ Babae 3 ☐ Walang particular na kasarian 4 ☐ Ayaw ipaalam

B06. CIVIL STATUS 1 ☐ Single 2 ☐ Kasal 3 ☐ Nagsasama 4 ☐ Balo/Hiwalay

B07. ANTAS NG EDUKASYON NA NAKAMIT 1 ☐ Walang pormal na edukasyon 2 ☐ Elementarya 3 ☐ High School
4 ☐ Kolehiyo 5 ☐ Post-graduate

B08. LAKI NG SAMBAHAYAN 1 ☐ 1-3 miyembro 2 ☐ 4-6 miyembro 3 ☐ 7-9 miyembro 4 ☐ ≥10 miyembro

PART C. KABUWANG BUWANANG KITA SA SAMBAHAYAN

C01. INAASAHAANG KABUWANG BUWANANG KITA BAGO TANGGAPIN ANG BANGKA
!!! ISAMA ANG KITA MULA SA LAHAT NG PINAGKUKUNAN

1 ☐ ≤P10K 2 ☐ >P10K to P20K 3 ☐ >P20K-P30K
4 ☐ >P30K-P40K 5 ☐ >P40K-P50K 6 ☐ >P50K

C02. KASALUKUYANG KABUWANG BUWANANG KITA MULA SA PANGINGISDA

1 ☐ ≤P10K 2 ☐ >P10K-P20K 3 ☐ >P20K-P30K
4 ☐ >P30K-P40K 5 ☐ >P40K-P50K 6 ☐ >P50K

C03: PANGALAWANG PINAGKUKUNAN NG KITA
!!! LAGYAN NG TSEK ANG LAHAT NG POSSIBLENG SAGOT

COL.1 ☐ Wala COL.2 ☐ Negosyo COL.3 ☐ Skilled labor
COL.4 ☐ Farming COL.5 ☐ Pension COL.6 ☐ Iba pa: _____

C04. KABUWANG BUWANANG KITA MULA SA PANGALAWANG PINAGKUKUNAN

1 ☐ ≤P10K 2 ☐ >P10K-P20K 3 ☐ >P20K-P30K
4 ☐ >P30K-P40K 5 ☐ >P40K-P50K 6 ☐ >P50K

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PARA SA MGA SUMUSUNOD NA ITEM, I-TSEK (✓) ANG KAHON NA TUGMA SA INYONG SAGOT

D. TALAAAN NG MGA PAG-AARI

D1. PAG-AARING SASAKYAN

BAGO MATANGGAP ANG BANGKA
(Kung meron, tukuyin ang bilang o dami)

SA KASALUKUYAN
(Kung meron, tukuyin ang bilang o dami)

- D1.1 Bisikleta
D1.2 Motorsiklo
D1.3 Tricycle
D1.4 Kotse
D1.5 Jeep
D1.6 Truck
D1.7 Iba pa: _____

- D1.1A ☐ Wala ☐ Meron : _____
D1.2A ☐ Wala ☐ Meron : _____
D1.3A ☐ Wala ☐ Meron : _____
D1.4A ☐ Wala ☐ Meron : _____
D1.5A ☐ Wala ☐ Meron : _____
D1.6A ☐ Wala ☐ Meron : _____
D1.7A ☐ Wala ☐ Meron : _____

- D1.1B ☐ Wala ☐ Meron : _____
D1.2B ☐ Wala ☐ Meron : _____
D1.3B ☐ Wala ☐ Meron : _____
D1.4B ☐ Wala ☐ Meron : _____
D1.5B ☐ Wala ☐ Meron : _____
D1.6B ☐ Wala ☐ Meron : _____
D1.7B ☐ Wala ☐ Meron : _____

D2. KAGAMITAN SA BAHAY

- D2.1 Television Set
D2.2 DVD Player
D2.3 Washing Machine
D2.4 Air-conditioner
D2.5 Electric Fan
D2.6 Refrigerator/Freezer
D2.7 Gas/Electric Stove
D2.8 Electric Heater
D2.9 Home Furniture
D2.10 Iba pa: _____

- D2.1A ☐ Wala ☐ Meron : _____
D2.2A ☐ Wala ☐ Meron : _____
D2.3A ☐ Wala ☐ Meron : _____
D2.4A ☐ Wala ☐ Meron : _____
D2.5A ☐ Wala ☐ Meron : _____
D2.6A ☐ Wala ☐ Meron : _____
D2.7A ☐ Wala ☐ Meron : _____
D2.8A ☐ Wala ☐ Meron : _____
D2.9A ☐ Wala ☐ Meron : _____
D2.10A ☐ Wala ☐ Meron : _____

- D2.1B ☐ Wala ☐ Meron : _____
D2.2B ☐ Wala ☐ Meron : _____
D2.3B ☐ Wala ☐ Meron : _____
D2.4B ☐ Wala ☐ Meron : _____
D2.5B ☐ Wala ☐ Meron : _____
D2.6B ☐ Wala ☐ Meron : _____
D2.7B ☐ Wala ☐ Meron : _____
D2.8B ☐ Wala ☐ Meron : _____
D2.9B ☐ Wala ☐ Meron : _____
D2.10B ☐ Wala ☐ Meron : _____

D3. KAGAMITANG TEKNOLOHIYA

- D3.1 Mobile/Cell Phone
D3.2 Landline/telephone
D3.3 Computer/Laptop
D3.4 Others: _____

- D3.1A ☐ Wala ☐ Meron : _____
D3.2A ☐ Wala ☐ Meron : _____
D3.3A ☐ Wala ☐ Meron : _____
D3.4A ☐ Wala ☐ Meron : _____

- D3.1B ☐ Wala ☐ Meron : _____
D3.2B ☐ Wala ☐ Meron : _____
D3.3B ☐ Wala ☐ Meron : _____
D3.4B ☐ Wala ☐ Meron : _____

D4. KAGAMITAN SA
PANGKABUHAYAN

- D4.1 Fishing Boat
D4.2 Tractor
D4.3 Spading Forks
D4.4 Wagons
D4.5 Sprinklers
D4.6 Plows
D4.7 Others: _____

- D4.1A ☐ Wala ☐ Meron : _____
D4.2A ☐ Wala ☐ Meron : _____
D4.3A ☐ Wala ☐ Meron : _____
D4.4A ☐ Wala ☐ Meron : _____
D4.5A ☐ Wala ☐ Meron : _____
D4.6A ☐ Wala ☐ Meron : _____
D4.7A ☐ Wala ☐ Meron : _____

- D4.1B ☐ Wala ☐ Meron : _____
D4.2B ☐ Wala ☐ Meron : _____
D4.3B ☐ Wala ☐ Meron : _____
D4.4B ☐ Wala ☐ Meron : _____
D4.5B ☐ Wala ☐ Meron : _____
D4.6B ☐ Wala ☐ Meron : _____
D4.7B ☐ Wala ☐ Meron : _____

E. KALAGAYAN NG PAMUMUHAY/ESTILO NG PAMUMUHAY

BAGO MATANGGAP ANG BANGKA

SA KASALUKUYAN

- E1. Pangunahing pinagkukunan ng inuming tubig
E2. Tubig para sa iba pang gamit sa bahay
E3. Suplay ng kuryente
!!! LAGYAN NG TSEK ANG LAHAT NG POSIBLENG SAGOT
E4. Pangunahing ginagamit sa pagluluto
E5. Subskripsyon sa internet na postpaid

- 1 ☐ Gripo 2 ☐ Mineral water
3 ☐ Balon 4 ☐ Iba pa: _____
1 ☐ Balon 2 ☐ Water district
3 ☐ Ilog/lawa 4 ☐ Iba pa: _____
0 ☐ Wala 1 ☐ Solar Power
2 ☐ Komersyal 3 ☐ Iba pa: _____
1 ☐ Kahoy 2 ☐ Uling
3 ☐ LPG 4 ☐ Iba pa: _____
0 ☐ Hindi 1 ☐ Oo : _____

- 1 ☐ Gripo 2 ☐ Mineral water
3 ☐ Balon 4 ☐ Iba pa: _____
1 ☐ Balon 2 ☐ Water district
3 ☐ Ilog/lawa 4 ☐ Iba pa: _____
0 ☐ Wala 1 ☐ Solar Power
2 ☐ Komersyal 3 ☐ Iba pa: _____
1 ☐ Kahoy 2 ☐ Uling
3 ☐ LPG 4 ☐ Iba pa: _____
0 ☐ Hindi 1 ☐ Oo : _____

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PART F. ARI-ARIAN

	BAGO MATANGGAP ANG BANGKA	SA KASALUKUYAN
F1. Bahay at Lupa		
F1.1 Estado ng Pag-mamayari	1 ☐ Renta I-SKIP ANG F1.2 3 ☐ Iba pa: _____ I-SKIP ANG F1.2	2 ☐ Pag-aari 1 ☐ Renta I-SKIP ANG F1.2 3 ☐ Iba pa: _____ I-SKIP ANG F1.2
F1.2 Uri ng Akwisisyon	1 ☐ Binayaran sa buong halaga 3 ☐ Minana 2 ☐ Inutang/Hulugan 4 ☐ Iba pa: _____	1 ☐ Binayaran sa buong halaga 3 ☐ Minana 2 ☐ Inutang/Hulugan 4 ☐ Iba pa: _____
F1.3 Uri ng Gusali	1 ☐ Isang palapag (bungalow), gawa sa bato/semento 2 ☐ Dalawang palapag, gawa sa bato/semento 3 ☐ Apartment-type, gawa sa bato/semento 4 ☐ Bahay kubo/ di sementado / iba pa: _____	1 ☐ Isang palapag (bungalow), gawa sa bato/semento 2 ☐ Dalawang palapag, gawa sa bato/semento 3 ☐ Apartment-type, gawa sa bato/semento 4 ☐ Bahay kubo/ di sementado / iba pa: _____
F2. Iba pang ari-arian	1 ☐ Bahay at lupa 3 ☐ Kondominyum 2 ☐ Bahay lang 4 ☐ Iba pa: _____	1 ☐ Bahay at lupa 3 ☐ Kondominyum 2 ☐ Bahay lang 4 ☐ Iba pa: _____

PART G. MIYEMBRO NG INSURANCE

	BAGO MATANGGAP ANG BANGKA	SA KASALUKUYAN
G1. SSS	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo
G2. GSIS	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo
G3. Philhealth	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo
G4. Pre-need Insurance	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo
G5. Life Insurance	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo
G6. Health Insurance	0 ☐ Hindi 1 ☐ Oo	0 ☐ Hindi 1 ☐ Oo

PART H. ACCESS SA SERBISYONG PANG-SOCIAL WELFARE SERVICES

H1. Pantawid Pamilyang Pilipino Program or 4Ps	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H. Retirement Pension	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H3. SPES or Special Program for the Employment of Students	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H4. Any livelihood program (excluding SLP)	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H5. Any free training program for business or employment	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H6. Any feeding program for children	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H7. Any scholarship program	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado
H8. Any employment or cash-for-work program	1 ☐ Oo	2 ☐ Hindi	3 ☐ Di sigurado

PART I. KARANASAN BILANG MANGINGISDA

I1. Gaano ka na katagal sa pangangisda?

1 ☐ Wala pang isang taon 2 ☐ 1 hanggang 5 years
3 ☐ 6 hanggang 10 taon 4 ☐ Mahigit 10 taon

I2. Miyembro ka ba ng anumang asosasyon o club sa pangangisda?

1 ☐ Hindi 2 ☐ Oo, mangyaring tukuyin: _____

I3. Kabilang ka, ilan ang mangingisda sa iyong sambahayan?

Pakilagay ang eksaktong bilang: _____

I4. Alin sa mga sumusunod ang naglalarawan ng lawak o antas ng pangangisda na sangkot ka at ang iyong pamilya?

1 ☐ Ang artisinal o tradisyunal na pangangisda ay ang paggamit ng simpleng kagamitan sa pangangisda at/o maliit na bangka
2 ☐ Pagpapatakbo ng medium-scale commercial fishing gamit ang mga bangka may motor
3 ☐ Malawakang komersyal/industriyal na pangangisda

I5. Anong uri ng sasakyang pandagat ang karaniwang ginagamit mo?

1 ☐ Wala, hindi ako gumagamit ng anumang sasakyang pandagat
2 ☐ Hindi-motorisadong kano o bangka
3 ☐ Maliit na motorisadong bangka na may mababang makina
4 ☐ Saktong sukat na motorisadong bangka
5 ☐ Malaking bangka / Barko / Fishing Trawler

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16. Uri ng kagamitan sa pangangisda na iyong ginagamit?

!!! LAGYAN NG TSEK ANG LAHAT NG POSSIBLENG SAGOT

- 1 ☐ Gill net/pante 2 ☐ Fish trap / bubu 3 ☐ Push and pull net
4 ☐ Hook and line/kawil 5 ☐ Iba pa, mangyaring tukuyin: _____

17. Anong mga kagamitang pangkaligtasan ang iyong dala sa pangangisda?

!!! LAGYAN NG TSEK ANG LAHAT NG POSSIBLENG SAGOT

- 1 ☐ Life jacket 2 ☐ Life buoy 3 ☐ Flashlight
4 ☐ Cellphone 5 ☐ Iba pa, mangyaring tukuyin: _____

18. Anong mga problema o isyu ang iyong naranasan sa pangangisda?

!!! LAGYAN NG TSEK ANG LAHAT NG POSSIBLENG SAGOT

- 1 ☐ Masamang kondisyon ng panahon 4 ☐ Kompetisyon sa pangangisda
2 ☐ Bawal mangisda 5 ☐ Iba pa: _____
3 ☐ Problema sa makina ng bangka

PART J. PAGLAHOK SA PROGRAMANG F/B PAGBABAGO (PARA SA MGA BENEPISYARYO NG PROGRAMA LAMANG)

PETSA NG PAGTANGGAP NG BANGKA:

- -

BUHAY AGOS TACH

KUNDISYON NG BANGKA:

☐ F ☐ NF

F = FUNCTIONAL / NF = NOT FUNCTIONAL

J1. Ano ang kasunduan sa paggamit ng bangka?

- 1 ☐ Sariling pag-aari 2 ☐ May kasama sa pagmamay-ari
Kung gayon, ilan kayong gumagamit ng bangka? _____

J2. Anong uri ng bangka ang iyong naatanggap?

- 1 ☐ May motor 2 ☐ Walang motor

J3. Ano ang disenyo ng bangka?

- 1 ☐ FRP Pakura 2 ☐ FRP Speed boat 3 ☐ FRP Tuna
4 ☐ FRP Fishing boat 5 ☐ FRP Basnigan and Multi-purpose boat
6 ☐ Iba pa, mangyaring tukuyin: _____

J4. Nasa pinakamainam na kondisyon ba ng bangka sa kasalukuyan?

- 1 ☐ Oo 2 ☐ Hindi
Kung hindi, itala ang dahilan:
1 ☐ Nasira dahil sa bagyo
2 ☐ Nasira ang motor
3 ☐ Nasa pagsasaayos
4 ☐ Iba pa: _____

PARA SA SUMUSUNOD NA MGA ITEM, I-TSEK (✓) ANG KAHON PARA SA PAG RATE NG IYONG ANTAS NG PAGSANG-AYON

J5. KAHALAGAHAN NG PROGRAMA

	LUBOS NA SUMASANG-AYON	5	4	3	2	1	LUBOS NA HINDI SUMASANG-AYON
J5.1 Ako ay pamilyar sa F/B Pagbabago Livelihood Program, kahit bago pa ako naging isa sa mga benepisyaryo ng bangka.		☐	☐	☐	☐	☐	
J5.2 Napansin ko ang positibong pagbabago at/o mga nakabubuting epekto bilang resulta ng F/B Livelihood Program.		☐	☐	☐	☐	☐	
J5.3 Patuloy akong nakakaranas ng mga hamon at/o kakulangan kaugnay ng aking partisipasyon sa programa.		☐	☐	☐	☐	☐	
J5.4 Sa palagay ko, epektibo ang F/B Pagbabago Program sa pagtataguyod ng matatag na kabuhayan para sa mga mangangisda.		☐	☐	☐	☐	☐	
J5.5 Inaakala ko na ang aking pakikilahok sa F/B Pagbabago Program ay tunay na nagpapabuti sa aking pinagkukunan ng kabuhayan.		☐	☐	☐	☐	☐	
J5.6 Ang bangka na aking natanggap mula sa programa ay hindi gaanong nagpabuti sa aking produktibidad sa pangisdaan at kalagayan sa ekonomiya.		☐	☐	☐	☐	☐	
J5.7 Isusulong ko ang F/B Pagbabago Program sa iba pang mga miyembro ng aking lokal na komunidad ng mangangisda.		☐	☐	☐	☐	☐	

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J6. PERSEPSYONG PANLIPUNAN NG PROGRAMA

	LUBOS NA SUMASANG- AYON	5	4	3	2	1	LUBOS NA HINDI SUMASANG- AYON
J6.1 Ang mga fiberglass na bangka na ipinamahagi sa pamamagitan ng F/B Pagbabago Program ay nagtataguyod ng kaligtasan at kagalingan sa mga benepisyaryong mangingingisda.		☐	☐	☐	☐	☐	
J6.2 Ang dami ng isdang nahuhuli ay dumami sa paggamit ng bangkang mula sa F/B Pagbabago Program.		☐	☐	☐	☐	☐	
J6.3 Sa pangkalahatan, pinabuti ng F/B Pagbabago Program ang antas ng kooperasyon at pakikipagtulungan ng mga mangingingisda.		☐	☐	☐	☐	☐	

	LUBOS NA SUMASANG- AYON	5	4	3	2	1	LUBOS NA HINDI SUMASANG- AYON
J6.4 Ang pakikilahok sa F/B Pagbabago Program ay nagpabuti sa sosyal na katayuan ng mga mangingingisda sa aking lokal na komunidad.		☐	☐	☐	☐	☐	
J6.5 Ang F/B Pagbabago Program ay tumutulong sa pagpapalaganap ng pagsasalin ng kaalaman at kasanayan sa mga mangingingisda.		☐	☐	☐	☐	☐	
J6.6 Ang F/B Pagbabago Program ay naglalaan ng kontribusyon sa pangkalahatang pagpapabuti ng sosyal na kagalingan at kalagayan sa ekonomiya ng mga benepisyaryong mangingingisda.		☐	☐	☐	☐	☐	

J7. PAGPAPANATILI NG PROGRAMA

	LUBOS NA SUMASANG- AYON	5	4	3	2	1	LUBOS NA HINDI SUMASANG- AYON
J7.1 Ang pamamahagi ng mga fiberglass na bangka ay nagdudulot ng negatibong epekto sa ekosistema ng mga anyong tubig at sa dami ng isda.		☐	☐	☐	☐	☐	
J7.2 Ang mga fiberglass na bangkang ipinamahagi sa pamamagitan ng programa ay may mataas na kalidad, abot-kaya na presyo, at ligtas para sa pangmatagalang paggamit.		☐	☐	☐	☐	☐	
J7.3 Ang implementasyon ng F/B Pagbabago Program sa lokal, rehiyonal, at pambansang antas ay organisado at may layuning obhetibo.		☐	☐	☐	☐	☐	
J7.4 Ang F/B Pagbabago Program ay nagtataguyod ng isang maayos na sistema na sumusuporta sa pangmatagalang kaunlaran ng industriyang pangisdaan.		☐	☐	☐	☐	☐	
J7.5 Ang F/B Pagbabago Program ay dapat itigil dahil sa kakulangan ng suporta mula sa mga lokal na pamahalaan.		☐	☐	☐	☐	☐	

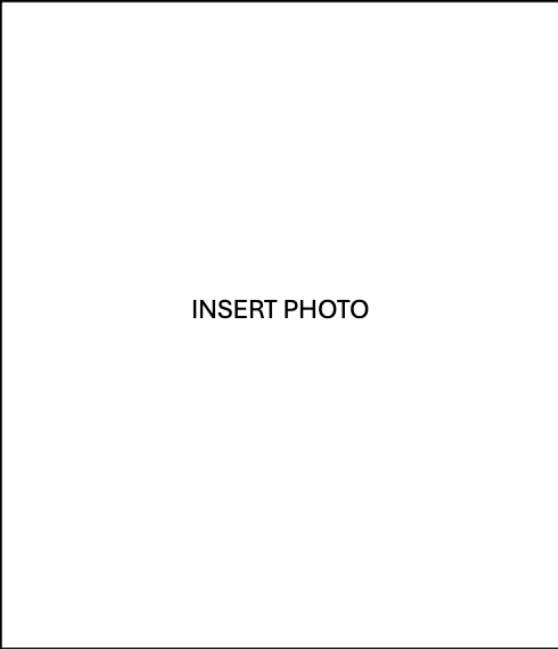
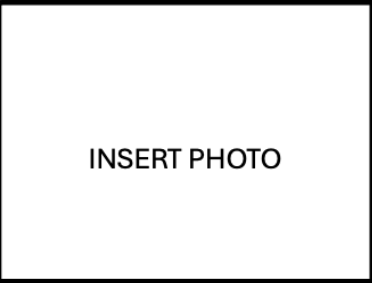
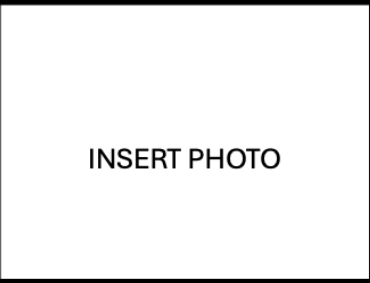
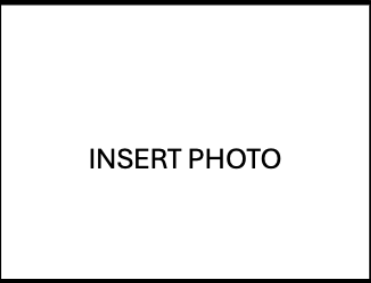
PART K. KARAGDAGAANG MGA KOMENTO AT MUNGKAHI

-- end of questionnaire --

APPENDIX C. PHOTODOCUMENTATION

		
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APPENDIX D. SUPPLEMENTARY STATISTICS
